OPTIMAL TRANSPORTS ON THE SPHERICAL DOMAINS WITH BOUNDARY AND THEIR APPLICATIONS

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Abstract. This paper is concerned with the existence and global smoothness of optimal transports between two domains Ω and Ω^* in the unit sphere $\mathbb{S}^n$, with the cost $c(x,y) = F(d(x,y))$, where $d(x,y)$ is the spherical distance of $x, y \in \mathbb{S}^n$. The function $F(\theta)$ is defined only in a subinterval $[0, d_0)$ of $[0, \pi]$ with $\lim_{\theta\to d_0} F(\theta) = \lim_{\theta\to d_0} F'(\theta) = -\infty$, and $c(x,y)$ takes $-\infty$ when $d(x,y) \geq d_0$. In general, such cost function could not be defined on the entire $\Omega \times \Omega^*$ and it would happen that $\limsup_{y\in\Omega^*\cap B_{d_0}(x)} |\nabla_x c(x,y)| = \infty$ for $x \in \Omega$, so the classical argument in optimal transport can not be applied directly. Some sharp conditions are found for the existence of optimal maps and their smoothness. In particular, when $F(\theta) = \log(\kappa\cos\theta - 1)$ ($\kappa > 1$), or $F(\theta) = \log\cos\theta$, the related optimal transportation problem is respectively the refractor problem in geometric optics, and the Aleksandrov problem of hypersurfaces with boundary in convex geometry.

1. Introduction

The optimal transportation problem is to move one distribution of mass onto another as efficiently as possible. This problem was originally proposed by Monge [39] in 1781. Major advances were made in 1940's by Kantorovich [23, 24], who introduced a relax functional and the duality in optimal transportation. In the last three decades a wide range of applications has been found in differential geometry, fluid mechanics, image recognition, and geometric optics [11, 15, 21, 45, 46, 48]. These applications have made the subject a focus of attention.

It was discovered that the far field reflection problem [48] and refraction problem [20] are both optimal transportation on the unit sphere $\mathbb{S}^n$. See also the recent book [21] for optimal transport and geometric optics. Let κ be the qoutient of the refraction indexes of two different media. Gutiérrez-Huang [20] showed that constructing a refractor in the case $\kappa > 1$ is equivalent to finding an optimal map with respect to the cost function

$$c(x,y) = \log(\kappa \cos d(x,y) - 1), \tag{1.1}$$

where $d(x,y)$ is the geodesic distance of $x, y \in \mathbb{S}^n$. On the other hand, it was proved by Oliker [40] that Aleksandrov's problem of prescribing integral Gauss curvature of closed

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convex hypersurface [1, 2] is also a mass transfer problem on $\mathbb{S}^n$ with the cost function

$$c(x,y) = \log\cos d(x,y). \tag{1.2}$$

Let Ω, Ω^* be domains in $\mathbb{S}^n$, and $f \in L^1(\Omega), g \in L^1(\Omega^*)$ be two nonnegative functions satisfying the mass balance condition

$$\int_\Omega f d\gamma_{\mathbb{S}^n} = \int_{\Omega^*} g d\gamma_{\mathbb{S}^n}, \tag{1.3}$$

where $d\gamma_{\mathbb{S}^n}$ is the standard volume element of $\mathbb{S}^n$. Denote by $\mathcal{S}_{f,g}$ the set of all measure preserving maps from (Ω, f) to (Ω^*, g). Namely $\Phi \in \mathcal{S}_{f,g}$ if it satisfies

$$\int_\Omega f h \circ \Phi d\gamma_{\mathbb{S}^n} = \int_{\Omega^*} h g d\gamma_{\mathbb{S}^n}, \quad \forall h \in C(\Omega^*). \tag{1.4}$$

The optimal transportation problem is to searching an optimal map $T \in \mathcal{S}_{f,g}$ such that

$$\mathcal{C}(T) = \sup_{\Phi \in \mathcal{S}_{f,g}} \mathcal{C}(\Phi), \tag{1.5}$$

where the total cost of a map $\Phi \in \mathcal{S}_{f,g}$ is given by

$$\mathcal{C}(\Phi) = \int_\Omega c(x, \Phi(x)) f(x) d\gamma_{\mathbb{S}^n}. \tag{1.6}$$

By the classical approach in mass transport theory [4, 8, 22], if $c = c(x,y)$ satisfies

(A1) $c \in C^1(\mathcal{U})$ for some open set $\mathcal{U} \supseteq \bar{\Omega} \times \bar{\Omega}^*$, and

$$\|c\|_{\mathrm{Lip}(\mathcal{U})} < \infty, \tag{1.7}$$

(A2) $\nabla_x c(x, \cdot) : \Omega^* \to T_x\mathbb{S}^n$ is injective for every $x \in \Omega$,

then the optimal map of (1.5) exists.

However cost functions (1.1) and (1.2) go to negative infinity as $d(x,y) \to \arccos(1/\kappa)$ and $d(x,y) \to \pi/2$ respectively, and so do not satisfy (1.7) in general. Moreover, these two cost functions even fail to be defined on the entire $\Omega \times \Omega^*$, e.g. when the $\frac{\pi}{2}$-neighborhood of Ω is a proper subset of Ω^*. In [20] the authors solved the refraction problem under a constraint on Ω and Ω^*: there exists a $\delta > 0$ such that

$$\inf_{x \in \Omega, y \in \Omega^*} x \cdot y \geq 1/\kappa + \delta. \tag{1.8}$$

Under this condition, cost function (1.1) is bounded on $\Omega \times \Omega^*$ and satisfies (1.7). Therefore the classical method [4, 8, 22] works. Since Aleksandrov's problem concerns closed hypersurface without boundary, a constraint in the type of (1.8) cannot be imposed. Aleksandrov [1, 2] finds the following condition for the existence of solution:

$$\int_\omega f d\gamma_{\mathbb{S}^n} < \int_{\omega_{\frac{\pi}{2}}} d\gamma_{\mathbb{S}^n} \ \forall \text{ non-empty and convex } \omega \subset \mathbb{S}^n, \tag{1.9}$$

where f is the prescribed function of Aleksandrov's integral Gauss curvature, and $\omega_{\pi/2}$ is the $\pi/2$ neighbourhood of ω. It turns out that Aleksandrov's condition (1.9) is also necessary.

In this paper, we study optimal transports on $\mathbb{S}^n$ with cost functions satisfying some structural conditions which include refraction problem (1.1) and Aleksandrov's problem (1.2) as examples. Our motivation is twofold: one is to weaken the constraint (1.8) for the refraction problem, and another is to resolve the Aleksandrov's problem of prescribing integral Gauss curvature for hypersurfaces that are not necessarily closed.

For a $d_0 \in (0, \pi]$, let $\mathscr{C}_{d_0}$ be a class of cost functions $c(x, y)$ on $\mathbb{S}^n \times \mathbb{S}^n$ in the form

$$c(x,y) = \begin{cases} F(d(x,y)), & \text{if } d(x,y) < d_0, \\ -\infty, & \text{if } d(x,y) \geq d_0, \end{cases} \tag{1.10}$$

where $F \in C^1([0, d_0))$ is strictly concave and strictly decreasing, and satisfies

$$\lim_{\theta \to d_0} F(\theta) = -\infty \text{ and } \int_0^{d_0} F(\theta) > -\infty. \tag{1.11}$$

Clearly cost functions (1.1) and (1.2) belong to $\mathscr{C}_{d_0}$ with $d_0 = \arccos(1/\kappa)$ and $d_0 = \pi/2$ respectively.

A fundamental relation in mass transport theory is that maximizing the total cost functional (1.5) can be converted to minimizing the following Kantorovich's functional

$$\mathcal{J}(\varphi, \psi) = \int_\Omega \varphi f d\gamma_{\mathbb{S}^n} + \int_{\Omega^*} \psi g d\gamma_{\mathbb{S}^n}, \tag{1.12}$$

constrained within the admissible set

$$\mathcal{K} = \left\{ (\varphi, \psi) \in C(\bar{\Omega}) \times C(\bar{\Omega}^*) : \varphi(x) + \psi(y) \geq c(x,y),\ \forall (x,y) \in \bar{\Omega} \times \bar{\Omega}^* \right\}. \tag{1.13}$$

We say $(w, v) \in \mathcal{K}$ is a pair of potential functions if

$$\mathcal{J}(w, v) = \inf_{(\varphi,\psi) \in \mathcal{K}} \mathcal{J}(\varphi, \psi). \tag{1.14}$$

By [4, 8, 22], if cost function satisfies condition (A1) then the potential functions exist; and if condition (A2) holds then the c-normal map of the potential function yields the optimal map of (1.5).

As cost functions in $\mathscr{C}_{d_0}$ do not meet (A1), we propose an Aleksandrov type condition similar to (1.9) for the existence of potential functions. Let us introduce this condition and some notions that are used later. We say a non-empty set $\omega \subset \mathbb{S}^n$ is d_0-*convex* if

$$\omega = \bigcap_{y \in \mathcal{I}} V_{y, d_0}, \tag{1.15}$$

where $\mathcal{I}$ is a subset of $\mathbb{S}^n$ and $V_{y,d_0} = \{x \in \mathbb{S}^n : d(x,y) \geq d_0\}$. Extending f and g on $\mathbb{S}^n$ by zero outside $\bar{\Omega}$ and $\bar{\Omega}^*$, we always assume in the paper f and g are nonnegative and integrable on $\mathbb{S}^n$ with $\operatorname{supp}(f) \subseteq \bar{\Omega}$ and $\operatorname{supp}(g) \subseteq \bar{\Omega}^*$. We say (Ω, f) and (Ω^*, g) are *Aleksandrov related* if they satisfy (a) and (b) below:

(a) f and g satisfy the mass balance condition (1.3),

(b) for all d_0-convex set $\omega \neq \emptyset$ with $\mu_f(\Omega \setminus \omega) > 0$ and $\mu_g^*(\omega_{d_0}) > 0$, there holds

$$\mu_f(\omega) < \mu_g^*(\omega_{d_0}), \tag{1.16}$$

where $d\mu_f = f d\gamma_{\mathbb{S}^n}, d\mu_g^* = g d\gamma_{\mathbb{S}^n}$ are spherical measures, and $\omega_{d_0} = \cup_{x\in\omega} B_{d_0}(x)$.

Our first main result is concerned with the existence of optimal maps when the cost function lies in the class $\mathscr{C}_{d_0}$.

Theorem 1.1. *Let $c \in \mathscr{C}_{d_0}$, Ω, Ω^* be two domains in $\mathbb{S}^n$, and f, g be two nonnegative integrable functions on $\mathbb{S}^n$ such that $\operatorname{supp}(f) \subseteq \bar{\Omega}$ and $\operatorname{supp}(g) \subseteq \bar{\Omega}^*$. Suppose*

$$\begin{aligned} &\Omega \cap B_{d_0}(y) \neq \emptyset, \quad \forall y \in \bar{\Omega}^*, \\ &\Omega^* \cap B_{d_0}(x) \neq \emptyset, \ \forall x \in \bar{\Omega}, \end{aligned} \tag{1.17}$$

and for any open sets $\mathcal{O} \subset \Omega$ and $\mathcal{O}^* \subset \Omega^*$,

$$\int_{\mathcal{O}} f d\gamma_{\mathbb{S}^n} > 0 \ \textit{and} \ \int_{\mathcal{O}^*} g d\gamma_{\mathbb{S}^n} > 0. \tag{1.18}$$

Then the following are equivalent

(i) *(Ω, f) and (Ω^*, g) are Aleksandrov related;*

(ii) *there is a unique (up to additive constants) pair of potential functions of (1.14);*

(iii) *there is a unique optimal map of (1.5).*

For the refraction problem ($\kappa > 1$), if the constraint (1.8) holds, then Ω, Ω^* satisfy the position condition (1.17), and $(\Omega, f), (\Omega^*, g)$ are Aleksandrov related, because $\Omega^* \subseteq \omega_{d_0}$ for every d_0-convex $\omega \cap \Omega \neq \emptyset$. Hence the existence of refractors under assumption (1.8) is a consequence of Theorem 1.1.

When $\Omega = \Omega^* = \mathbb{S}^n$, Theorem 1.1 was first obtained in [34]. If in addition restricting to the cost function (1.2), the related optimal transportation problem is equivalent to the Gauss image problem which was introduced by Böröczky et al. [3] as an extension of Aleksandrov's classical problem [1, 2]. They proved that if μ and μ^* are two measures on $\mathbb{S}^n$ such that $\mu^* \ll \gamma_{\mathbb{S}^n}$, $\mu(\mathbb{S}^n) = \mu^*(\mathbb{S}^n)$, and

$$\mu(\omega) < \mu^*(\omega_{\frac{\pi}{2}}), \text{ for every spherical convex } \omega,$$

then there is a closed hypersurface $\mathcal{M}$ (no boundary) in $\mathbb{R}^{n+1}$, containing the origin in the interior, whose radial Gauss map $\mathscr{A}_{\mathcal{M}}$ satisfies

$$\mu^*(\mathscr{A}_{\mathcal{M}}(\omega)) = \mu(\omega), \text{ for each Borel } \omega \subset \mathbb{S}^n.$$

Our Theorem 1.1 deals with optimal transports between spherical subdomains and as a by-product solves the Gauss image problem of hypersurface with non-empty boundary.

On the other hand, Theorem 1.1 finds the Aleksandrov type condition for a wide class of cost functions in $\mathscr{C}_{d_0}$.

In this paper we also study the regularity of the potential functions of Theorem 1.1. Let us recall some notions from optimal transportation [38]. A function $w \in C(\bar{\Omega})$ is called *c-convex* if for each $x \in \bar{\Omega}$ there is $y \in B_{d_0}(x)$ such that

$$w(x') \geq c(x', y) - c(x, y) + w(x), \quad \forall x' \in \bar{\Omega}. \tag{1.19}$$

The *c-normal map* of a c-convex function $w \in C(\bar{\Omega})$ is defined as

$$\chi_w(x) := \{y \in B_{d_0}(x) : w(x') \geq c(x', y) - c(x, y) + w(x),\ \forall x' \in \bar{\Omega}\}, \quad \text{for } x \in \Omega, \tag{1.20}$$

and for the boundary point $\bar{x} \in \bar{\Omega}$, we define

$$\chi_w(\bar{x}) := \{y : y = \lim_{k\to\infty} y_k,\ \text{where } y_k \in \chi_w(x_k) \text{ with } \{x_k\} \subset \Omega \text{ and } x_k \to \bar{x}\}.$$

Let (w, v) be a pair of potential functions of (1.14) and T be the optimal map of (1.5). By the proof of Theorem 1.1, one will see that w is c-convex and

$$T(x) = \chi_w(x),\ \text{for a.e. } x \in \Omega. \tag{1.21}$$

If $w \in C^1(\Omega)$, then $\chi_w(x)$ is singleton and

$$\nabla_x c(x, \chi_w(x)) = \nabla w(x). \tag{1.22}$$

Since $\chi_w(x) \in B_{d_0}(x)$, the left-hand side of (1.22) is well-defined. Differentiating (1.22) and using (1.21), we find by $T \in \mathcal{S}_{f,g}$ that w satisfies the Monge-Ampère equation

$$\det(\nabla^2 w(x) - \nabla_x^2 c(x, \chi_w(x))) = \frac{f}{g \circ \chi_w} \det \nabla_{x,y}^2 c(x, \chi_w(x)), \quad x \in \Omega, \tag{1.23}$$

with a natural boundary condition

$$\chi_w(\Omega) = \Omega^*, \tag{1.24}$$

where ∇ in (1.22) and (1.23) denotes the derivatives of $\mathbb{S}^n$ w.r.t. an orthonormal frame.

For the regularity of w, the solution to (1.23) and (1.24), some conditions on the cost function and the geometry of domains are necessary [38, 42, 31]. Denote

$$\mathscr{C}_{d_0}^\infty = \left\{c \in \mathscr{C}_{d_0} :\ c \text{ is of the form (1.10) whose } F \text{ satisfies conditions (F1) and (F2)}\right\},$$

where

(F1) $\tilde{F}(s) := F \circ \arccos s \in C^\infty((s_0, 1])$, where $s_0 = \cos d_0$;
(F2) $F''(\theta) < 0$ for all $\theta \in [0, d_0)$.

Note that $\nabla_x d(\cdot, y)$ is not C^1 at $x = y$. Since $d(x, y) = \arccos(x \cdot y)$, condition (F1) guarantees that $c(x, y)$ is C^∞ in a neighbourhood of $(x, y) \in \mathbb{S}^n \times \mathbb{S}^n$ if $d(x, y) < d_0$. We will see in Section 4 later that condition (F2) yields

$$\det \nabla_{x,y}^2 c(x, y) > 0 \text{ for all } x, y \in \mathbb{S}^n \text{ such that } d(x, y) < d_0. \tag{1.25}$$

This shows that (1.23) is elliptic. For $c \in \mathscr{C}_{d_0}$, it can bee seen that $\nabla_x c(x,\cdot) : B_{d_0}(x) \to T_x\mathbb{S}^n$ is a homeomorphism. Consider a $(0,2)$-tensor on $T\mathbb{S}^n$ below

$$\mathcal{A}(x,Z) := \nabla^2_{xx} c\big(x, [\nabla_x c(x,\cdot)]^{-1}(Z)\big), \text{ for } x \in \mathbb{S}^n \text{ and } Z \in T_x\mathbb{S}^n. \tag{1.26}$$

We say a cost function $c \in \mathscr{C}^\infty_{d_0}$ satisfies the MTW condition, if for every $x \in \mathbb{S}^n$ and $Z \in T_x\mathbb{S}^n$, there holds

$$\frac{d^2}{dt^2}\mathcal{A}|_{(x,Z+t\eta)}(\xi,\xi) \geq 0, \ \forall \xi,\eta \in T_x\mathbb{S}^n \text{ with } \langle \xi,\eta\rangle = 0. \tag{1.27}$$

This condition was first proposed by Ma, Trudinger and Wang in [38] (see also [42]), and was found by Loeper to be necessary for the interior regularity [31].

For the global regularity, we recall the notion of c-convexity for domains. See also Definition 8.1 later. Assume Ω, Ω^* satisfy (1.17), and are also C^2-smooth domains of $\mathbb{S}^n$. Given $y \in \bar\Omega^*$ so that $\partial\Omega \cap B_{d_0}(y) \neq \emptyset$, or $x \in \bar\Omega$ so that $\partial\Omega^* \cap B_{d_0}(x) \neq \emptyset$, let

$$(\partial\Omega)^\wedge_y = \nabla_y c((\partial\Omega)_y, y) \subset T_y\mathbb{S}^n, \quad \text{where } (\partial\Omega)_y = \partial\Omega \cap B_{d_0}(y),$$
$$(\partial\Omega^*)^\wedge_x = \nabla_x c(x, (\partial\Omega^*)_x) \subset T_x\mathbb{S}^n, \ \text{where } (\partial\Omega^*)_x = \partial\Omega^* \cap B_{d_0}(x).$$

One sees that $(\partial\Omega)^\wedge_y$ and $(\partial\Omega^*)^\wedge_x$ are hypersurfaces of $\mathbb{R}^n \simeq T_x\mathbb{S}^n$ and $\mathbb{R}^n \simeq T_y\mathbb{S}^n$ respectively, with the metrics $\hat g_y$ and $\hat g^*_x$ induced from the Euclidean space. Denote by $\hat{I\!I}_y$ and $\hat{I\!I}^*_x$ the second fundamental forms of $(\partial\Omega)^\wedge_y$ and $(\partial\Omega^*)^\wedge_x$ respectively. We say Ω is uniformly c-convex w.r.t. Ω^* if it is c-convex and moreover, for any small $\delta > 0$ there is a $\delta' > 0$ so that for each $y \in \bar\Omega^*$ with $(\partial\Omega)_y \neq \emptyset$, there holds

$$\hat{I\!I}_y|_{\nabla_y c(x,y)} \geq \nabla^2_{x,y} c(\vec{\boldsymbol{n}}|_x, \hat\nu|_{\nabla_y c(x,y)}) \cdot \delta' \hat g_y|_{\nabla_y c(x,y)}, \quad \forall x \in (\partial\Omega)_y \cap B_{d_0-\delta}(y), \tag{1.28}$$

where $\vec{\boldsymbol{n}}$ and $\hat\nu$ are respectively the unit outer normals of $\partial\Omega$ and $(\partial\Omega)^\wedge_y$. Analogously Ω^* is uniformly c-convex w.r.t. Ω if it is c-convex and moreover, for any small $\delta > 0$, there is a $\delta' > 0$ so that for each $x \in \bar\Omega$ with $(\partial\Omega^*)_x \neq \emptyset$, there holds

$$\hat{I\!I}^*_x|_{\nabla_x c(x,y)} \geq \nabla^2_{x,y} c(\hat\nu^*|_{\nabla_x c(x,y)}, \vec{\boldsymbol{n}}^*_y) \cdot \delta' \hat g^*_x|_{\nabla_x c(x,y)}, \quad \forall y \in (\partial\Omega^*)_x \cap B_{d_0-\delta}(x), \tag{1.29}$$

where $\vec{\boldsymbol{n}}^*$ and $\hat\nu^*$ are respectively the unit outer normals of $\partial\Omega^*$ at x and $(\partial\Omega^*)^\wedge_x$. Such convexity concept for cost functions defined in an open neighbourhood of $\Omega \times \Omega^*$ was introduced in [38, 42]. Our notions (1.28) and (1.29) are modifications of theirs, as $c \in \mathscr{C}^\infty_{d_0}$ are not defined on the entire $\Omega \times \Omega^*$ in general, and $(\partial\Omega)^\wedge_y, (\partial\Omega^*)^\wedge_x$ are non-compact hypersurfaces. We will show (1.28) and (1.29) are equivalent to (8.7) and (8.8), which are actually used in the a priori estimates. See more discussions in Section 8.

Stated below is the second main result of our paper.

Theorem 1.2. *Let $c \in \mathscr{C}^\infty_{d_0}$, Ω and Ω^* be two C^4-smooth domains in $\mathbb{S}^n$, and $f \in C^2(\bar\Omega)$ and $g \in C^2(\bar\Omega^*)$ be two positive functions. Suppose that*

- *cost function c satisfies the MTW condition (1.27);*
- *Ω, Ω^* satisfy condition (1.17), and are uniformly c-convex to each other;*
- *(Ω, f), (Ω^*, g) are Aleksandrov related.*

Then problem (1.23) *and* (1.24) *admit a unique (up to additive constants) uniformly c-convex solution $w \in C^{3,\alpha}(\bar{\Omega})$, where $\alpha \in (0,1)$. As a result, $T = \chi_w$ is a $C^{2,\alpha}$-diffeomorphism maximizing* (1.6).

Furthermore, if Ω, Ω^*, f, g are C^∞-smooth, then $w \in C^\infty(\bar{\Omega})$ and $\chi_w \in C^\infty(\bar{\Omega}; \bar{\Omega}^*)$.

The regularity of potential functions is a fundamental problem in optimal transportation, and has been studied in many influential works [5, 6, 7, 9, 12, 43] for quadratic cost, in [38, 42, 16, 31, 36, 37] for general costs satisfying the MTW condition, and in [13, 14, 17, 27, 32, 33] for optimal transport on Riemannian manifold. The main difficulty in the proof of Theorem 1.2 is that $c(x, \chi_w(x))$ has no bounded derivatives in advance for cost function $c \in \mathscr{C}^\infty_{d_0}$, unless it has been shown that $\chi_w(x) \subset B_{d_0-\varepsilon_0}(x)$ for all $x \in \bar{\Omega}$, with some $\varepsilon_0 > 0$. This relies on a gradient estimate for potential function w, which will be proved by using the Kantorovich's functional and the condition (1.16).

We employ a parabolic approach and show the existence of smooth solutions to (1.23) and (1.24). The flow method for optimal transport with bounded cost function was studied in [28] for compact manifolds without boundary and in [29] for Euclidean domains with boundary. In addition to the fact that the cost functions in Theorem 1.2 could be unbounded without proper derivative estimates, another difficulty when invoking parabolic argument is that an admissible initial data is needed for starting the flow. See Definition 4.2 later for admissible initial datas. The existence of such nice initial data was posed as an assumption in [28, 29]. We overcome this by constructing a family of approximating problems.

The paper is organized as follows. In Section 2, we study the Aleksandrov relation and show a crucial oscillation estimate for every minimizing sequence of (1.14). In Section 3, we prove Theorem 1.1 through a synthesis of the essential oscillation estimates and a classical variational argument in optimal transport. As a by-product, we obtain the existence of weak solutions to (1.23) and (1.24). For the regularity of potential functions and optimal maps, a parabolic flow towards the smooth solutions to (1.23) and (1.24) is introduced in Section 4. We also show the monotonicity of Kantorovich's functional and establish the global C^0 and C^1 estimates for the solution to the flow in this section. As a result, our flow deforms every admissible initial data to a smooth and uniformly c-convex solution of (1.23) and (1.24), provided the flow enjoys the global second derivatives estimates. Such a priori estimates are the technical part of the paper, whose proof is inspired by [43, 38, 42, 29] and will be postponed until in Sections 7-10. In Section 5, we prove Theorem 1.2 by constructing admissible initial datas for approximating problems. In Section 6, some applications of our theorems are discussed. We provide the details of the global second derivative estimates in remaining sections: determinant estimates in Section 7, analytical aspects of c-convex domains in Section 8, obliqueness estimates in Section 9, and completing the a priori interior and boundary C^2-estimates in Section 10. Strict c-convexity and $C^{1,\alpha}$-regularity for potential functions are used in one step of our construction in Section 5 (see Lemma 5.4). Such property can be proved by a slight modification of arguments in [10, 16, 19, 44]. We include a discussion for it in Section 11 as an appendix for readers' convenience.

2. Kantorovich's functional meeting Aleksandrov condition: an oscillation estimate

2.1. Strengthening Aleksandrov condition.

We first prove some basic properties of d_0-convex sets in $\mathbb{S}^n$. See (1.15) for definition. Let ω be a nonempty subset of $\mathbb{S}^n$. Its d_0-dual $\omega^\sharp$ is defined as

$$\omega^\sharp = \bigcap_{y\in\omega} V_{y,d_0}.$$

Denote by $B_r(x)$ and $\bar{B}_r(x)$ the open and closed geodesic balls centered at x with radius r respectively. We sometimes omit x when the center is not important in the discussion.

Lemma 2.1. *Suppose $\omega \neq \emptyset$ and $\omega \subset \bar{B}_{\pi-d_0}$. We have the following properties:*

(i) *$\omega^\sharp$ is d_0-convex and $\omega \subset (\omega^\sharp)^\sharp$;*
(ii) *if ω is d_0-convex, then $\omega = (\omega^\sharp)^\sharp$;*
(iii) *$(\omega^\sharp)^\sharp = \langle\omega\rangle$, where $\langle\omega\rangle$ is the smallest d_0-convex set containing ω;*
(iv) *$\omega^\sharp = \mathbb{S}^n \setminus \omega_{d_0}$, where $\omega_{d_0} = \{x \in \mathbb{S}^n : d(x,y) < d_0$ for some $y \in \omega\}$.*

Proof. Since $\omega \subset \bar{B}_{\pi-d_0}$, we have $\omega^\sharp \neq \emptyset$. Suppose $x \in \omega$. Then $d(x,y) \geq d_0$ for all $y \in \omega^\sharp$. Namely $x \in V_{y;d_0}$ for all $y \in \omega^\sharp$. Hence $\omega \subset (\omega^\sharp)^\sharp$. This proves (i).

Assume that ω is d_0-convex. For (ii), it suffices to show $(\omega^\sharp)^\sharp \subset \omega$. We write

$$\omega = \bigcap_{y\in\mathcal{I}} V_{y,d_0}.$$

Observe that $d(x,y) \geq d_0$ for all $x \in \omega$ and $y \in \mathcal{I}$. Hence $\mathcal{I} \subset \omega^\sharp$. Given any $z \in (\omega^\sharp)^\sharp$, one has $d(z,y) \geq d_0$ for all $y \in \omega^\sharp$. In particular, $d(z,y) \geq d_0$ for all $y \in \mathcal{I}$. Therefore $z \in \omega$. This verifies (ii).

Clearly $\omega \subset \langle\omega\rangle \subset (\omega^\sharp)^\sharp$. Taking d_0-duals twice and using (ii), we obtain $(\omega^\sharp)^\sharp \subset \langle\omega\rangle \subset (\omega^\sharp)^\sharp$. So $\langle\omega\rangle = (\omega^\sharp)^\sharp$, and therefore (iii) holds.

Property (iv) follows by the calculation below

$$\mathbb{S}^n \setminus \omega_{d_0} = \mathbb{S}^n \setminus \Big(\bigcup_{y\in\omega} B_{d_0}(y)\Big) = \bigcap_{y\in\omega}(\mathbb{S}^n \setminus B_{d_0}(y)) = \bigcap_{y\in\omega} V_{y,d_0} = \omega^\sharp.$$

∎

The Aleksandrov relation (1.16) is symmetric in (Ω, f) and (Ω^*, g).

Lemma 2.2. *Let f, g be two nonnegative integrable functions supported on $\bar{\Omega}$ and $\bar{\Omega}^*$, respectively. Suppose (1.18) holds. If (Ω, f) and (Ω^*, g) are Aleksandrov related, then for any d_0-convex $\omega \neq \emptyset$ with $\mu_g^*(\Omega^* \setminus \omega) > 0$ and $\mu_f(\omega_{d_0}) > 0$,*

$$\mu_g^*(\omega) < \mu_f(\omega_{d_0}). \tag{2.1}$$

Proof. We claim that $\mu_f(\Omega \setminus \omega^\sharp) > 0$ and $\mu_g^*((\omega^\sharp)_{d_0}) > 0$. By virtue of (1.18), this is equivalent to $\Omega \setminus \omega^\sharp \neq \emptyset$ and $(\omega^\sharp)_{d_0} \cap \Omega^* \neq \emptyset$. Assume by contradiction argument that $\Omega \setminus \omega^\sharp = \emptyset$. Using (iv) in Lemma 2.1, we have $\Omega \cap \omega_{d_0} = \emptyset$, which contradicts $\mu_f(\omega_{d_0}) > 0$. On the other hand, if $(\omega^\sharp)_{d_0} \cap \Omega^* = \emptyset$, then, by (iv) in Lemma 2.1 again, we see that $\emptyset = \Omega^* \setminus (\omega^\sharp)^\sharp = \Omega^* \setminus \omega$, which contradicts $\mu_g^*(\Omega^* \setminus \omega) > 0$.

Applying (1.16) to $\omega^\sharp$ gives $\mu_f(\omega^\sharp) < \mu_g^*((\omega^\sharp)_{d_0})$. It then follows from Lemma 2.1 that

$$\mu_f(\omega_{d_0}) = \mu_f(\mathbb{S}^n) - \mu_f(\omega^\sharp) > \mu_g^*(\mathbb{S}^n) - \mu_g^*((\omega^\sharp)_{d_0}) = \mu_g^*((\omega^\sharp)^\sharp) = \mu_g^*(\omega).$$

□

We next strengthen the Aleksandrov relation. The following lemma is a generalization of [3, Lemma 5.11] where the case of $d_0 = \pi/2$ and $\Omega = \Omega^* = \mathbb{S}^n$ was discussed.

Lemma 2.3. *Let f, g be two nonnegative integrable functions supported on $\bar{\Omega}$ and $\bar{\Omega}^*$, respectively. Suppose (1.17) and (1.18) hold. If (Ω, f) and (Ω^*, g) are Aleksandrov related, then for any $\varepsilon_0 > 0$, there are small constants $\delta, \alpha > 0$, depending only on $\Omega, \Omega^*, f, g, \varepsilon_0$, such that for any compact $\omega \subset \bar{B}_{\pi - d_0}$ with $\omega \cap \Omega \neq \emptyset$ and $\mu_f(\Omega \setminus \langle\omega\rangle) \geq \varepsilon_0$,*

$$\mu_f(\omega) < (1-\delta)\mu_g^*(\omega_{d_0 - \alpha}). \tag{2.2}$$

Recall that $\langle\omega\rangle = (\omega^\sharp)^\sharp$ is the smallest d_0-convex set containing ω as in Lemma 2.1.

Proof. For the sake of (2.2), it suffices to show the existence of $\delta > 0$ such that

$$\mu_f(\omega) < (1-\delta)\mu_g^*(\omega_{d_0}). \tag{2.3}$$

Once this has been proved, the existence of α in (2.2) follows from the absolute continuity of μ_g^*, changing a smaller δ, and the positive lower bound of $\{\mu_g^*(B_{d_0}(x))\}_{x \in \bar{\Omega}}$ as argued in [3, Lemma 5.11]. Note that $\inf_{x \in \bar{\Omega}}\{\mu_g^*(B_{d_0}(x))\} > 0$. Otherwise, there exists $\{x_i\}_{i \in \mathbb{N}} \subset \bar{\Omega}$ such that $\lim_{i \to \infty} \mu_g^*(B_{d_0}(x_i)) = 0$. By passing to subsequence, $B_{d_0}(x_i)$ converges in the Hausdorff metric to $B_{d_0}(x_0)$ for some $x_0 \in \bar{\Omega}$, and $\mu_g^*(B_{d_0}(x_0)) = 0$, a contradiction.

It remains to show (2.3). Suppose on the contrary there exists a sequence of compact sets $\omega_i \subset \bar{B}_{\pi - d_0}$, satisfying $\omega_i \cap \Omega \neq \emptyset$ and $\mu_f(\Omega \setminus \langle\omega_i\rangle) \geq \varepsilon_0$, but

$$\lim_{i \to \infty} \frac{\mu_f(\omega_i)}{\mu_g^*((\omega_i)_{d_0})} = 1. \tag{2.4}$$

By virtue of Lemma 2.1 and the d_0-convexity of $\omega_i^\sharp$, we have

$$(\omega_i)_{d_0} = \mathbb{S}^n \setminus \omega_i^\sharp = \mathbb{S}^n \setminus \langle\omega_i\rangle^\sharp = \langle\omega_i\rangle_{d_0}.$$

Therefore $\mu_g^*((\omega_i)_{d_0}) = \mu_g^*(\langle\omega_i\rangle_{d_0})$. It is clear that $\mu_f(\omega_i) \le \mu_f(\langle\omega_i\rangle)$. Hence

$$\frac{\mu_f(\omega_i)}{\mu_g^*((\omega_i)_{d_0})} \le \frac{\mu_f(\langle\omega_i\rangle)}{\mu_g^*(\langle\omega_i\rangle_{d_0})}, \text{ for all } i \in \mathbb{N}. \tag{2.5}$$

Using (1.17) and (1.18), $\mu_g^*(\langle\omega_i\rangle_{d_0}) > 0$. Since $\mu_f(\Omega \setminus \langle\omega_i\rangle) \ge \varepsilon_0$, we obtain by (1.16)

$$\mu_f(\langle\omega_i\rangle) < \mu_g^*(\langle\omega_i\rangle_{d_0}), \text{ for all } i \in \mathbb{N}.$$

This together with (2.4) and (2.5) yields

$$\lim_{i\to\infty} \frac{\mu_f(\langle\omega_i\rangle)}{\mu_g^*(\langle\omega_i\rangle_{d_0})} = 1.$$

It then follows by Blaschke selection theorem, after passing to a subsequence if necessary, $\langle\omega_i\rangle$ converges in the Hausdorff metric to a d_0-convex set ω_∞ satisfying

$$\omega_\infty \cap \bar\Omega \ne \emptyset, \quad \mu_f(\Omega \setminus \omega_\infty) \ge \varepsilon_0, \quad \text{and } \mu_f(\omega_\infty) = \mu_g^*((\omega_\infty)_{d_0}). \tag{2.6}$$

By (1.17) and (1.18), $\mu_g^*((\omega_\infty)_{d_0}) > 0$. If $\omega_\infty \cap \Omega \ne \emptyset$, the last identity in (2.6) contradicts (1.16). If $\omega_\infty \cap \bar\Omega \subset \partial\Omega$, then $\mu_g^*((\omega_\infty)_{d_0}) = \mu_f(\omega_\infty) = 0$, a contradiction again. □

2.2. **An oscillation estimate.**

Our main goal in Section 2 is the oscillation estimate in Lemma 2.7. Let us first collect some basic notions from optimal transport theory. Recall that the *c-transform* of $w \in C(\bar\Omega)$ and *c-transform* of $v \in C(\bar\Omega^*)$ are given by

$$\begin{aligned} w^c(y) &:= \sup_{x\in\bar\Omega}\{c(x,y) - w(x)\}, \quad \forall y \in \bar\Omega^*, \\ v^c(x) &:= \sup_{y\in\bar\Omega^*}\{c(x,y) - v(y)\}, \quad \forall x \in \bar\Omega. \end{aligned} \tag{2.7}$$

Since Ω and Ω^* satisfy (1.17), we have

$$d_1 := \max_{x\in\bar\Omega} \min_{y\in\bar\Omega^*} d(x,y) < d_0. \tag{2.8}$$

Therefore the c-transform is finite-valued. The following dual property is well-known. We provide a proof for readers' convenience.

Lemma 2.4. *Assume $c \in \mathscr{C}_{d_0}$ and (1.17) holds. If $w \in C(\bar\Omega)$ and $w = v^c$ for some $v \in C(\bar\Omega^*)$, then $(w^c)^c = w$.*

Proof. By (2.7), $w(x) \ge c(x,y) - v(y)$ for all $x \in \bar\Omega$ and $y \in \bar\Omega^*$. So $v \ge w^c$ on $\bar\Omega^*$. Then

$$\begin{aligned} w(x) &= \sup_{y\in\bar\Omega^*}\{c(x,y) - v(y)\} \\ &\le \sup_{y\in\bar\Omega^*}\{c(x,y) - w^c(y)\} = (w^c)^c(x). \end{aligned} \tag{2.9}$$

On the other hand, $w^c(y) \ge c(x,y) - w(x)$ for all $x \in \bar\Omega$ and all $y \in \bar\Omega^*$. Therefore $w \ge (w^c)^c$ on $\bar\Omega$. This together with (2.9) gives $(w^c)^c = w$. □

Definition 2.1. *A pair $(w,v) \in \mathcal{K}$ is called* tightened *if $w = v^c$ and $v = w^c$.*

Recall that $\mathcal{K}$ is given by (1.13).

Lemma 2.5. *Assume $c \in \mathscr{C}_{d_0}$ and (1.17) holds. Let $(w, v) \in \mathcal{K}$ be tightened. Then*

(i) *if w is differentiable at $x \in \Omega$, then $\chi_w(x) \in \Omega^*$ is single-valued and (1.22) holds;*

(ii) *the identity below holds a.e. in Ω,*

$$w(x) + v(\chi_w(x)) = c(x, \chi_w(x)); \tag{2.10}$$

(iii) *the following estimate holds*

$$d_w := \sup\{d(x, y) : x \in \bar{\Omega}, y \in \chi_w(x) \cap \bar{\Omega}^*\} < d_0. \tag{2.11}$$

Proof. Since (1.17) holds and $w = v^c$, for any $x \in \bar{\Omega}$, there is $y_x \in \bar{\Omega}^* \cap B_{d_0}(x)$ so that

$$w(\cdot) \geq c(\cdot, y_x) - v(y_x) \text{ on } \bar{\Omega},$$

and the equality holds at x. Hence $w(\cdot) - c(\cdot, y_x)$ attains its global minimum on $\bar{\Omega}$ at x. By (1.20), $y_x \in \chi_w(x)$. Since $c \in \mathscr{C}_{d_0}$, we find that $\nabla_x c(x, \cdot) : B_{d_0}(x) \to T_x\mathbb{S}^n$ is a homeomorphism. Consequently, $\chi_w(x) = y_x$ is single-valued if w is differentiable at x, as in this case $\nabla_x c(x, y_x) = \nabla w(x)$. Therefore (2.10) follows at where w is differentiable. Since c-convex function is semi-convex, by Aleksandrov's theorem, w is twice-differentiable a.e. in Ω. Hence (2.10) holds almost everywhere.

Suppose on the contrary (2.11) fails. Then there is a sequence $(x_i, y_i) \in \bar{\Omega} \times \bar{\Omega}^*$ such that $y_i \in \chi_w(x_i) \cap \bar{\Omega}^*$, $w(x_i) + v(y_i) = c(x_i, y_i)$, and $d(x_i, y_i) \to d_0$ (as $i \to \infty$). By passing to a subsequence, we assume $x_i \to x_\infty \in \bar{\Omega}$ and $y_i \to y_\infty \in \bar{\Omega}^*$. Hence

$$\lim_{i\to\infty} c(x_i, y_i) = w(x_\infty) + v(y_\infty) > -\infty,$$

contradicting with $d(x_i, y_i) \to d_0$. $\square$

Lemma 2.6. *Assume $c \in \mathscr{C}_{d_0}$ and (1.17) holds. If $w \in C(\bar{\Omega})$ is c-convex, then*

$$\|w\|_{C^{0,1}(\bar{\Omega})} \leq \max_{0\leq\theta\leq d_w} |F'(\theta)| + \frac{4}{d_0 - d_w} \max_{\bar{\Omega}} |w|. \tag{2.12}$$

where d_w is given by (2.11).

Proof. Given $x_1, x_2 \in \bar{\Omega}$, let $y_i \in \chi_w(x_i) \cap \bar{\Omega}^*$, we need to estimate $\frac{|w(x_1)-w(x_2)|}{d(x_1,x_2)}$.

Case 1: $d(x_1, x_2) < \frac{1}{2}(d_0 - d_w)$**.** In this case $x_i \in B_{d_0}(y_j)$ for each $i, j \in \{1, 2\}$. Let $v = w^c$. Then

$$\begin{aligned} 0 &= (w(x_2) + v(y_2) - c(x_2, y_2)) - (w(x_1) + v(y_2) - c(x_1, y_2)) \\ &\quad + (w(x_1) + v(y_2) - c(x_1, y_2)) \\ &\geq w(x_2) - w(x_1) + c(x_1, y_2) - c(x_2, y_2) \\ &= w(x_2) - w(x_1) - \int_0^{d(x_1,x_2)} \frac{d}{dt} c(\gamma(t), y_2) dt, \end{aligned}$$

where $\gamma(t)$ is the normal minimal geodesic from x_1 to x_2, and

$$\begin{aligned} 0 &= (w(x_2)+v(y_1)-c(x_2,y_1))-(w(x_1)+v(y_1)-c(x_1,y_1))) \\ &\quad -(w(x_2)+v(y_1)-c(x_2,y_1)) \\ &\le w(x_2)-w(x_1)+c(x_1,y_1)-c(x_2,y_1) \\ &= w(x_2)-w(x_1)-\int_0^{d(x_1,x_2)}\frac{d}{dt}c(\gamma(t),y_1)dt. \end{aligned}$$

Combining the two inequalities,

$$-d(x_2,x_1)\max_{0\le\theta\le d_w}|F'(\theta)| \le w(x_2)-w(x_1) \le d(x_2,x_1)\max_{0\le\theta\le d_w}|F'(\theta)|. \tag{2.13}$$

Case 2: $d(x_1,x_2)\ge\frac{1}{2}(d_0-d_w)$**.** Then

$$\frac{|w(x_1)-w(x_2)|}{d(x_1,x_2)} \le \frac{4}{d_0-d_w}\max_{\bar\Omega}|w|. \tag{2.14}$$

Putting (2.13) and (2.14) together, we obtain (2.12). ☐

Every $(w,v)\in\mathcal{K}$ yields a tightened pair $((w^c)^c,w^c)\in\mathcal{K}$. Since $v\ge w^c$ and $w\ge(w^c)^c$,

$$\mathcal{J}((w^c)^c,w^c)\le\mathcal{J}(w,v).$$

For any $A\in\mathbb{R}$, let $\mathcal{K}_A$ be the subset of $\mathcal{K}$ given by

$$\mathcal{K}_A=\big\{(w,v)\in\mathcal{K}:\ (w,v) \text{ is tightened and } \mathcal{J}(w,v)\le A\big\}.$$

The main result of this section is the following oscillation estimate. Our proof is inspired by [3] where the authors studied the cost (1.2) with the case $\Omega=\Omega^*=\mathbb{S}^n$.

Lemma 2.7. *Let $c\in\mathscr{C}_{d_0}$, and f,g be two nonnegative integrable functions supported on $\bar\Omega$ and $\bar\Omega^*$, respectively. Suppose (1.17) and (1.18) hold, and (Ω,f) and (Ω^*,g) are Aleksandrov related. Then for any $A>0$, there exists a positive constant $C=C_A$, depending on A, such that*

$$\sup_{(w,v)\in\mathcal{K}_A}\big(\operatorname{osc}_{\bar\Omega}w+\operatorname{osc}_{\bar\Omega^*}v\big)\le C. \tag{2.15}$$

Proof. By the symmetry, we only need to prove

$$\sup_{(w,v)\in\mathcal{K}_A}\operatorname{osc}_{\bar\Omega}w\le C. \tag{2.16}$$

Since Aleksandrov relation is symmetric (Lemma 2.2), the argument for proving (2.16) can be applied to v and thus gives the estimate of $\sup_{(w,v)\in\mathcal{K}_A}\operatorname{osc}_{\bar\Omega}v$.

Suppose on the contrary that there exists a sequence $\{(w_j,v_j)\}_{j\ge1}\subset\mathcal{K}_A$ such that

$$\lim_{j\to\infty}\operatorname{osc}_{\bar\Omega}w_j=\infty. \tag{2.17}$$

Let $\bar{w}_j := w_j + \min_{\bar{\Omega}^*} v_j - c_0$ and $\bar{v}_j := v_j - \min_{\bar{\Omega}^*} v_j + c_0$ for some $c_0 > 0$ (independent of j) to be specified later. Observe that

$$\min_{\bar{\Omega}^*} \bar{v}_j = c_0 > 0. \tag{2.18}$$

Clearly $(\bar{w}_j, \bar{v}_j) \in \mathcal{K}$ is tightened. By the mass balance condition (1.3),

$$\limsup_{j\to\infty} \mathcal{J}(\bar{w}_j, \bar{v}_j) = \limsup_{j\to\infty} \mathcal{J}(w_j, v_j) \leq A. \tag{2.19}$$

Assume $\bar{w}_j$ achieves its maximum on $\bar{\Omega}$ at $\bar{x}_j$. Take a $\bar{y}_j \in \chi_{\bar{w}_j}(\bar{x}_j) \cap \bar{\Omega}^*$. Then

$$F(0) \geq c(\bar{x}_j, \bar{y}_j) = \bar{w}_j(\bar{x}_j) + \bar{v}_j(\bar{y}_j) \geq \max_{\bar{\Omega}} \bar{w}_j + c_0.$$

This together with (2.17) implies that

$$s_j := \min_{\bar{\Omega}} \bar{w}_j \to -\infty. \tag{2.20}$$

In what follows, we show $\mathcal{J}(\bar{w}_j, \bar{v}_j) \to \infty$ via (2.20), which contradicts with (2.19).

Step 1: Estimate $\int_{\Omega^*} \bar{v}_j g$.

Assume $s_j = \bar{w}_j(x_j)$ for some $x_j \in \bar{\Omega}$. Then $\bar{v}_j(y) \geq c(x_j, y) - s_j$ for all $y \in \Omega^*$. Hence

$$\begin{aligned}\int_{\Omega^*} g\bar{v}_j &= \int_{B_{d_0}(x_j)} g\bar{v}_j + \int_{\Omega^*\setminus B_{d_0}(x_j)} g\bar{v}_j \\ &\geq \int_{B_{d_0}(x_j)} g(y)F(d(x_j,y))d\gamma_{\mathbb{S}^n}(y) - s_j\mu_g^*(B_{d_0}(x_j)) + c_0\mu_g^*(\Omega^* \setminus B_{d_0}(x_j)),\end{aligned}$$

where we use $\bar{v}_j \geq c_0$ on Ω^* in the second inequality. Using (1.11), we further obtain

$$\int_{\Omega^*} g\bar{v}_j \geq -\sigma_0 s_j - C \geq -\frac{1}{2}\sigma_0 s_j, \tag{2.21}$$

provided j is sufficiently large so that $s_j \leq -\frac{2C}{\sigma_0}$. Here $\sigma_0 := \min_{x\in\bar{\Omega}}\{\mu_g^*(B_{d_0}(x))\}$. By (1.17) and (1.18), $\sigma_0 > 0$.

Step 2: Let $v_j(y_j) = \min_{\bar{\Omega}^*} v_j$ for some $y_j \in \bar{\Omega}^*$, and suppose $y_j \to y_\infty$ by passing to a subsequence. We estimate $\int_{B_{d_0}(y_\infty)} f\bar{w}_j$.

Since $\gamma_{\mathbb{S}^n}(B_{d_0}(y_\infty)\Delta B_{d_0}(y_j)) = o(1)$, we conclude that

$$\begin{aligned}\int_{B_{d_0}(y_\infty)} f\bar{w}_j &\geq \int_{B_{d_0}(y_j)} f\bar{w}_j - \Big|\int_{B_{d_0}(y_j)} f\bar{w}_j - \int_{B_{d_0}(y_\infty)} f\bar{w}_j\Big| \\ &\geq \int_{B_{d_0}(y_j)} f\bar{w}_j + o(1)s_j.\end{aligned}$$

Clearly $\bar{v}_j(y_j) = \min_{\bar{\Omega}^*} \bar{v}_j$. Using (2.18), we have $\bar{w}_j(\cdot) \geq c(\cdot, y_j) - c_0$ on $\bar{\Omega}$. By (1.11),

$$\int_{B_{d_0}(y_\infty)} f\bar{w}_j \geq -C - \mu_f(B_{d_0}(y_j))c_0 + o(1)s_j \geq -C + o(1)s_j. \tag{2.22}$$

Step 3: Estimate $\int_{V_{y_\infty,d_0}} f\bar{w}_j$.

We aim to show there exist two constants $\delta \in (0,1)$ and $C > 0$ such that

$$\int_{V_{y_\infty,d_0}} f\bar{w}_j \geq (\delta-1)\int_{\Omega^*} g\bar{v}_j - C. \tag{2.23}$$

For this end, we assume $\gamma_{\mathbb{S}^n}(V_{y_\infty,d_0}\cap\Omega) > 0$. Otherwise (2.23) holds immediately, as the left-hand side of (2.23) is zero but the right-hand side is negative. For each j, divide V_{y_∞,d_0} into a collection of compact subsets $\{V_{j,k}\}_{k=1}^{m_j}$: $V_{y_\infty,d_0} = \bigcup_{k=1}^{m_j} V_{j,k}$, such that

$$\max_{1\leq k\leq m_j} \operatorname{diam} V_{j,k} \leq \exp(s_j), \tag{2.24}$$

and $V_{j,k}$ are almost disjoint, namely

$$\mu_f\left(V_{j,k}\cap(\cup_{l\neq k}V_{j,l})\right) = 0,\ \forall k = 1,\dots,m_j. \tag{2.25}$$

By reordering $\{V_{j,k}\}_{k=1}^{m_j}$, we assume

$$V_{j,k}\cap\Omega \neq \emptyset,\ \text{iff } k = 1,\dots,n_j, \tag{2.26}$$

for some $n_j \leq m_j$. Select a collection of points $x_{j,k} \in V_{j,k}\cap\Omega$, for $1 \leq k \leq n_j$, such that

$$s_{j,k} = \bar{w}_j(x_{j,k}) = \min_{V_{j,k}\cap\bar{\Omega}} \bar{w}_j,\ \text{for } k = 1,\dots,n_j. \tag{2.27}$$

By reindexing k if necessary, we assume

$$s_j \leq s_{j,1} \leq s_{j,2} \leq \cdots \leq s_{j,n_j} < 0. \tag{2.28}$$

To see $s_{j,n_j} < 0$, we take $y_{j,n_j} \in \chi_{\bar{w}_j}(x_{j,n_j})\cap\bar{\Omega}^*$ and select $c_0 > F(0)$ so that

$$s_{j,n_j} = \bar{w}_j(x_{j,n_j}) = c(x_{j,n_j}, y_{j,n_j}) - \bar{v}_j(y_{j,n_j}) \leq F(0) - c_0 < 0.$$

By (1.17) and (1.18), we deduce that, for any $1 \leq k \leq n_j$,

$$\mu_f(\Omega\setminus\langle\cup_{l=1}^k V_{j,l}\rangle) \geq \mu_f(\Omega\setminus V_{y_\infty,d_0}) = \mu_f(\Omega\cap B_{d_0}(y_\infty)) =: \varepsilon_0 > 0. \tag{2.29}$$

It then follows by (2.26) and using Lemma 2.3 that

$$\mu_f\big(\cup_{l=1}^k V_{j,l}\big) < (1-\delta)\mu_g^*\big((\cup_{l=1}^k V_{j,l})_{d_0-\alpha}\big),\ \forall 1 \leq k \leq n_j. \tag{2.30}$$

We remark that δ and α depend on ε_0 in (2.29) and are independent of j. Set

$$W_{j,k} := (V_{j,k})_{d_0-\alpha} \setminus \Big(\bigcup_{l=1}^{k-1}(V_{j,l})_{d_0-\alpha}\Big).$$

Then, by (2.25) and (2.30), for any $1 \leq k \leq n_j$,

$$\sum_{l=1}^k \mu_f(V_{j,l}) = \mu_f\big(\cup_{l=1}^k V_{j,l}\big) < (1-\delta)\mu_g^*\big((\cup_{l=1}^k V_{j,l})_{d_0-\alpha}\big) = (1-\delta)\sum_{l=1}^k \mu_g^*(W_{j,l}). \tag{2.31}$$

By virtue of (2.25), (2.26) and (2.27), we have

$$\begin{aligned}\int_{V_{y_\infty,d_0}} f\bar{w}_j &= \sum_{k=1}^{n_j}\int_{V_{j,k}} f\bar{w}_j \geq \sum_{k=1}^{n_j} s_{j,k}\mu_f(V_{j,k}) \\ &= s_{j,n_j}\sum_{k=1}^{n_j}\mu_f(V_{j,k}) + \sum_{k=1}^{n_j-1}\Big((s_{j,k}-s_{j,k+1})\sum_{l=1}^{k}\mu_f(V_{j,l})\Big)\end{aligned}$$

Using (2.28) and (2.31), we further obtain,

$$\begin{aligned}\int_{V_{y_\infty,d_0}} f\bar{w}_j &\geq (1-\delta)\Big[s_{j,n_j}\sum_{k=1}^{n_j}\mu_g^*(W_{j,k}) + \sum_{k=1}^{n_j-1}\Big((s_{j,k}-s_{j,k+1})\sum_{l=1}^{k}\mu_g^*(W_{j,l})\Big)\Big] \\ &= (1-\delta)\sum_{k=1}^{n_j}\int_{W_{j,k}} g s_{j,k}. \qquad (2.32)\end{aligned}$$

On the other hand, for any $y \in (V_{j,k})_{d_0-\alpha}\cap\Omega^*$, we have

$$s_{j,k} = \bar{w}_j(x_{j,k}) \geq c(x_{j,k},y) - \bar{v}_j(y),\ \forall 1\leq k\leq n_j.$$

By (2.24), $d(x_{j,k},y) < d_0-\alpha/2$ for large j. Therefore, for any $y\in W_{j,k}\cap\Omega^*$

$$s_{j,k} \geq F(d_0-\alpha/2) - \bar{v}_j(y),\ \forall 1\leq k\leq n_j.$$

Plugging this into (2.32), we obtain

$$\int_{V_{y_\infty,d_0}} f\bar{w}_j \geq (\delta-1)\int_{\bigcup_{k=1}^{n_j} W_{j,k}} g\bar{v}_j - C.$$

This together with (2.18) gives (2.23).

Step 4: Show $\mathcal{J}(\bar{w}_j,\bar{v}_j)\to\infty$ as $j\to\infty$.

Employing (2.21), (2.22) and (2.23), we obtain

$$\begin{aligned}\mathcal{J}(\bar{w}_j,\bar{v}_j) &= \int_{B_{d_0}(y_\infty)} f\bar{w}_j + \int_{V_{y_\infty,d_0}} f\bar{w}_j + \int_{\Omega^*} g\bar{v}_j \\ &\geq \int_{B_{d_0}(y_\infty)} f\bar{w}_j + \delta\int_{\Omega^*} g\bar{v}_j - C \\ &\geq o(1)s_j - \frac{\delta}{2}\sigma_0 s_j - C \to \infty.\end{aligned}$$

This contradicts with (2.19). We complete the proof. □

As a corollary of Lemma 2.7, we prove the following stay-away property (2.33) and Lipschitz estimate (2.34) for functions in $\mathcal{K}_A$.

Lemma 2.8. *Let $c\in\mathscr{C}_{d_0}$, and f,g be two nonnegative integrable functions supported on $\bar{\Omega}$ and $\bar{\Omega}^*$, respectively. Suppose (1.17) and (1.18) hold, and (Ω,f) and (Ω^*,g) are*

Aleksandrov related. Then for any $A > 0$, there are positive constants $C = C_A$ and small $\varepsilon = \varepsilon_A$, depending on A, such that for all $(w, v) \in \mathcal{K}_A$

$$\max\{d_w, d_v\} \leq d_0 - \varepsilon, \tag{2.33}$$

where d_w is give by (2.11) and $d_v = \sup\{d(x, y) : y \in \bar{\Omega}^*,\ x \in \chi_v(y) \cap \bar{\Omega}\}$. Moreover

$$\|\bar{w}\|_{C^{0,1}(\bar{\Omega})} + \|\bar{v}\|_{C^{0,1}(\bar{\Omega}^*)} \leq C, \tag{2.34}$$

where $\bar{w} := w + \min_{\bar{\Omega}^*} v$ and $\bar{v} := v - \min_{\bar{\Omega}^*} v$.

Proof. In view of Lemma 2.7, we see that

$$0 \leq \bar{v}(y) \leq C,\ \forall y \in \bar{\Omega}^*. \tag{2.35}$$

Since f and g satisfy the mass balance condition (1.3), we have

$$A \geq \mathcal{J}(w, v) = \mathcal{J}(\bar{w}, \bar{v}) \geq \mu_f(\Omega) \min_{\bar{\Omega}} \bar{w}.$$

It follows that $\min_{\bar{\Omega}} \bar{w} \leq C$, and by using Lemma 2.7 again

$$\max_{\bar{\Omega}} \bar{w} = \min_{\bar{\Omega}} \bar{w} + \operatorname{osc}_{\bar{\Omega}} \bar{w} \leq C. \tag{2.36}$$

On the other hand,

$$\min_{\bar{\Omega}} \bar{w} = \min_{x \in \bar{\Omega}} \max_{y \in \bar{\Omega}^*} \{c(x, y) - \bar{v}(y)\} \geq F(d_1) - \max_{\bar{\Omega}^*} \bar{v} \geq -C, \tag{2.37}$$

where d_1 is defined by (2.8).

By virtue of (2.35) and (2.37), for any $x \in \bar{\Omega}$ and $y \in \chi_w(x) \cap \bar{\Omega}^*$,

$$F(d(x, y)) = w(x) + v(y) = \bar{w}(x) + \bar{v}(y) \geq \min_{\bar{\Omega}} \bar{w} + \min_{\bar{\Omega}^*} \bar{v} \geq -C.$$

Therefore $d_w \leq d_0 - \varepsilon$ for some $\varepsilon > 0$. Similarly $d_v \leq d_0 - \varepsilon$. This proves (2.33). The Lipschitz estimate (2.34) follows by employing Lemma 2.6 and (2.33). □

3. Proof of Theorem 1.1

In this section, we prove Theorem 1.1 by the a priori estimate (2.4). We first present some well-known results in optimal transportation.

Given a tightened pair $(w, v) \in \mathcal{K}$, by Lemma 2.5, there is a map $T_w : \Omega \to \Omega^*$ solving

$$w(x) + v(T_w(x)) = c(x, T_w(x)),\ \text{a.e in } \Omega. \tag{3.1}$$

For $c \in \mathscr{C}_{d_0}$, since F is C^1-smooth and strictly concave and decreasing, we infer that $\nabla_x c(x, \cdot) : B_{d_0}(x) \to T_x\mathbb{S}^n$ is a homeomorphism. Hence (3.1) is uniquely solved by

$$\nabla_x c(x, T_w(x)) = \nabla w(x),\ \text{a.e in } \Omega.$$

The following conclusion is classical in optimal transportation and can be proved by the variational argument in [8, 22, 46].

Lemma 3.1. *Let $c \in \mathscr{C}_{d_0}$, and f, g be two nonnegative integrable functions supported on domains $\bar\Omega, \bar\Omega^* \subset \mathbb{S}^n$ respectively. Suppose (1.17) and (1.18) hold. Let $(w, v) \in \mathcal{K}$ be a tightened pair. Then the following statements are equivalent*

(i) *$(w, v) \in \mathcal{K}$ is a pair of potential functions of (1.14);*
(ii) *$T_w \in \mathcal{S}_{f,g}$;*
(iii) *T_w is an optimal map of (1.5).*

Moreover, if T is an optimal map, then $T = T_w$ a.e. in Ω, whenever (w, v) is a pair of potential functions. Namely the optimal map is unique.

Proof. We only sketch the main ideas. The readers may consult [8, 22, 46] for details.

Assume (w, v) is a pair of potential functions. For any $h \in C(\bar\Omega^*)$ and $\varepsilon \in (0, 1)$, let

$$v_\varepsilon(y) = v(y) + \varepsilon h(y) \text{ and } w_\varepsilon(x) = \sup_{y\in\bar\Omega^*}\{c(x, y) - v_\varepsilon(y)\}.$$

If w is differentiable at $x_0 \in \Omega$, by Lemma 2.5, $\chi_w(x_0)$ is single-valued. We have

$$w_\varepsilon(x_0) = w(x_0) - \varepsilon h(\chi_w(x_0)) + o(\varepsilon).$$

By (3.1), $T_w = \chi_w$ a.e. in Ω. Since w is a.e. differentiable, we have

$$w_\varepsilon(x) = w(x) - \varepsilon h(T_w(x)) + o(\varepsilon), \text{ a.e. } x \in \Omega.$$

As a result

$$0 = \frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} \mathcal{J}(w_\varepsilon, v_\varepsilon) = -\int_\Omega f h \circ T_w d\gamma_{\mathbb{S}^n} + \int_{\Omega^*} hg d\gamma_{\mathbb{S}^n}.$$

Hence $T_w \in \mathcal{S}_{f,g}$. Therefore (i) implies (ii).

It is not hard to verify the following

$$\inf_{(\varphi,\psi)\in\mathcal{K}} \mathcal{J}(\varphi, \psi) \geq \sup_{\Phi\in\mathcal{S}_{f,g}} \mathcal{C}(\Phi). \tag{3.2}$$

Assume that (w, v) is tightened and $T_w \in \mathcal{S}_{f,g}$. By (1.4) and (3.1), we obtain

$$\mathcal{J}(w, v) = \int_\Omega wf + \int_{\Omega^*} vg = \int_\Omega wf + \int_\Omega fv \circ T_w = \mathcal{C}(T_w).$$

Combining this with (3.2), we have

$$\mathcal{C}(T_w) = \sup_{\Phi\in\mathcal{S}_{f,g}} \mathcal{C}(\Phi).$$

Hence (ii) implies (iii).

Let (w, v) be tightened and T_w be an optimal map. By (3.1) and (3.2), we obtain

$$\inf_{(\varphi,\psi)\in\mathcal{K}} \mathcal{J}(\varphi, \psi) \leq \mathcal{J}(w, v) = \mathcal{C}(T_w) = \sup_{\Phi\in\mathcal{S}_{f,g}} \mathcal{C}(\Phi) \leq \inf_{(\varphi,\psi)\in\mathcal{K}} \mathcal{J}(\varphi, \psi). \tag{3.3}$$

Hence (w, v) is a pair of potential function, and thus (iii) implies (i).

Suppose $T \in \mathcal{S}_{f,g}$ is an optimal map and (w, v) is a pair of potential functions. Since $\mathcal{C}(T_w) = \mathcal{C}(T)$, we deduce by the first equality in (3.3) and $T \in \mathcal{S}_{f,g}$

$$\int_\Omega f(x)c(x, T(x)) = \mathcal{C}(T) = \mathcal{J}(w, v) = \int_\Omega f(x)(w(x) + v \circ T(x)).$$

It then follows from (1.18) that

$$w(x) + v(T(x)) = c(x, T(x)), \ a.e.\ x \in \Omega.$$

By Lemma 2.5 and (3.1), we see that $T = T_w$ a.e. on Ω. $\square$

If (w_i, v_i), $i = 1, 2$, are two pairs of potential functions. In view of Lemma 3.1, $T_{w_1} = T_{w_2}$ a.e. on Ω. Since (3.1) holds for both T_{w_1} and T_{w_2} at their differentiable points, $\nabla w_1 = \nabla w_2$ almost everywhere. Then $w_1 = w_2 + C$ and $v_1 = v_2 - C$ for some constant C. We summarize such uniqueness of potential functions below.

Lemma 3.2. *Let $c \in \mathscr{C}_{d_0}$, and f, g be two nonnegative integrable functions supported on domains $\bar\Omega, \bar\Omega^* \subset \mathbb{S}^n$ respectively. Suppose (1.17) and (1.18) hold. If (w_1, v_1) and (w_2, v_2) are potential functions, then $w_1 = w_2 + C$ and $v_1 = v_2 - C$ for some C.*

The following conclusion is crucial in our proof of Theorem 1.1.

Lemma 3.3. *Let $c \in \mathscr{C}_{d_0}$, and f, g be two nonnegative integrable functions supported on domains $\bar\Omega, \bar\Omega^* \subset \mathbb{S}^n$ respectively. Suppose (1.17) and (1.18) hold. If (Ω, f) and (Ω^*, g) are Aleksandrov related, then there exists a pair of potential functions of (1.14).*

Proof. Let $\{(w_j, v_j)\}_{j\geq 1}$ be tightened minimizing sequence of $\mathcal{J}$, namely

$$\lim_{j\to\infty} \mathcal{J}(w_j, v_j) = \inf_{(\varphi,\psi)\in\mathcal{K}} \mathcal{J}(\varphi, \psi). \tag{3.4}$$

Hence there exists $A > 0$ such that $(w_j, v_j) \in \mathcal{K}_A$ for all j. Consider

$$\bar w_j(x) = w_j(x) + \min_{\bar\Omega^*} v_j, \ x \in \bar\Omega,$$
$$\bar v_j(y) = v_j(x) - \min_{\bar\Omega^*} v_j, \ y \in \bar\Omega^*.$$

As argued in Lemma 2.8 (see (2.35), (2.36) and (2.37) therein), we conclude that

$$\max_{\bar\Omega} |\bar w_j| + \max_{\bar\Omega^*} |\bar v_j| \leq C, \text{ for all } j \geq 1,$$

for some $C > 0$ depends only on A (so independent of j). By Lemma 2.8, we also have

$$\|\bar w_j\|_{C^{0,1}(\bar\Omega)} + \|\bar v_j\|_{C^{0,1}(\bar\Omega^*)} \leq C, \text{ for all } j \geq 1.$$

By the Arzelà-Ascoli theorem, $(\bar w_j, \bar v_j) \to (w, v) \in \mathcal{K}$ uniformly along a subsequence. We deduce from (3.4) that (w, v) is a pair of potential functions. $\square$

We next show that the potential functions are the Aleksandrov weak solutions to (1.23) and (1.24) under the condition (1.18), as discussed in [38, Section 3].

Definition 3.1. *We call a c-convex function $w \in C(\bar{\Omega})$ an Aleksandrov weak solution of* (1.23) *and* (1.24) *if*

$$\mu_f(\omega) = \mu_g^*(\chi_w(\omega)), \quad \forall \text{ Borel set } \omega \subset \Omega. \tag{3.5}$$

$$\Omega^* \subset \chi_w(\bar{\Omega}), \ \gamma_{\mathbb{S}^n}(\{x \in \Omega : f(x) > 0 \text{ and } \chi_w(x) \setminus \bar{\Omega}^* \neq \emptyset\}) = 0. \tag{3.6}$$

The existence of weak solutions were proved in [38] for cost function satisfying (A1) and (A2) conditions as stated in the section of introduction. Their argument is also adapted to the following theorem. We provide the proof for readers' convenience.

Theorem 3.1. [38, Section 3] *Let $c \in \mathscr{C}_{d_0}$, and f, g be two nonnegative integrable functions supported on domains $\bar{\Omega}, \bar{\Omega}^* \subset \mathbb{S}^n$ respectively. Suppose* (1.17) *and* (1.18) *hold. If tightened $(w, v) \in \mathcal{K}$ is a pair of potential functions of* (1.14)*, then w is an Aleksandrov weak solution of* (1.23) *and* (1.24).

Proof. Since $T_w = \chi_w$ a.e. in Ω and $T_w \in \mathcal{S}_{f,g}$ as shown in Lemma 3.1, we conclude

$$\mu_f(\chi_w^{-1}(\omega)) = \mu_g^*(\omega), \text{ for any Borel } \omega \subset \bar{\Omega}^*. \tag{3.7}$$

Let $E_v = \{y \in \bar{\Omega}^* : v \text{ is not twice differentiable at } y\}$. Since v is semi-convex, by Aleksandrov's theorem, $\gamma_{\mathbb{S}^n}(E_v) = 0$. By (3.7), we infer that

$$\mu_f(\chi_w^{-1}(E_v)) = \mu_g^*(E_v) = 0. \tag{3.8}$$

Note that for any $\omega \subset \bar{\Omega}$,

$$\omega \subset \chi_w^{-1}(\chi_w(\omega)) \text{ and } \chi_w^{-1}(\chi_w(\omega)) \setminus \chi_w^{-1}(E_v) \subset \omega. \tag{3.9}$$

It follows by (3.8) and (3.9) that

$$\begin{aligned} \mu_f(\chi_w^{-1}(\chi_w(\omega))) &\geq \mu_f(\omega) \\ &\geq \mu_f(\chi_w^{-1}(\chi_w(\omega)) \setminus \chi_w^{-1}(E_v)) \\ &= \mu_f(\chi_w^{-1}(\chi_w(\omega))), \text{ for all Borel set } \omega \subset \mathbb{S}^n. \end{aligned}$$

As a result, $\mu_f(\omega) = \mu_f(\chi_w^{-1}(\chi_w(\omega)))$. This together with (3.7) yields

$$\mu_f(\omega) = \mu_g^*(\chi_w(\omega)).$$

We assert $\Omega^* \subset \chi_w(\bar{\Omega})$. Otherwise, since Ω^* is open and $\chi_w(\bar{\Omega})$ is closed, we find by (1.18) that $\mu_g^*(\Omega^* \setminus \chi_w(\bar{\Omega})) > 0$. Note that

$$\mu_g^*(\chi_w(\bar{\Omega})) \geq \mu_g^*(\chi_w(\Omega)) = \mu_f(\Omega) = \mu_g^*(\Omega^*) = \mu_g^*(\Omega^* \setminus \chi_w(\bar{\Omega})) + \mu_g^*(\chi_w(\bar{\Omega})).$$

This implies that

$$\mu_g^*(\Omega^* \setminus \chi_w(\bar{\Omega}) = 0,$$

arriving at a contradiction.

Let $F_w = \{x \in \Omega : f(x) > 0 \text{ and } \chi_w(x) \setminus \bar{\Omega}^* \neq \emptyset\}$. By (3.7), we have

$$0 = \mu_g^*(\chi_w(F_w) \setminus \bar{\Omega}^*) = \mu_f\big(\chi_w^{-1}\big(\chi_w(F_w) \setminus \bar{\Omega}^*\big)\big) \geq \mu_f(F_w).$$

Hence $\gamma_{\mathbb{S}^n}(F_w) = 0$. This completes the proof. □

We are ready to prove Theorem 1.1.

Proof of Theorem 1.1. By Lemmas 3.2 & 3.3, we see (i) $\Longrightarrow$ (ii). It follows by Lemma 3.1 that (ii) $\Longrightarrow$ (iii). Hence it remains to show (iii) $\Longrightarrow$ (i).

Suppose $T \in \mathcal{S}_{f,g}$ is an optimal map of (1.5). Lemma 3.1 asserts the existence of potential functions (w, v) such that $T = T_w$ a.e. on $\bar{\Omega}$. In view of Theorem 3.1, w is an Aleksandrov weak solution of (1.23) and (1.24). Hence

$$\mu_f(\Omega) = \mu_g^*(T_w(\Omega)) = \mu_g^*(\Omega^*).$$

Suppose Borel set ω satisfies $\mu_f(\Omega \setminus \omega) > 0$ and $\mu_g^*(\omega_{d_0}) > 0$. By (3.5) again, we see that

$$\mu_f(\omega) = \mu_g^*(T_w(\omega)) < \mu_g^*(\omega_{d_0}).$$

The last inequality is because $T_w(x) \subset B_{d_0}(x)$ for all x. □

4. A parabolic flow towards smooth potential functions

In this section, we study (1.23) and (1.24) in smooth category, and aim to show the existence of smooth potential functions to (1.14) and smooth optimal maps to (1.5). From now on, we always assume that $f \in C^2(\bar{\Omega})$ and $g \in C^2(\bar{\Omega}^*)$ are positive.

4.1. **Preliminaries.**

Recall that $\tilde{F}(s) = F \circ \arccos s$. Since $x \cdot y = \cos d(x, y)$, we have $F(d(x, y)) = \tilde{F}(x \cdot y)$. Given $(x, y) \in \mathbb{S}^n \times \mathbb{S}^n$ such that $d(x, y) < d_0$, let $\{\boldsymbol{e}_i\}_{i=1}^n$ and $\{\bar{\boldsymbol{e}}_i\}_{i=1}^n$ be smooth frames around x and y, respectively. We have

$$\begin{aligned} c_i(x, y) &:= \nabla_{\boldsymbol{e}_i} c(x, y) = \tilde{F}' \langle \boldsymbol{e}_i, y \rangle, \\ c_{\bar{i}}(x, y) &:= \nabla_{\bar{e}_i} c(x, y) = \tilde{F}' \langle x, \bar{\boldsymbol{e}}_i \rangle, \end{aligned} \tag{4.1}$$

Differentiating (4.1) repeatedly and using condition (F1) of cost functions in $\mathscr{C}_{d_0}^\infty$, we see the smoothness of $c(x, y)$ as stated in the lemma below.

Lemma 4.1. *Cost $c \in \mathscr{C}_{d_0}^\infty$ is C^∞ in a neighbourhood of $(x, y) \in \mathbb{S}^n \times \mathbb{S}^n$ if $d(x, y) < d_0$.*

If c-convex function w is twice-differentiable at x, then (1.19) implies that

$$\mathcal{B}[w](x) := \nabla^2 w(x) - \nabla_x^2 c(x, \chi_w(x)) \geq 0. \tag{4.2}$$

When a c-convex $w \in C^2(\bar{\Omega})$ satisfies the strict (4.2) on $\bar{\Omega}$, it is called *uniformly c-convex*.

Let $w \in C^2(\bar{\Omega})$ be a uniformly c-convex potential function. By Lemma 3.1, $\chi_w(x) \in B_{d_0}(x) \cap \bar{\Omega}^*$ and (1.21) holds. That is

$$w_i(x) := \boldsymbol{e}_i w(x) = c_i(x, \chi_w(x)). \tag{4.3}$$

Differentiating this equation again, we obtain

$$\begin{aligned} w_{ij} &= \boldsymbol{e}_j \boldsymbol{e}_i w - (\nabla_{\boldsymbol{e}_j} \boldsymbol{e}_i) w \\ &= \nabla_{xx}^2 c|_{(x, \chi_w(x))}(\boldsymbol{e}_i, \boldsymbol{e}_j) + \nabla_{x,y}^2 c|_{(x, \chi_w(x))}(\boldsymbol{e}_i, D_{\boldsymbol{e}_j} \chi_w(x)). \end{aligned}$$

It follows that

$$w_{ij}(x) - \mathcal{A}_{ij}(x, \nabla w) = c_{i\bar{k}}\sigma^{\bar{k}\bar{l}}\langle \bar{\boldsymbol{e}}_l, D_{\boldsymbol{e}_j}\chi_w(x)\rangle \tag{4.4}$$

where $\sigma^{\bar{i}\bar{j}}$ is the inverse of matrix $\sigma_{\bar{i}\bar{j}} = \langle \bar{\boldsymbol{e}}_i, \bar{\boldsymbol{e}}_j\rangle$ and $\mathcal{A}_{ij}(x, \nabla w) = \mathcal{A}|_{(x,\nabla w)}(\boldsymbol{e}_i, \boldsymbol{e}_j)$ is given by (1.26). By (4.4), the determinant of the gradient matrix $D\chi_w$ is

$$\begin{aligned} \mathrm{Jac}(\chi_w) &= \sqrt{\det\langle D_{\boldsymbol{e}_i}\chi_w, D_{\boldsymbol{e}_j}\chi_w\rangle} \\ &= \sqrt{\det(\langle D_{\boldsymbol{e}_i}\chi_w, \bar{\boldsymbol{e}}_k\rangle\sigma^{\bar{k}\bar{l}}\langle D_{\boldsymbol{e}_j}\chi_w, \bar{\boldsymbol{e}}_l\rangle)} \\ &= \det(w_{ij} - \mathcal{A}_{ij}(x, \nabla w))\frac{\sqrt{\det \sigma_{\bar{i}\bar{j}}}}{\det c_{i\bar{j}}}. \end{aligned} \tag{4.5}$$

Since $\chi_w \in \mathcal{S}_{f,g}$, we deduce by the area formula that

$$\det(w_{ij}(x) - \mathcal{A}_{ij}(x, \nabla w)) = \frac{f(x)}{g(\chi_w(x))}\frac{\sqrt{\det \sigma_{ij}}}{\sqrt{\det \sigma_{\bar{i}\bar{j}}}}\det c_{i\bar{j}}(x, \chi_w(x)). \tag{4.6}$$

We remark that (4.6) is invariant under the change of coordinates. This verifies (1.23).

To study the regularity of w, we expect this is an elliptic equation with non-degenerate right-hand side. Namely we need (1.25). Differentiating (4.1) gives

$$c_{i\bar{j}}(x, y) = \nabla^2_{x,y}c|_{(x,y)}(\boldsymbol{e}_i, \bar{\boldsymbol{e}}_j) = \tilde{F}''\langle \boldsymbol{e}_i, y\rangle\langle x, \bar{\boldsymbol{e}}_j\rangle + \tilde{F}'\langle \boldsymbol{e}_i, \bar{\boldsymbol{e}}_j\rangle. \tag{4.7}$$

Take $\xi \in \mathbb{S}^n$ such that $x \cdot \xi = 0$ and

$$y = \cos\theta \cdot x + \sin\theta \cdot \xi, \text{ where } \theta = d(x, y).$$

If x and y are collinear, then $\sin\theta = 0$ and ξ can be arbitrary. Suppose $\{\boldsymbol{e}_i\}_{i=1}^n$ is orthonormal at $x \in \mathbb{S}^n$. Take $\bar{\boldsymbol{e}}_i(y)$ to be the parallel transport of $\boldsymbol{e}_i(x)$ (for all $i = 1, \ldots, n$) along the geodesic $\gamma(t) = \exp_x(t\xi)$ at $y = \gamma(\theta)$. Then, $\{\bar{\boldsymbol{e}}_i(y)\}_{i=1}^n$ is also orthonormal. Without loss of generality, we may assume $\boldsymbol{e}_1(x) = \xi$. Consequently,

$$\bar{\boldsymbol{e}}_1(y) = \dot{\gamma}(y) = \cos\theta \cdot \xi - \sin\theta \cdot x, \quad \bar{\boldsymbol{e}}_i(y) = \boldsymbol{e}_i(x) \text{ for all } i \geq 2.$$

Plugging this into (4.7), we obtain

$$c_{i\bar{j}}(x, y) = \begin{cases} -\tilde{F}''\sin^2\theta + \tilde{F}'\cos\theta, & i = j = 1, \\ \tilde{F}'\delta_{ij} & i, j \geq 2, \\ 0, & \text{otherwise.} \end{cases} \tag{4.8}$$

Direct computation shows

$$\tilde{F}'(s) = -\frac{F'(\theta)}{\sin\theta} \text{ and } \tilde{F}''(s) = \frac{F''(\theta) - F'(\theta)\cot\theta}{\sin^2\theta}. \tag{4.9}$$

Combining this with (4.8) gives

$$\frac{\det c_{i\bar{j}}(x, y)}{\sqrt{\det \sigma_{ik}(x)\det \sigma_{\bar{k}\bar{j}}(y)}} = -F''(\theta)\Big(\frac{-F'(\theta)}{\sin\theta}\Big)^{n-1}, \text{ where } \theta = d(x, y). \tag{4.10}$$

Lemma 4.2. *Suppose $c \in \mathscr{C}^\infty_{d_0}$. Then (1.25) holds for $(x, y) \in \mathbb{S}^n \times \mathbb{S}^n$ if $d(x, y) < d_0$.*

Proof. By condition (F2),

$$\lim_{\theta \to 0} \frac{F'(\theta)}{\sin\theta} = F''(0) < 0.$$

Therefore $F'(\theta)/\sin\theta < 0$ for all $\theta \in [0, d_0)$. Hence (1.25) follows by (4.10). $\square$

Lemma 4.3. *Let $c \in \mathscr{C}^\infty_{d_0}$ and $w \in C^2(\bar\Omega)$ be uniformly c-convex. Then*

$$\chi_w(x) = G(x, \nabla w(x)), \tag{4.11}$$

where $G : T\mathbb{S}^n \to \mathbb{S}^n$ is given below with $\theta(Z) := [F']^{-1}(-|Z|)$,

$$G(x, Z) = -\frac{\sin\theta(Z)}{F'(\theta(Z))} Z + \cos\theta(Z) \cdot x. \tag{4.12}$$

Proof. By (4.3) and using in the first equations in (4.1) and in (4.9), we have

$$\chi_w(x) = -\frac{\sin d}{F'(d)} \nabla w + \cos d \cdot x, \ \text{ where } d = d(x, \chi_w(x)). \tag{4.13}$$

Since $\chi_w(x) \in \mathbb{S}^n$, we find that $F'(d) = -|\nabla w|$. As F is strictly concave, we obtain

$$d(x, \chi_w(x)) = \theta(Z).$$

This together with (4.13) yields (4.11). $\square$

Lemma 4.4. *Let $c \in \mathscr{C}^\infty_{d_0}$. Then the $(0, 2)$-tensor (1.26) can be written as*

$$\mathcal{A}(x, Z) = \frac{F''(\theta) - F'(\theta)\cot\theta}{[F'(\theta)]^2} Z^* \otimes Z^* + F'(\theta)\cot\theta \sigma_{\mathbb{S}^n}(x), \tag{4.14}$$

where $\theta = \theta(Z)$ is the function in Lemma 4.3, Z^* is the 1-form such that $Z^*(\xi) = \langle Z, \xi\rangle$ for all $\xi \in T_x\mathbb{S}^n$, and $\sigma_{\mathbb{S}^n}$ is the standard metric of $\mathbb{S}^n$. As a result,

$$\nabla_x \mathcal{A}(x, Z) = 0, \ \forall (x, Z) \in T\mathbb{S}^n. \tag{4.15}$$

Proof. Differentiating the first equation in (4.1) yields

$$\begin{aligned} c_{ij}(x, y) &= \boldsymbol{e}_j \boldsymbol{e}_i c - (\nabla_{\boldsymbol{e}_j} \boldsymbol{e}_i) c \\ &= \tilde F'' \langle \boldsymbol{e}_i, y\rangle \langle \boldsymbol{e}_j, y\rangle + \tilde F' \langle D_{\boldsymbol{e}_j} \boldsymbol{e}_i - \nabla_{\boldsymbol{e}_j} \boldsymbol{e}_i, y\rangle \\ &= \tilde F'' \langle \boldsymbol{e}_i, y\rangle \langle \boldsymbol{e}_j, y\rangle - \tilde F' \langle x, y\rangle \sigma_{ij}(x). \end{aligned}$$

where D is the derivative in $\mathbb{R}^{n+1}$ and $\sigma_{ij} = \langle \boldsymbol{e}_i, \boldsymbol{e}_j\rangle$. By (4.9) and (4.12), we obtain

$$\mathcal{A}_{ij}(x, Z) = c_{ij}(x, G(x, Z)) = \frac{F''(\theta) - F'(\theta)\cot\theta}{[F'(\theta)]^2} Z_i Z_j + F'(\theta)\cot\theta \sigma_{ij}(x).$$

$\square$

Proposition 4.1. *Let $c \in \mathscr{C}^\infty_{d_0}$ and $w \in C^2(\bar\Omega)$ be uniformly c-convex. Then w is a potential function of (1.14) if and only if*

$$\begin{aligned} \det(w_{ij} - \mathcal{A}_{ij}(x, \nabla w)) &= \Psi(x, \nabla w) \det \sigma_{ij}(x), \ \forall\, x \in \Omega, \\ \chi_w(\Omega) &= \Omega^*. \end{aligned} \tag{4.16}$$

where $\Psi : T\mathbb{S}^n \to \mathbb{S}^n$ is defined below with $G(x, Z)$ and $\theta = \theta(Z)$ given in Lemma 4.3,

$$\Psi(x, Z) = -F''(\theta)\Big(\frac{-F'(\theta)}{\sin\theta}\Big)^{n-1} \frac{f(x)}{g(G(x, Z))}. \tag{4.17}$$

Proof. If w is a potential function, then (4.16) follows by (4.6), (4.10) and Lemma 4.3.

On the other hand, if $w \in C^2(\bar\Omega)$ is a solution of (4.16), then its c-normal map $\chi_w \in \mathcal{S}_{f,g}$. By virtue of Lemma 3.1, w is a potential of (1.14). □

4.2. **A parabolic flow towards optimal transportation.**

For the purpose of Theorem 1.2, we employ a parabolic approach. Let us consider the following flow whose limit, if exists, leads to a solution of (4.16)

$$\begin{cases} \partial_t w(x,t) = -\dfrac{\Psi(x, \nabla w)}{\det(\nabla^2 w - \mathcal{A}(x, \nabla w))} + 1, & (x,t) \in \Omega \times [0, T), \\ \varrho^*(\chi_w(x,t)) = 0, & (x,t) \in \partial\Omega \times [0,T), \\ w(x, 0) = w_0(x), & x \in \bar\Omega, \end{cases} \tag{4.18}$$

where ϱ^* is a defining function for Ω^*, and $w_0 \in C^\infty(\bar\Omega)$ is an admissible initial data in the sense of Definitions 4.1 and 4.2 below. For convenience, we always use the notation

$$\det(\nabla^2 w - \mathcal{A}(x, \nabla w)) = \frac{\det(w_{ij} - \mathcal{A}_{ij}(x, \nabla w))}{\det \sigma_{ij}},$$

so that the left-hand side above is independent of the choice of frames (also coordinates).

Definition 4.1. *We say $\varrho^* \in C^2(\bar\Omega^*)$ is a defining function for Ω^* if $\varrho^* < 0$ in Ω^*, $\varrho^*|_{\partial\Omega^*} = 0$ and $\nabla\varrho^*|_{\partial\Omega^*} = \nu^*$ on $\partial\Omega^*$, where ν^* is the unit outer normal of $\partial\Omega^*$.*

Definition 4.2. *Let $c \in \mathscr{C}^\infty_{d_0}$ and Ω, Ω^* be two smooth domains in $\mathbb{S}^n$. We say $w_0 \in C^4(\bar\Omega)$ is admissible if it is uniformly c-convex and $\chi_{w_0} : \bar\Omega \to \bar\Omega^*$ is a C^3-diffeomorphism.*

Our flow (4.18) is different from the ones introduced in [28, 29], where a logarithmic type parabolic optimal transport equation has been studied. The advantage of (4.18) is that (4.24) holds along the flow. Such normalization condition is useful in deriving the C^0 and C^1 estimates of $w(\cdot, t)$. Since $c \in \mathscr{C}^\infty_{d_0}$ and its derivatives are unbounded beforehand, the C^0 and C^1 estimates are crucial in our argument.

Short-time existence of (4.18) follows by the argument as in [28, 29]. We show some elementary properties of (4.18) below.

Lemma 4.5. *Suppose Ω and Ω^* are c-convex w.r.t. each other. Let $w \in C^{4,2}_{x,t}(\bar\Omega \times [0, T))$ be a uniformly c-convex solution to (4.18) with an admissible initial data w_0. Then for each $t \in (0, T)$, $w(\cdot, t) \in C^4(\bar\Omega)$ satisfies*

(i) *$w(\cdot,t)$ is strictly c-convex.*
(ii) *$\chi_{w(\cdot,t)}:\bar\Omega\to\bar\Omega^*$ is a C^3-diffeomorphism.*

Proof. Suppose by contradiction argument $w(\cdot,t_0)$ is not strictly c-convex, which means that there exists a c-function $\phi_{y_0,a}(\cdot):=c(\cdot,y_0)+a$ supporting $w(\cdot,t_0)$ from below at two distinct points $x_0,x_1\in\Omega$. Take $\hat w(\cdot,t_0)=w(\cdot,t_0)-\phi_{y_0,a}(\cdot)$. Denote by $\{x_s\}_{s\in[0,1]}$ the c-segment w.r.t. y_0 from x_0 to x_1, namely $\nabla_y c(x_s,y_0)=(1-t)\eta_0+t\eta_1$ with $\eta_0=\nabla_y c(x_0,y_0)$ and $\eta_1=\nabla_y c(x_1,y_0)$. Let $\phi_s^*(\cdot)=c(x_s,\cdot)-c(x_s,y_0)$. Take $y_s=\chi_w(x_s)$. Then

$$\begin{aligned}\hat w(x_s,t_0) &= w(x_s,t_0)-\big(c(x_s,y_0)-c(x_0,y_0)+w(x_0,t_0)\big)\\ &\le w(x_s,t_0)-\big(c(x_s,y_0)-c(x_0,y_0)\big)-\big(c(x_0,y_s)-c(x_s,y_s)+w(x_s,t_0)\big)\\ &= \phi_s^*(y_s)-\phi_0^*(y_s).\end{aligned}$$

Arguing similarly with x_0 replacing by x_1, we have $\hat w(x_s,t_0)\le\phi_s^*(y_s)-\phi_1^*(y_s)$. Combining this with Loeper's maximum principle [26, 31, 41] (see also Section 11.1), we find that

$$0\le\hat w(x_s,t_0)\le\phi_s^*(y_s)-\max\{\phi_0^*(y_s),\phi_1^*(y_s)\}\le 0.$$

Hence $\hat w(x_s,t_0)=0$ for $s\in(0,1)$. However, the uniform c-convexity of $w(\cdot,t)$ implies that $\hat w(\cdot,t)$ attains its strict local minimums at x_0 and x_1, namely $\hat w(x_s,t_0)>0$ for very small s. We reach a contradiction.

As a result of strict c-convexity of $w(\cdot,t)$, we see that $\chi_{w(\cdot,t)}$ is injective. One can derive (ii) by following the topological argument in [29, Corollary 5.5]. $\square$

Lemma 4.6. *Let $w\in C^{4,2}_{x,t}(\bar\Omega\times[0,T))$ be a uniformly c-convex solution of (4.18), and $v(y,t)$ be its c-transform. Then $v\in C^{4,2}_{y,t}(\bar\Omega^*\times[0,T))$ satisfies*

$$\partial_t v(y,t)=\partial_t w(\chi_{v(\cdot,t)}(y),t), \tag{4.19}$$

and is a uniformly c-convex solution to

$$\begin{cases}\partial_t v(y,t)=\dfrac{\det(\nabla^2 v-\mathcal{A}(y,\nabla v))}{\Psi^*(y,\nabla v)}-1, & (y,t)\in\Omega^*\times[0,T),\\ \varrho(\chi_v(y,t))=0, & (y,t)\in\partial\Omega^*\times[0,T),\\ v(y,0)=v_0(y), & x\in\bar\Omega,\end{cases} \tag{4.20}$$

where v_0 is the c-transform of w_0, ϱ is a defining function for Ω, and

$$\Psi^*(y,Z)=-F''(\theta)\Big(\frac{-F'(\theta)}{\sin\theta}\Big)^{n-1}\frac{g(y)}{f(G(y,Z))}.$$

Proof. Denote $Y(x,t)=\chi_{w(\cdot,t)}(x)$ for convenience. By Lemma 2.5, we have

$$w(x,t)+v(Y(x,t),t)=c(x,Y(x,t)),\quad \forall x\in\Omega.$$

Differentiating this equation w.r.t. t gives

$$\partial_t w(x,t)+\partial_t v(Y,t)+\langle\nabla v(Y,t),\partial_t Y\rangle=\tilde F\langle x,\partial_t Y\rangle. \tag{4.21}$$

On the other hand, since $c(x,\cdot)-v(\cdot,t)$ attains its maximum at $y=Y(x,t)$, we conclude

$$\nabla v(Y,t)=\tilde{F}'\sum_i\langle x,\bar{\boldsymbol{e}}_i\rangle\bar{\boldsymbol{e}}_i,$$

where $\{\bar{\boldsymbol{e}}_i\}_{i=1}^n$ is an orthonormal basis at $T_y\mathbb{S}^n$. Since $\langle\partial_t Y,Y\rangle=0$, we see that

$$\langle\nabla v,\partial_t Y\rangle=\tilde{F}'\sum_i\langle x,\bar{\boldsymbol{e}}_i\rangle\langle\partial_t Y,\bar{\boldsymbol{e}}_i\rangle=\tilde{F}'\langle x,\partial_t Y\rangle. \tag{4.22}$$

Inserting (4.22) into (4.21), we get

$$\partial_t v(Y,t)=-\partial_t w(x,t).$$

In view of Lemma 2.5, if $y=Y(x,t)$ then $x=\chi_{v(\cdot,t)}(y)$. Hence (4.19) follows.

As $\chi_{w(\cdot,t)}\circ\chi_{v(\cdot,t)}=id_{\Omega^*}$ and $\chi_{v(\cdot,t)}\circ\chi_{w(\cdot,t)}=id_{\Omega}$, we infer that

$$\mathrm{Jac}(\chi_{w(\cdot,t)})\big|_{X(y,t)}=\frac{1}{\mathrm{Jac}(\chi_{v(\cdot,t)})}\big|_{(y,t)},$$

where $X(y,t)=\chi_{v(\cdot,t)}(y)$ for convenience. This together with (4.5) implies

$$\frac{\det(\nabla^2 w(X,t)-\nabla^2_{xx}c(X,y))}{\det\nabla^2_{x,y}c(X,y)}=\frac{\det\nabla^2_{x,y}c(X,y)}{\det(\nabla^2 v-\nabla^2_{yy}c(X,y))}$$

in orthonormal frame. It then follows by (4.19) and (4.18) that

$$\begin{aligned}\partial_t v(y,t)&=\frac{f(X)\det\nabla^2_{x,y}c(X,y)}{g(y)\det(\nabla^2 w(X,t)-\nabla^2_{xx}c(X,y))}-1\\&=\frac{f(X)\det(\nabla^2 v(y,t)-\nabla^2_{yy}c(X,y))}{g(y)\det\nabla^2_{x,y}c(X,y)}-1.\end{aligned}\tag{4.23}$$

Since $c(x,y)$ is a function of $d(x,y)$ and is symmetric in x and y, it can be verified that

$$\nabla^2_{yy}c(X,y)=\mathcal{A}(y,Z),\quad X(y,t)=G(y,\nabla v(y,t)),$$

and

$$\frac{g(y)}{f(X)}\det\nabla^2_{x,y}c(X,y)=\Psi^*(y,\nabla v).$$

Substituting the above relations into (4.23), we obain (4.20). □

Lemma 4.7. *Let $w\in C^{4,2}_{x,t}(\bar{\Omega}\times[0,T))$ be a uniformly c-convex solution of (4.18), and $v(y,t)$ be its c-transform. Suppose (1.3) holds. Then*

$$\int_{\Omega^*}g(y)v(y,t)d\gamma_{\mathbb{S}^n}=constant,\ \forall t\in[0,T), \tag{4.24}$$

and the Kantorovich's functional (1.12) is non-increasing under the flow

$$\frac{d}{dt}\mathcal{J}(w(\cdot,t),v(\cdot,t))\le 0,\ \forall t\in[0,T), \tag{4.25}$$

with equality holds at some $t_0\in[0,T)$ if and only if $w(\cdot,t_0)$ is a solution of (1.23).

Proof. Denote $X(y,t)=\chi_{v(\cdot,t)}(y)$ and $Y(x,t)=\chi_{w(\cdot,t)}(x)$. By Lemma 4.6, we have

$$\begin{aligned}\frac{d}{dt}\int_{\Omega^*} gv d\gamma_{\mathbb{S}^n}(y) &= \int_{\Omega^*} f(X)\mathrm{Jac}(X)d\gamma_{\mathbb{S}^n}(y)-\int_{\Omega^*} gd\gamma_{\mathbb{S}^n}\\ &= \int_{\Omega} fd\gamma_{\mathbb{S}^n}-\int_{\Omega^*} gd\gamma_{\mathbb{S}^n}\\ &= 0,\end{aligned}$$

where the area formula is used in the second line. Hence (4.24) follows.

In view of (4.19) and (4.18),

$$\begin{aligned}\frac{d}{dt}\mathcal{J}(w(\cdot,t),v(\cdot,t)) &= \int_{\Omega} f\partial_t w d\gamma_{\mathbb{S}^n}-\int_{\Omega^*} g\partial_t w(X,t)d\gamma_{\mathbb{S}^n}\\ &= \int_{\Omega} f\partial_t w d\gamma_{\mathbb{S}^n}-\int_{\Omega} g(Y)\partial_t w(x,t)\mathrm{Jac}(Y)d\gamma_{\mathbb{S}^n}\\ &= -\int_{\Omega} g\circ Y\mathrm{Jac}(Y)\Big(-\frac{f}{g\circ Y\mathrm{Jac}(Y)}+1\Big)\partial_t w(x,t)d\gamma_{\mathbb{S}^n}\\ &= -\int_{\Omega} g\circ Y\mathrm{Jac}(Y)[\partial_t w(x,t)]^2 d\gamma_{\mathbb{S}^n}\\ &\le 0.\end{aligned}$$

Since $g\circ Y\mathrm{Jac}(Y)$ is positive, we find that if $\frac{d}{dt}\mathcal{J}=0$ at $t=t_0$, then $\partial_t w(\cdot,t_0)=0$. It then follows by (4.18) that $w(\cdot,t_0)$ satisfies (1.23). □

4.3. C^0 and C^1 estimates of the flow.

In this subsection we prove the crucial spatial gradient estimates for $w(\cdot,t)$. As a result, $c(x,\chi_{w(\cdot,t)})$ has bounded derivatives on $\bar\Omega$.

Lemma 4.8. *Let Ω,Ω^* be two domains satisfying (1.17). Suppose $w\in C^{4,2}_{x,t}(\bar\Omega\times[0,T))$ be a uniformly c-convex solution of (4.18). If (Ω,f) and (Ω^*,g) are Aleksandrov related, then there exists $C>0$ that depends on w_0 but is independent of T such that*

$$\sup_{\bar\Omega\times[0,T)}|w|\le C. \tag{4.26}$$

Moreover $v(\cdot,t)$, the c-transform of $w(\cdot,t)$, satisfies

$$\sup_{\bar\Omega^*\times[0,T)}|v|\le C. \tag{4.27}$$

Proof. By (4.25) in Lemma 4.7, we find that

$$\big(w(\cdot,t),v(\cdot,t)\big)\in\mathcal{K}_A,\ \forall t\in[0,T),$$

where $v(\cdot,t)$ is the c-transform of $w(\cdot,t)$, and

$$A=\mathcal{J}(w_0,v_0).$$

In view of Lemma 2.7, there exists $C > 0$ depending on w_0 such that

$$\sup_{t\in[0,T)} \Big(\mathrm{osc}_{\bar\Omega}\, w(\cdot,t) + \mathrm{osc}_{\bar\Omega^*}\, v(\cdot,t) \Big) \le C. \tag{4.28}$$

By (4.24) in Lemma 4.7, we obtain the uniform L^∞-norm of v, namely (4.27).

We also deduce by (4.24) and (4.25)

$$\int_\Omega w(\cdot,t) f d\gamma_{\mathbb{S}^n} \le A' := \int_\Omega w_0 f d\gamma_{\mathbb{S}^n}. \tag{4.29}$$

As a consequence,

$$\min_{\bar\Omega} w(\cdot,t) \le A' \Big(\int_\Omega f \Big)^{-1}.$$

This together with (4.28) yields

$$\max_{\bar\Omega} w(\cdot,t) \le \min_{\bar\Omega} w(\cdot,t) + \mathrm{osc}_{\bar\Omega} w(\cdot,t) \le C, \ \forall t \in [0,T). \tag{4.30}$$

On the other hand, by (4.27),

$$\min_{\bar\Omega} w(\cdot,t) = \min_{x\in\bar\Omega} \max_{y\in\bar\Omega^*} \{c(x,y) - v(y,t)\} \ge \min_{x\in\bar\Omega} \max_{y\in\bar\Omega^*} c(x,y) - C \ge F(d_1) - C. \tag{4.31}$$

where d_1 is defined by (2.8). By (4.30) and (4.31), one sees (4.26). $\square$

Lemma 4.9. *Under the assumptions of Lemma 4.8, there are $\varepsilon_0 > 0$ and $C > 0$ that depend on w_0 but are both independent of T, such that*

$$d(x, \chi_{w(\cdot,t)}(x)) < d_0 - \varepsilon_0, \quad d(\chi_{v(\cdot,t)}(y), y) < d_0 - \varepsilon_0, \tag{4.32}$$

and

$$\sup_{\bar\Omega\times[0,T)} |\nabla w| + \sup_{\bar\Omega^*\times[0,T)} |\nabla v| \le C. \tag{4.33}$$

Proof. By Lemma 2.5 and estimates (4.26), (4.27),

$$c(x, \chi_{w(\cdot,t)}(x)) = w(x,t) + v(\chi_{w(\cdot,t)}(x), t) \ge -C.$$

This implies the first estimate in (4.32). Similarly we get the second one.

The gradient estimate (4.33) follows by

$$|\nabla w(\cdot,t)| = |\nabla_x c(x, \chi_{w(\cdot,t)}(x))| \text{ and } |\nabla v(\cdot,t)| = |\nabla_y c(\chi_{v(\cdot,t)}(y), y)|,$$

and using (4.32). $\square$

The next conclusion is a direct consequence of (4.32).

Corollary 4.1. *Under the assumptions of Lemma 4.8, for every $(x,t) \in \bar\Omega \times [0,T)$, there is a neighbourhood $\mathcal{U}$ of $(x, \chi_{w(\cdot,t)}(x))$ in $\mathbb{S}^n \times \mathbb{S}^n$ such that $c(x,y)$ is smooth in $\mathcal{U}$.*

4.4. **Convergence of the flow.**

Theorem 4.1. *Let $c \in \mathscr{C}^\infty_{d_0}$ be a cost function satisfying the MTW condition, Ω and Ω^* be two C^4-smooth domains in $\mathbb{S}^n$ that are uniformly c-convex to each other, and $f \in C^2(\bar{\Omega})$ and $g \in C^2(\bar{\Omega}^*)$ be two positive functions. Suppose that condition (1.17) holds, (Ω, f) and (Ω^*, g) are Aleksandrov related, and $w_0 \in C^4(\bar{\Omega})$ is an admissible initial data. Then there exists a uniformly c-convex solution $w \in C^{4,2}_{x,t}(\bar{\Omega} \times [0,\infty))$ to the flow (4.18).*

In addition $w(\cdot,t)$ converges along a sequence of times t_i in $C^{3,\alpha}$ to a uniformly c-convex function $w^\infty \in C^{3,\alpha}(\bar{\Omega})$ as $t_i \to \infty$ which satisfies (1.23) and (1.24).

Theorem 4.1 follows by the a priori estimates given below.

Proposition 4.2. *Under the conditions of Theorem 4.1, if $w \in C^4(\bar{\Omega} \times [0,T))$ is a uniformly convex solution to the flow (4.18), then*

$$\sup_{\bar{\Omega}} |\nabla^2 w(\cdot,t)| \leq C,$$

where $C > 0$ depends on $f, g, F|_{[0,d_0-\varepsilon_0]}, w_0$ and the modulus of c-convexity of Ω, Ω^*, but is independent of T.

We say δ_0 is a *modulus of c-convexity* of Ω w.r.t. Ω^* if (1.28) holds for δ_0. The modulus of c-convexity of Ω^* w.r.t. Ω is defined in a symmetric way.

Proposition 4.2 is a consequence of Lemma 7.2, Lemma 10.3 and Lemma 10.5, which are deferred to the subsequent sections. Let us provide a proof of Theorem 4.1.

Proof of Theorem 4.1. By Proposition 4.2, we find that (4.18) is uniformly parabolic. The Krylov's regularity theory for parabolic equations (see Lieberman [35], Chapter 14) gives us $C^{2,\alpha}(\bar{\Omega})$ estimates (independent of T) for w, and the long-time existence of (4.18) follows from a standard argument using Arzelà-Ascoli theorem. Applying the Schauder theorey for linear uniformly parabolic equation satisfied by $\nabla_x w$ which subject to linear oblique boundary condition, we conclude that $w \in C^{3,\alpha}(\bar{\Omega})$ (independent of time). The readers may consult [35, Chapter 5].

By Lemma 4.8, we have for all $t \geq 0$,

$$\mathcal{J}_{w_0}(t) := \mathcal{J}(w(\cdot,t), v(\cdot,t)) \geq -C.$$

As a result,

$$\int_0^\infty \mathcal{J}'_{w_0}(t)dt \geq \liminf_{T\to\infty} \mathcal{J}_{w_0}(T) - \mathcal{J}_{w_0}(0) \geq -C. \tag{4.34}$$

By (4.25), $\mathcal{J}'_{w_0}(t) \leq 0$. Hence (4.34) implies that there is a sequence $t_i \to \infty$ so that

$$-\int_\Omega g \circ Y \operatorname{Jac}(Y)[\partial_t w(x,t_i)]^2 d\gamma_{\mathbb{S}^n} = \mathcal{J}'_{w_0}(t_i) \to 0.$$

Passing to a subsequence, we obtain by the a priori estimates $\|w(\cdot,t_i)\|_{C^{3,\alpha}(\bar{\Omega})} \leq C$ that $w(\cdot,t_i) \to w^\infty$ in $C^{3,\alpha}(\bar{\Omega})$-topology and w^∞ satisfies (1.23) and (1.24). $\square$

5. Proof of Theorem 1.2

In this section we prove Theorem 1.2 by employing Theorem 4.1. The main difficulty is the existence of admissible initial data for starting our flow (4.18). It seems not easy to construct an admissible initial data directly for the given domains Ω, Ω^*. Our idea is to design a family of approximation problems for (1.23) and (1.24) and apply Theorem 4.1 to each one. We will construct admissible initial data for every single approximation problem, and Theorem 1.2 follows by a limiting process. Our argument consists of three main steps below.

Step 1. Construct a uniformly c-convex function $w_0 \in C^\infty(\bar{\Omega}_\delta)$ such that $\chi_{w_0}(\Omega_\delta)$ lies in a small neighbourhood of Ω^*_δ, where $\delta > 0$ is small,

$$\Omega_\delta = \{x \in \Omega : d(x, \partial\Omega) > \delta\} \text{ and } \Omega^*_\delta = \{y \in \Omega^* : d(y, \partial\Omega^*) > \delta\}. \tag{5.1}$$

Step 2. Extend w_0 to a c-convex function $w_1 \in C(\bar{\Omega}^\delta)$ such that $\chi_{w_1}(\bar{\Omega}^\delta) = \bar{\Omega}^*$, where

$$\Omega^\delta = \{x \in \mathbb{S}^n : d(x, \Omega) < \delta\}. \tag{5.2}$$

Step 3. Construct a smooth perturbation $w_\sigma \in C^\infty(\bar{\Omega})$ of w_1 such that w_σ are uniformly c-convex and $\Omega^{*,\sigma} := \chi_{w_\sigma}(\Omega)$ is a smooth perturbation of Ω^* and so Ω and $\Omega^{*,\sigma}$ are uniformly c-convex w.r.t. each other.

The three steps above lead to the following conclusion which is crucial in our proof of Theorem 1.2.

Proposition 5.1. *Let Ω and Ω^* be two C^4-smooth domains satisfying (1.17), $f \in C^2(\bar{\Omega})$ and $g \in C^2(\bar{\Omega}^*)$ be two positive functions. Suppose Ω and Ω^* are uniformly c-convex to each other, and $c \in \mathscr{C}^\infty_{d_0}$ satisfies the MTW condition. Then for every small $\sigma > 0$, there is a uniformly c-convex function $w_\sigma \in C^\infty(\bar{\Omega})$ such that $\Omega^{*,\sigma} := \chi_{w_\sigma}(\Omega)$ converges in C^4-topology to Ω^* as $\sigma \to 0$. As a result, Ω and $\Omega^{*,\sigma}$ are uniformly c-convex to each other with a uniform modulus of c-convexity for all small σ.*

The proof of Proposition 5.1 is presented in Section 5.4, which is based on the results of Step 1 (in Section 5.2) and Step 2 (in Section 5.3). Steps 2&3 are proved by using the idea of Trudinger and Wang [42]. But Step 1 is a consequence of some new observations: Lemmas 5.2 and 5.3. Since these two technical lemmas would be of independent interest, we include them in Section 5.1 separately.

For the sake of completing Theorem 1.2, we also need the following result.

Lemma 5.1. *Let $f \in C^2(\bar{\Omega})$, $g \in C^2(\bar{\Omega}^*)$ be positive functions, and Λ, Λ^* be two smooth domains such that $\max\{d_H(\Lambda, \Omega), d_H(\Lambda^*, \Omega^*)\} \le \varepsilon$, where d_H is the Hausdorff distance. Suppose (1.17) holds and (Ω, f) and (Ω^*, g) are Aleksandrov related. Then for every sufficiently small $\varepsilon > 0$ there are positive functions $f_\varepsilon \in C^\infty(\bar{\Lambda})$ and $g_\varepsilon \in C^\infty(\bar{\Lambda}^*)$ such that (Λ, f_ε) and $(\Lambda^*, g_\varepsilon)$ are Aleksandrov related.*

Proof. Given $\varepsilon > 0$, take a positive function $g_\varepsilon \in C^\infty(\bar{\Lambda}^*)$ such that

$$\mu^*_{g_\varepsilon}(\Lambda^*) = \mu_f(\Omega) \ \text{ and } \ \|g_\varepsilon - g\|_{L^1(\Omega^* \cup \Lambda^*)} \leq \sigma_\varepsilon, \tag{5.3}$$

where $\sigma_\varepsilon \to 0$ as $\varepsilon \to 0$. We regard f, g, g_ε as functions on $\mathbb{S}^n$ by the zero extension outside $\bar{\Omega}, \bar{\Omega}^*, \bar{\Lambda}^*$ respectively. We assert that (Ω, f) and $(\Lambda^*, g_\varepsilon)$ are Aleksandrov related provided $\sigma_\varepsilon > 0$ is sufficiently small. For this purpose, it suffices to prove

$$\mu_f(\omega) < \mu^*_{g_\varepsilon}(\omega_{d_0}), \ \text{for all } \omega \in \mathcal{D}, \tag{5.4}$$

where $\mathcal{D} = \{\text{non-empty } d_0\text{-convex set } \omega : \mu_f(\omega) > 0,\ \mu_f(\Omega \setminus \omega) > 0,\ \mu^*_{g_\varepsilon}(\omega_{d_0}) > 0\}$.

Given $\sigma_1 > 0$, let $\mathcal{D}_{\sigma_1} = \{\omega \in \mathcal{G} : \mu_f(\Omega \setminus \omega) \leq \sigma_1\}$. By (1.17), $\bar{\Omega}^* \subset \Omega_{d_0}$. Hence if σ_1 is very small, then $\Lambda^* \subset\subset \omega_{d_0}$ for all $\omega \in \mathcal{D}_{\sigma_1}$. Fix such σ_1. We see from (5.3) that

$$\mu_f(\omega) < \mu_f(\Omega) = \mu^*_{g_\varepsilon}(\Lambda^*) = \mu^*_{g_\varepsilon}(\omega_{d_0}),\ \forall \omega \in \mathcal{D}_{\sigma_1},$$

where the last equality uses $g_\varepsilon \equiv 0$ outside $\bar{\Lambda}^*$. This verifies (5.4) among $\mathcal{D}_{\sigma_1}$.

By Lemma 2.3, there exists $\delta > 0$ depending on σ_1 such that

$$\mu_f(\omega) < (1 - \delta)\mu^*_g(\omega_{d_0}),\ \forall \omega \in \mathcal{D} \setminus \mathcal{D}_{\sigma_1}. \tag{5.5}$$

For each $\omega \in \mathcal{D}$, since $\mu_f(\omega) > 0$, we derive from (1.17) that

$$\mu^*_g(\omega_{d_0}) \geq c_0 := \inf_{x \in \bar{\Omega}} \mu^*_g(B_{d_0}(x)) > 0.$$

If $\varepsilon > 0$ is sufficiently small so that $0 < \sigma_\varepsilon \leq c_0\delta$, where $\delta > 0$ is given in (5.5), then

$$\mu^*_g(\omega_{d_0}) \geq \sigma_\varepsilon / \delta, \ \text{for all } \omega \in \mathcal{D}. \tag{5.6}$$

Employing (5.5), (5.3) first and then (5.6), we deduce that, for each $\omega \in \mathcal{D} \setminus \mathcal{D}_{\sigma_1}$,

$$\begin{aligned} \mu^*_{g_\varepsilon}(\omega_{d_0}) - \mu_f(\omega) &\geq \mu^*_g(\omega_{d_0}) - \mu_f(\omega) - |\mu^*_{g_\varepsilon}(\omega_{d_0}) - \mu^*_g(\omega_{d_0})| \\ &> \delta\mu^*_g(\omega_{d_0}) - \sigma_\varepsilon \\ &\geq 0. \end{aligned}$$

This verifies (5.4) among $\mathcal{D} \setminus \mathcal{D}_{\sigma_1}$. Hence (Ω, f) and $(\Lambda^*, g_\varepsilon)$ are Aleksandrov related.

Since the Aleksandrov relation is symmetric (Lemma 2.2), running the above argument again, there is $f_\varepsilon \in C^\infty(\bar{\Lambda})$ so that (Λ, f_ε) and $(\Lambda^*, g_\varepsilon)$ are Aleksandrov related. ☐

It is the position to complete Theorem 1.2.

Proof of Theorem 1.2. Given a small $\sigma > 0$, let $w_\sigma \in C^\infty(\bar{\Omega})$ be the uniformly c-convex function in Proposition 5.1. By Lemma 5.1, there is a smooth and positive function g_σ such that (i) $\|g_\sigma - g\|_{C^2(\mathbb{S}^n)} \to 0$ as $\sigma \to 0$ and (ii) (Ω, f) and $(\Omega^{*,\sigma}, g_\sigma)$ are Aleksandrov related. Since (1.17) holds for all small σ, we infer by Theorem 4.1 that flow (4.18) with initial data w_σ converges along a sequence of times to a uniformly c-convex function $w_{\sigma,\infty}$ which satisfies (1.23) with g replaced by g_σ, and the boundary condition

$$\chi_{w_{\sigma,\infty}}(\Omega) = \Omega^{*,\sigma}.$$

Since the modulus of c-convexity of $\Omega^{*,\sigma}$ is controlled by that of Ω^*, and $\|g_\sigma - g\|_{C^2(\mathbb{S}^n)}$ is arbitrary small as $\sigma \to 0$, we see form Proposition 4.2 that

$$\|w_{\sigma,\infty}\|_{C^{3,\alpha}(\bar\Omega)} \le C$$

for some C which is independent of σ. Sending $\sigma \to 0$, $w_{\sigma,\infty}$ converges in $C^{3,\alpha}$-topology to a uniformly c-convex function w on $\bar\Omega$. Then w solves (1.23) and (1.24), and so Theorem 1.2 is proved. □

5.1. **Technical lemmas: enhancement of smoothness and c-convexity.**

For the purpose of Step 1, we need two lemmas: one is used to smoothen a c-convex function, while another is to strengthen the c-convexity.

Lemma 5.2. *Let $w \in C(\bar\Omega)$ be a c-convex function. There is a smooth approximation $w_\varepsilon \in C^\infty(\bar\Omega)$ of w with following properties:*

(i) $\|w_\varepsilon - w\|_{C^{0,1}(\bar\Omega)} \to 0$ *as* $\varepsilon \to 0$*;*

(ii) *there is $\beta_w > 0$ depending on d_w given by (2.11), but independent of ε, such that for any small $\delta > 0$ and $\varepsilon \in (0, \delta/2)$,*

$$\nabla^2 w_\varepsilon(x) - \mathcal{A}(x, \nabla w_\varepsilon(x)) \ge -(\varepsilon + \mathit{diam}(\chi_w(x)))\beta_w \sigma_{\mathbb{S}^n}, \ \forall\, x \in \Omega,$$

where $\sigma_{\mathbb{S}^n}$ the metric of $\mathbb{S}^n$.

Proof. Fix a small $\delta > 0$ and let $\Omega^\delta = \{x \in \mathbb{S}^n : \ d_\Omega(x) < \delta\}$ be its δ-neighbourhood, where $d_\Omega(\cdot)$ is the spherical distance to Ω. Extend w to Ω^δ by

$$\tilde w(x) = \sup\{\phi(x) : \ \phi \text{ is a } c\text{-support function of } w \text{ in } \Omega\}, \ x \in \Omega^\delta,$$

where a c-support function of w in Ω means a function $\phi(\cdot) = c(\cdot, y) + a$ with fixed $y \in \mathbb{S}^n$ and constant a such that $\phi \le w$ in Ω and $\phi(x) = w(x)$ at some $x \in \Omega$. By (2.11) in Lemma 2.5, $\tilde w$ is finite-valued and c-convex if δ is small. For $\varepsilon \in (0, \delta/2)$, let $\phi_\varepsilon \in C_c^\infty(\mathbb{R}^n)$ be a nonnegative mollifier with $\|\phi_\varepsilon\|_{L^1(\mathbb{R}^n)} = 1$ and supported in the Euclidean ball $\mathbf{B}_\varepsilon$ at the origin of $\mathbb{R}^n$. Consider the mollification

$$w_\varepsilon(x) = \int_{\mathbb{R}^n} \phi_\varepsilon(z)\tilde w(\exp_x z)dz, \ x \in \bar\Omega. \tag{5.7}$$

As c-convex functions are Lipschitz (see Lemma 2.6), we have $\|w_\varepsilon - w\|_{C^{0,1}(\bar\Omega)} \to 0$.

Let $\gamma : [0,1] \to \bar\Omega$ be a C^1-curve with constant speed $m_\gamma = |\dot\gamma|$. Denote $\gamma_t = \gamma(t)$. For each $t \in [0,1]$, if $y_{\gamma_t} \in \chi_w(\gamma_t)$ and $\{\gamma_s\}_{s\in[0,1]} \subset B_{d_0}(y_{\gamma_t})$, then

$$\begin{aligned} \tilde w(\gamma_t) &= (1-t)\big(c(\gamma_0, y_{\gamma_t}) + a_{\gamma_t}\big) + t\big(c(\gamma_1, y_{\gamma_t}) + a_{\gamma_t}\big) + \omega_c(\gamma_{0,1,t}; y_{\gamma_t}) \\ &\le (1-t)\tilde w(\gamma_0) + t\tilde w(\gamma_1) + \omega_c(\gamma_{0,1,t}; y_{\gamma_t}), \end{aligned} \tag{5.8}$$

where $a_{\gamma_t} = \tilde w(\gamma_t) - c(\gamma_t, y_{\gamma_t})$, and function ω_c is defined as

$$\omega_c(\gamma_{0,1,t}; y_{\gamma_t}) := c(\gamma_t, y_{\gamma_t}) - (1-t)c(\gamma_0, y_{\gamma_t}) - tc(\gamma_1, y_{\gamma_t}).$$

Denote $\dot\gamma_t = m_\gamma e_{\gamma_t}$, $c_e(\gamma_t, y) = \langle \nabla_x c|_{(\gamma_t,y)}, e_{\gamma_t}\rangle$ and $c_{ee}(\gamma_t, y) = \nabla_x^2 c|_{(\gamma_t,y)}(e_{\gamma_t}, e_{\gamma_t})$. Then

$$\begin{aligned}\omega_c(\gamma_{0,1,t}; y_{\gamma_t}) &= \Big((1-t)\int_0^t c_e(\gamma_\tau, y_{\gamma_t})d\tau - t\int_t^1 c_e(\gamma_\tau, y_{\gamma_t})d\tau\Big)m_\gamma \\ &= -(1-t)t\Big(\fint_0^t \int_\tau^t c_{ee}(\gamma_s, y_{\gamma_t})dsd\tau + \fint_t^1\int_t^\tau c_{ee}(\gamma_s, y_{\gamma_t})dsd\tau\Big)m_\gamma^2. \end{aligned} \tag{5.9}$$

Given any $x_* \in \bar\Omega$ and any unit vector $e \in T_{x_*}\mathbb{S}^n$, let

$$x_0 = \exp_{x_*}(-re) \text{ and } x_1 = \exp_{x_*} re,\ \ r > 0.$$

Denote by $\{x_t\}_{t\in[0,1]}$ the minimal geodesic from x_0 to x_1. Then $x_{1/2} = x_*$. Take $y_* \in \chi_w(x_*)$. By parallel transport, we identify each tangent space $T_{x_t}\mathbb{S}^n$ with $T_{x_*}\mathbb{S}^n$, and take $\gamma^z(t) = \exp_{x_t} z$, where $z \in \mathbf{B}_\varepsilon$. When r and ε are small enough, we have $\{\gamma^z_t\}_{t\in[0,1]} \subset B_{d_0}(y_*)$. By (5.7) and (5.8), we see that

$$w_\varepsilon(x_*) = \int_{\mathbb{R}^n}\phi_\varepsilon(z)\tilde w(\gamma^z_{1/2})dz \le \frac12 w_\varepsilon(x_0) + \frac12 w_\varepsilon(x_1) + \int_{\mathbb{R}^n}\phi_\varepsilon(z)\omega_c(\gamma^z_{0,1,\frac12}; y_*)dz.$$

It follows that

$$\begin{aligned}\nabla_e^2 w_\varepsilon(x_*) &= \lim_{r\to0}\frac{w_\varepsilon(x_0)+w_\varepsilon(x_1)-2w_\varepsilon(x_*)}{r^2} \\ &\ge -\lim_{r\to0}\frac{2}{r^2}\int_{\mathbb{R}^n}\phi_\varepsilon(z)\omega_c(\gamma^z_{0,1,\frac12}; y_*)dz.\end{aligned} \tag{5.10}$$

Since $\gamma^z_{1/2} \in B_\varepsilon(x_*)$, we have $c_{ee}(\gamma^z_s, y_*) = c_{ee}(x_*, y_*) + O(r+\varepsilon)$. Hence, by (5.9),

$$\omega_c(\gamma^z_{0,1,\frac12}; y_*) = -\frac{r^2}{2}\Big(\nabla_e^2 c(x_*, y_*) + O\big(r+\varepsilon\big)\Big).$$

Plugging this into (5.10) and using $m_{\gamma^z} = 2r(1+O(\varepsilon))$, we deduce

$$\nabla_e^2 w_\varepsilon(x_*) \ge \nabla_e^2 c(x_*, y_*) + O(\varepsilon) = \mathcal{A}_{ee}(x_*, \nabla w_\varepsilon(x_*)) + O\big(\varepsilon + \operatorname{diam}(\chi_w(x))\big).$$

Since x_* and $e \in T_{x_*}\mathbb{S}^n$ are arbitrary, we complete the proof. $\square$

We next prove a lemma which gives an improvement of c-convexity. Recall that if a c-convex function w satisfies $\mathcal{B}[w] > 0$, see (4.2), then it is uniformly convex.

Lemma 5.3. *Suppose $c \in \mathscr{C}^\infty_{d_0}$ satisfies MTW condition (1.27). Given $w \in C^\infty(\bar\Omega)$, there is a smooth approximation w_ε of w such that*

$$\mathcal{B}[w_\varepsilon](x) \ge \mathcal{B}[w](x) + \big(c_0 - c_1\varepsilon\big)\varepsilon\sigma_{\mathbb{S}^n},\ \ \forall\ x \in \Omega, \tag{5.11}$$

where $c_0, c_1 > 0$ depends only on $\|w\|_{C^{0,1}(\bar\Omega)}$ and is independent of ε.

Proof. Let $u = e^w > 0$ and consider

$$\mathscr{B}[u] := e^w\mathcal{B}[w] = \nabla^2 u - \frac{du\otimes du}{u} - u\mathcal{A}(x, u^{-1}\nabla u).$$

Define $w_\varepsilon := \log(u + K(u)\varepsilon)$, where $K(s) = s^{-a}$ for some $a > 1$ to be specified. Then

$$\mathscr{B}[u_\varepsilon] = (1+K'\varepsilon)\nabla^2 u + K''\varepsilon du\otimes du - \frac{(1+K'\varepsilon)^2 du\otimes du}{u+K\varepsilon} - (u+K\varepsilon)\mathcal{A}\Big(x, \frac{(1+K'\varepsilon)\nabla u}{u+K\varepsilon}\Big). \tag{5.12}$$

By the Taylor expansions (w.r.t. ε), we see that

$$\frac{K''\varepsilon}{1+K'\varepsilon} - \frac{1+K'\varepsilon}{u+K\varepsilon} \geq \Big(\frac{K''u^2 - K'u + K}{u^2}\Big)\varepsilon - c_1\varepsilon^2,$$

and

$$\frac{u+K\varepsilon}{1+K'\varepsilon}\mathcal{A}\Big(x, \frac{(1+K'\varepsilon)\nabla u}{u+K\varepsilon}\Big) \geq u\mathcal{A}\Big(x, \frac{\nabla u}{u}\Big) + (K - K'u)\Big(\mathcal{A} - u^{-1}\nabla u\cdot D_Z\mathcal{A}\Big)\varepsilon - c_1\varepsilon^2.$$

Plugging the above two identities to (5.12), we obtain

$$\begin{aligned}\frac{\mathscr{B}[u_\varepsilon]}{1+K'\varepsilon} &\geq \mathscr{B}[u] + \varepsilon(K''u^2 - K'u + K)\frac{du\otimes du}{u^2} \\ &\quad -\varepsilon(K'u - K)\left(u^{-1}\nabla u\cdot D_Z\mathcal{A}_{ij} - \mathcal{A}\right) - c_1\varepsilon^2 \\ &=: \mathscr{B}[u] + \varepsilon\mathcal{M}(x, u, u^{-1}\nabla u) - c_1\varepsilon^2,\end{aligned}$$

where the $(0,2)$-tensor $\mathcal{M}$ is defined as

$$\begin{aligned}\mathcal{M}(x,u,Z) &= (K''u^2 - K'u + K)Z^*\otimes Z^* - (K'u - K)(Z\cdot D_Z\mathcal{A} - \mathcal{A}) \\ &= (a+1)^2u^{-a}Z^*\otimes Z^* + (a+1)u^{-a}(Z\cdot D_Z\mathcal{A} - \mathcal{A}).\end{aligned} \tag{5.13}$$

where Z^* is the 1-form dual to Z.

We next show that if $a > 1$ is selected properly large, then

$$\mathcal{M}(x, u, u^{-1}\nabla u) > 0 \quad \text{on } \bar{\Omega}. \tag{5.14}$$

Once this is proved, (5.11) follows by the compactness of $\bar{\Omega}$. By (4.14) and using condition (F2), we see that

$$\mathcal{M}(x,u,0) = -(a+1)u^{-a}\mathcal{A}(x,0) = -(a+1)u^{-\alpha}F''(0)\sigma_{\mathbb{S}^n} > 0,\ \forall x\in\bar{\Omega}. \tag{5.15}$$

Therefore we always assume $\nabla u(x) \neq 0$ in the following, otherwise it is done. Given $Z \neq 0$, for any $0 < t < 1$, we have by (5.13),

$$\frac{d}{dt}\mathcal{M}(x,u,tZ) = 2t(a+1)^2u^{-a}Z^*\otimes Z^* + (a+1)u^{-a}t\frac{d^2}{dt^2}\mathcal{A}(x,tZ).$$

This together with (5.15) yields

$$\begin{aligned}\mathcal{M}(x,u,Z) &= \mathcal{M}(x,u,0) + \int_0^1 \frac{d}{dt}\mathcal{M}(x,tZ)dt \\ &= -(a+1)u^{-\alpha}F''(0)\sigma_{\mathbb{S}^n} + (a+1)^2u^{-a}Z^*\otimes Z^* \\ &\quad +(a+1)u^{-a}\int_0^1 t\frac{d^2}{dt^2}\mathcal{A}(x,tZ)dt.\end{aligned} \tag{5.16}$$

We next split our discussion into two cases.

Case 1: we show $\mathcal{M}_{\xi\xi}(x,u,Z)>0$ for all unit ξ with $\langle\xi,Z\rangle=0$. By (5.16) and (1.27),

$$\begin{aligned}\mathcal{M}_{\xi\xi}(x,u,Z) &= -(a+1)u^{-\alpha}F''(0)+(a+1)u^{-a}\int_0^1 t\frac{d^2}{dt^2}\mathcal{A}_{\xi\xi}(x,tZ)dt\\ &\geq -(a+1)u^{-\alpha}F''(0)>0.\end{aligned}$$

Case 2: we show $\mathcal{M}_{\bar Z\bar Z}(x,u,Z)>0$, where $\bar Z:=Z/|Z|$. By (4.14) and $|F'(\theta)|=|Z|$,

$$\mathcal{A}_{\bar Z\bar Z}(x,Z)=\frac{F''(\theta)-F'(\theta)\cot\theta}{[F'(\theta)]^2}|Z|^2+F'(\theta)\cot\theta=F''(\theta). \tag{5.17}$$

For convenience, take $\rho(t)\in[0,d_0)$ so that $F'(\rho(t))=-t|Z|$. Direct computation gives

$$F''(\rho(t))=-\frac{|Z|}{\dot\rho(t)}. \tag{5.18}$$

This together with (5.17) implies that

$$\begin{aligned}\int_0^1 t\frac{d^2}{dt^2}\mathcal{A}_{\bar Z\bar Z}(x,tZ)dt &= -|Z|\int_0^1 t\frac{d^2}{dt^2}\Big(\frac{1}{\dot\rho(t)}\Big)dt\\ &= -|Z|\Big[t\frac{d}{dt}\Big(\frac{1}{\dot\rho(t)}\Big)-\frac{1}{\dot\rho(t)}\Big]\Big|_0^1.\end{aligned}$$

Differentiating (5.18) again, one sees that

$$\frac{d}{dt}\frac{1}{\dot\rho(t)}=\frac{F'''(\rho(t))}{F''(\rho(t))}.$$

As a consequence,

$$\begin{aligned}\int_0^1 t\frac{d^2}{dt^2}\mathcal{A}_{\bar Z\bar Z}(x,tZ)dt &= -|Z|\Big[t\frac{F'''(\rho(t))}{F''(\rho(t))}+\frac{F''(\rho(t))}{|Z|}\Big]\Big|_0^1\\ &= -|Z|\frac{F'''(\theta)}{F''(\theta)}-F''(\theta)+F''(0).\end{aligned}$$

Inserting this into (5.16), we obtain

$$\begin{aligned}\mathcal{M}_{\bar Z\bar Z}(x,u,Z) &= |Z|^2(a+1)^2u^{-a}-(a+1)u^{-a}\Big(|Z|\frac{F'''(\theta)}{F''(\theta)}+F''(\theta)\Big)\\ &\geq \Big(a|Z|^2-C_0|Z|+\delta_0\Big)au^{-a},\end{aligned}$$

where $\delta_0, C_0>0$ are positive constants only depending on $\|w\|_{C^{0,1}(\bar\Omega)}$. The positivity of δ_0 comes from condition (F2), the uniform concavity of F. For large $a>1$ (depending on $\|w\|_{C^{0,1}(\bar\Omega)}$), we obtain

$$\mathcal{M}_{\bar Z\bar Z}(x,u,Z)>0.$$

As every $\eta\in T_x\mathbb{S}^n$ can be decomposed as $\eta=\xi+\langle\eta,\bar Z\rangle\bar Z$, where $\xi\perp Z$, we are done. $\square$

5.2. **Step 1: smooth and uniformly c-convex w_0 on subdomain Ω_δ.**

Assume (Ω, f) and (Ω^*, g) satisfy conditions in Theorem 1.2. Let $\delta > 0$ be a small constant such that $\Omega_{\delta/4}$ and $\Omega^*_{\delta/4}$ given by (5.1) are uniformly c-convex and satisfy (1.17). By Lemma 5.1, $(\Omega_{\delta/4}, f_{\delta/4})$ and $(\Omega^*_{\delta/4}, g_{\delta/4})$ are Aleksandrov related for some positive and continuous functions $f_{\delta/4}$ and $g_{\delta/4}$. Employing Theorem 3.1, we find an Aleksandrov weak solution $\bar{w}_0 \in C(\bar{\Omega}_{\delta/4})$ in the sense of Definition 3.1 for the optimal transport from $(\Omega_{\delta/4}, f_{\delta/4})$ to $(\Omega^*_{\delta/4}, g_{\delta/4})$. By Theorem 11.1 in the appendix, $\bar{w}_0$ is strictly c-convex and $C^{1,\alpha}$-differentiable in the interior of $\Omega_{\delta/4}$. Lemma 5.4 below is then an immediate consequence. Such convexity and regularity were proved by many authors [10, 16, 19, 44]. Since $c \in \mathscr{C}^\infty_{d_0}$ takes negative infinity somewhere, it is cannot be defined on the entire $\Omega \times \Omega^*$ in general, and hence our situation is a little bit different from the existing references. Some details for the proof of Theorem 11.1 are given in the appendix for readers' convenience.

Lemma 5.4. *Assume $c \in \mathscr{C}^\infty_{d_0}$ satisfies the MTW condition. Then $\bar{w}_0$ introduced above is strictly c-convex and is of $C^{1,\alpha}(\bar{\Omega}_{\delta/2})$ for some $\alpha \in (0, 1)$.*

Lemma 5.5. *For every small $\delta > 0$, there is a uniformly c-convex function $w_0 \in C^\infty(\bar{\Omega}_\delta)$ such that $\chi_{w_0}(\Omega_\delta) \subset\subset \Omega^*$.*

Proof. Let $\bar{w}_0 \in C^{1,\alpha}(\bar{\Omega}_{\delta/2})$ be the strictly c-convex function in Lemma 5.4. Then

$$\chi_{\bar{w}_0}(\Omega_\delta) \subset \chi_{\bar{w}_0}(\Omega_{\delta/4}) = \Omega^*_{\delta/4} \subset\subset \Omega^*, \tag{5.19}$$

where the equality follows from the interior $C^{1,\alpha}$-regularity of $\bar{w}_0$. Applying Lemma 5.2 to $\bar{w}_0$, we obtain a smooth approximation $(\bar{w}_0)_\varepsilon$ such that

$$\mathcal{B}[(\bar{w}_0)_\varepsilon](x) \geq -\varepsilon^\alpha \beta \sigma_{\mathbb{S}^n}, \ \forall x \in \bar{\Omega}_\delta. \tag{5.20}$$

for some $\beta > 0$ depending on $\bar{w}_0$. The $C^{1,\alpha}$-regularity of $\bar{w}_0$ is used as this implies

$$\sup_{x \in \bar{\Omega}_\delta} \operatorname{diam}(\chi_{\bar{w}_0}(x)) \leq C\varepsilon^\alpha,$$

for some $C > 0$ depending on $\|\bar{w}_0\|_{C^{1,\alpha}(\bar{\Omega}_{\delta/2})}$. We next employ Lemma 5.3 to obtain a smooth approximation of $(\bar{w}_0)_\varepsilon$, denoted by $(\bar{w}_0)_{\varepsilon,\varepsilon'}$, such that

$$\mathcal{B}[(\bar{w}_0)_{\varepsilon,\varepsilon'}](x) \geq \mathcal{B}[(\bar{w}_0)_\varepsilon](x) + \big(c_0 - c_1\varepsilon'\big)\varepsilon' \sigma_{\mathbb{S}^n}, \ \forall \ x \in \bar{\Omega}_\delta, \tag{5.21}$$

where c_0 and c_1 depend on $\|(\bar{w}_0)_\varepsilon\|_{C^{0,1}(\bar{\Omega})}$ and in essence depend on $\bar{w}_0$. By (5.20) and (5.21), if taking $\varepsilon' = \varepsilon^{\alpha/2}$ and ε small, then

$$\mathcal{B}[(\bar{w}_0)_{\varepsilon,\varepsilon^{\alpha/2}}](x) \geq (c_0 - (\beta + c_1)\varepsilon^{\alpha/2})\varepsilon^{\alpha/2}\sigma_{\mathbb{S}^n} > 0, \ \forall x \in \bar{\Omega}_\delta.$$

We complete proof by letting $w_0 = (\bar{w}_0)_{\varepsilon,\varepsilon^{\alpha/2}}$ for a small but fixed ε. □

5.3. **Step 2: Extending w_0 to a c-convex w_1 on Ω^δ.**

We say $\phi = \phi_y : \Omega \to [-\infty, \infty)$ is a c-function if it is of the form $\phi(\cdot) = c(\cdot, y) + a$ for some $y \in \mathbb{S}^n$ and $a \in \mathbb{R}$. Let $\Omega_y = \Omega \cap B_{d_0}(y)$. Then $\phi_y(x) = -\infty$ if for all $x \in \Omega \setminus \Omega_y$. Denote by $\phi = \phi_{y;x_0}$ the c-function that is normalized at $x_0 \in \Omega$, namely $x_0 \in \Omega_y$ and

$$\phi_{y;x_0}(\cdot) = c(\cdot, y) - c(x_0, y). \tag{5.22}$$

Similarly we may define c-function $\phi^* = \phi^*_x : \Omega^* \to [-\infty, \infty)$ for some $x \in \mathbb{S}^n$, and the normalized c-function ϕ^*_{x,y_0} at $y_0 \in \Omega^*_x = \Omega^* \cap B_{d_0}(x)$. The following geometric property of c-function was shown in [42]. We provide a proof for readers' convenience.

Lemma 5.6 ([42])**.** *Let Ω be C^2-smooth domain and $w \in C^2(\bar{\Omega})$ be a c-convex function. Given $x_0 \in \partial\Omega$, let*

$$y = G(x_0, \nabla w(x_0) + s\nu_{x_0}), \ \textit{ for some } s > 0,$$

where ν_{x_0} is the unit outer normal of $\partial\Omega$ at x_0 and $G : T\mathbb{S}^n \to \mathbb{S}^n$ is the map in Lemma 4.3. If Ω is uniformly c-convex w.r.t. y and $c \in \mathscr{C}^\infty_{d_0}$ satisfies the MTW condition, then

$$w(x) > \phi_{y;x_0}(x) + w(x_0), \ \forall x \in \bar{\Omega} \setminus \{x_0\}. \tag{5.23}$$

Proof. Note that (5.23) holds automatically whenever $x \notin B_{d_0}(y)$. Hence we assume that $x \in B_{d_0}(y)$ in the following arguement. Consider the c-segment ℓ_y from x_0 to $x \in \bar{\Omega}_y$

$$\ell_y = \{x_t : t \in [0,1] \text{ and } \nabla_y c(x_t, y) = (1-t)W_0 + tW_1\}, \tag{5.24}$$

where $W_0 = \nabla_y c(x_0, y)$ and $W_1 = \nabla_y c(x, y)$. Since $x_0, x \in \bar{\Omega}_y$, W_0 and W_1 are well-defined. As Ω is c-convex w.r.t. y, we see $\ell_y \subset \bar{\Omega}_y$. Set

$$\mathcal{E}(t) = w(x_t) - c(x_t, y).$$

Then (5.23) is equivalent to $\mathcal{E}(1) > \mathcal{E}(0)$. We have

$$\begin{aligned} \dot{\mathcal{E}}(t) &= \langle \nabla w(x_t) - \nabla_x c(x_t, y), \dot{x}_t \rangle, \qquad (5.25) \\ \ddot{\mathcal{E}}(t) &= [\nabla^2 w(x_t) - \nabla^2_x c(x_t, y)](\dot{x}_t, \dot{x}_t) + \langle \nabla w(x_t) - \nabla_x c(x_t, y), \ddot{x}_t \rangle \\ &\geq [\mathcal{A}|_{(x_t, Z^t_0)} - \mathcal{A}|_{(x_t, Z^t_1)}](\dot{x}_t, \dot{x}_t) + \langle Z^t_0 - Z^t_1, \ddot{x}_t \rangle, \qquad (5.26) \end{aligned}$$

where $Z^t_0 = \nabla w(x_t)$ and $Z^t_1 = \nabla_x c(x_t, y)$. Note that the c-convexity of w is used to derive (5.26). Let $Z^t_\tau = \tau Z^t_1 + (1-\tau) Z^t_0 \in T_{x_t}\mathbb{S}^n$, and $\{y_\tau\}_{\tau \in [0,1]}$ be a c-segment $\ell^*_{x_t}$ from $G(x_t, \nabla w(x_t))$ to y, namely $\nabla_x c(x_t, y_\tau) = Z^t_\tau$. Then

$$\frac{d}{d\tau} \mathcal{A}|_{(x_t, Z^t_\tau)}(\dot{x}_t, \dot{x}_t) = \nabla_y \nabla^2_x c|_{(x_t, y_\tau)}(\dot{x}_t, \dot{x}_t, y'_\tau), \tag{5.27}$$

where $y'_\tau \in T_{y_\tau}\mathbb{S}^n$ is the vector such that

$$\nabla^2_{x,y} c|_{(x_t, y_\tau)}(Z, y'_\tau) = \langle Z, Z^t_1 - Z^t_0 \rangle, \ \forall Z \in T_{x_t}\mathbb{S}^n. \tag{5.28}$$

On the other hand, we deduce from (5.24) that

$$\nabla^2_{x,y} c|_{(x_t, y)}(\ddot{x}_t, W) + \nabla_y \nabla^2_x c|_{(x_t, y)}(\dot{x}_t, \dot{x}_t, W) = 0, \ \forall W \in T_y\mathbb{S}^n. \tag{5.29}$$

Successively employing (5.28), (5.29) and (5.27) in order, we obtain

$$\begin{aligned}\langle Z_1^t - Z_0^t, \ddot{x}_t\rangle &= \left[\nabla^2_{x,y} c|_{(x_t,y_\tau)}(\ddot{x}_t, y'_\tau)\right]\big|_{\tau=1} \\ &= -\nabla_y \nabla^2_x c|_{(x_t,y)}(\dot{x}_t, \dot{x}_t, y'_\tau|_{\tau=1}) \\ &= -\frac{d}{d\tau}\big|_{\tau=1} \mathcal{A}|_{(x_t,Z_\tau^t)}(\dot{x}_t, \dot{x}_t).\end{aligned}$$

Therefore (5.26) can be written as

$$\begin{aligned}\ddot{\mathcal{E}}(t) &\geq -\int_0^1 \frac{d}{d\tau}\mathcal{A}|_{(x_t,Z_\tau^t)}(\dot{x}_t, \dot{x}_t) + \frac{d}{d\tau}\big|_{\tau=1}\mathcal{A}|_{(x_t,Z_\tau^t)}(\dot{x}_t, \dot{x}_t) \\ &= \int_0^1 \int_\tau^1 \frac{d^2}{d\tau^2}\mathcal{A}|_{(x_t,Z_\tau^t)}(\dot{x}_t, \dot{x}_t). \end{aligned} \tag{5.30}$$

By (5.25), we see that

$$\langle \frac{d}{d\tau} Z_\tau^t, \dot{x}_t\rangle = 0 \text{ if and only if } \dot{\mathcal{E}}(t) = 0.$$

This together with (5.30) and the MTW condition yields

$$\ddot{\mathcal{E}}(t) \geq 0 \quad \text{whenever } \dot{\mathcal{E}}(t) = 0. \tag{5.31}$$

By the uniform c-convexity (w.r.t. y) of Ω,

$$\dot{\mathcal{E}}(0) = \langle \nabla w(x_0) - \nabla_x c(x_0, y), \dot{x}_0\rangle = -s\langle \nu|_{x_0}, \dot{x}_0\rangle > 0. \tag{5.32}$$

Using (5.31) and (5.32), we conclude that $\dot{\mathcal{E}}(t) \geq 0$ for all $t \in [0,1]$ and the inequality is strict for every $t \in [0, \delta)$, provided $\delta > 0$ is small. Hence $\mathcal{E}(1) > \mathcal{E}(0)$. $\square$

Let $w_0 \in C^\infty(\bar{\Omega}_\delta)$ be the function in Lemma 5.5. We extend it to a c-convex function on $\bar{\Omega}$ as the supremum of c-functions lying below w_0 with their c-normal images in Ω^*:

$$w_1(x) := \sup_{\phi \in \mathcal{G}_\delta} \phi(x), \text{ for } x \in \bar{\Omega}^\delta, \tag{5.33}$$

where Ω^δ is given by (5.2) and $\mathcal{G}_\delta$ is the collection of c-functions $\phi = \phi_y$ such that $y \in \bar{\Omega}^*$ and $\phi_y|_{\Omega_\delta} \leq w_0|_{\Omega_\delta}$. The lemma below was proved in [42] under the assumption that $\chi_{w_0}(\Omega_\delta)$ is c-convex w.r.t. Ω_δ. As pointed by the authors, only c-convexity of w_0 is needed to justify this. We provide a proof for readers' convenience.

Lemma 5.7 ([42]). *Suppose Ω and Ω^* are uniformly c-convex w.r.t. each other and $c \in \mathscr{C}^\infty_{d_0}$ satisfies the MTW condition. For each small $\delta > 0$, w_1 given by (5.33) is a c-convex extension of $w_0 \in C^\infty(\bar{\Omega}_\delta)$ to $\bar{\Omega}^\delta$ with $\chi_{w_1}(\Omega^\delta) = \Omega^*$, and satisfies the properties:*

- (i) *for any $x \in \Omega^\delta \setminus \bar{\Omega}_\delta$, there exist unique points $x_b \in \partial\Omega_\delta$ and $y_b \in \partial\Omega^*$ such that $\chi_{w_1}(\ell_{y_b} \setminus \{x_b\}) = \{y_b\}$, where ℓ_{y_b} is the c-segment w.r.t. y_b joining x to x_b;*
- (ii) *the resultant mappings $x \mapsto x_b$ and $x \mapsto y_b$ are C^2-diffeomorphisms from $\partial\Omega_r$ (or $\partial\Omega^r$) to $\partial\Omega_\delta$ and $\partial\Omega^*$, respectively, for any $0 < r < \delta$. Moreover, the mapping from $\partial\Omega_r$ to $\partial\Omega_\delta$ given by $x \mapsto x_b$ converges to the identity map as $r \to \delta$.*

Proof. Take a $\phi_y \in \mathcal{G}_\delta$ with $y \in \Omega^* \setminus \omega^*$, where $\omega^* = \chi_{w_0}(\Omega_\delta)$. By adding a constant, we assume ϕ_y supports w_0 on $\bar\Omega_\delta$ at a point x_b. Then x_b must lie in $\partial\Omega_\delta$ by the uniform c-convexity of w_0. The resultant c-support function is

$$\phi(x) = \phi_{y;x_b}(x) + w_0(x_b).$$

Observe that y lies on the c^*-segment $\ell^*_{x_b}$ starting from $y_{0,b} = \chi_{w_0}(x_b)$ and given by

$$\nabla_x c(x_b, \ell^*_{x_b}) = \{\nabla w_0(x_b) + t\nu_{x_b} : t \geq 0\}, \tag{5.34}$$

where $t > 0$ and ν_{x_b} is the unit outer normal of $\partial\Omega_\delta$ at x_b. Conversely, for any $x_b \in \partial\Omega_\delta$ and $y \in \ell^*_{x_b}$, since Ω_δ (for every small δ) is c-convex, Lemma 5.6 implies that

$$w_0(x) > \phi_{y;x_b}(x) + w_0(x_b),\ \forall x \in \bar\Omega_\delta \setminus \{x_b\}. \tag{5.35}$$

This proves that w_1 is a c-convex extension of w_0.

Since Ω^* is uniformly c-convex w.r.t. Ω, the intersection point y_b of $\ell^*_{x_b}$ and $\partial\Omega^*$ is unique. Denote $\omega^* = \chi_{w_0}(\Omega_\delta)$. We claim that $\ell^*_{x_b}$ intersects $\partial\omega^*$ only at $y_{0,b} = \chi_{w_0}(x_b)$. Suppose there exists another $y_{1,b} \in \partial\omega^* \cap \ell^*_{x_b}$. Then there exists $x_{1,b} \in \partial\Omega_\delta$ such that $y_{1,b} = \chi_{w_0}(x_{1,b})$. By Lemma 5.6,

$$w_0(x_{1,b}) > \phi_{y_{1,b};x_b}(x_{1,b}) + w_0(x_b).$$

This shows that if $\phi(\cdot) = \phi_{y_{1,b};x_b}(\cdot) + a$ supports w_0 at $x_{1,b}$, then $a > w_0(x_b)$. But in this case $\phi(x_b) = a > w_0(x_b)$, arriving a contradiction. We remark that the argument does not rely on the c-convexity of ω^*. The uniqueness of $y_{0,b}$ implies that the map $x_b \mapsto y_b$ is onto $\partial\Omega^*$. By Lemma 5.6 it is also one-to-one. It then follows from (5.34) that the map $x_b \mapsto y_b$ is a C^2-diffeomorphism from $\partial\Omega_\delta$ to $\partial\Omega^*$.

If B_r is a sufficiently small exterior tangent geodesic ball of $\partial\Omega_\delta$ at x_b, then B_r is uniformly c-convex w.r.t. Ω^*. Employing Lemma 5.6 again,

$$\phi_{y_b;x_b}(x) > \phi_{y;x_b}(x)\ \ \forall\ x \in B_r \text{ and } y \in \ell^*_{x_b}.$$

Therefore,

$$w_1(x) = \max_{x_b \in \partial\Omega_\delta} \left[\phi_{y_b;x_b}(x) + w_0(x_b)\right],\ \forall\ x \in \Omega^\delta \setminus \bar\Omega_\delta. \tag{5.36}$$

For each $x \in \Omega^\delta \setminus \Omega_\delta$, there exists an $x_b \in \partial\Omega_\delta$ such that the maximum in (5.36) is attained. We next show such x_b is unique.

Let v be the c-transform of w_1 on Ω^*. For $x_b \in \partial\Omega_\delta$ and $y \in \ell^*_{x_b}$, we infer by (5.35)

$$c(x,y) - w_1(x) = c(x,y) - w_0(x) \leq c(x_b,y) - w_0(x_b),\ \forall x \in \Omega_\delta,$$

and by the definition of the extension w_1, namely (5.33),

$$c(x,y) - w_1(x) \leq c(x,y) - \phi_{y;x_b}(x) - w_0(x_b) = c(x_b,y) - w_0(x_b),\ \forall x \in \Omega^\delta \setminus \Omega_\delta.$$

The two inequalities above become equalities at $x = x_b$. Hence

$$v(y) = c(x_b,y) - w_0(x_b),\ \forall y \in \ell^*_{x_b} \text{ and } x_b \in \partial\Omega_\delta. \tag{5.37}$$

As $v|_{\omega^*}$ is the c-transform of w_0, we see that v is smooth in $\bar{\Omega}^* \setminus \partial\omega^*$. Arguing as before, for any $x \in \Omega^\delta \setminus \bar{\Omega}_\delta$, there is a $y_b \in \partial\Omega^*$ such that

$$v(y) \geq \phi^*_{x;y_b}(y) + v(y_b) := c(x,y) - c(x,y_b) + v(y_b), \ \forall y \in \bar{\Omega}^*. \tag{5.38}$$

Moreover x lies on the c-segment ℓ_{y_b} given by

$$\nabla_y c(\ell_{y_b}, y_b) = \left\{\nabla v(y_b) + t\nu^*_{y_b} : \ t \in [0, \bar{\delta}]\right\}, \tag{5.39}$$

where ν^* is the unit outer normal of $\partial\Omega^*$. By (5.37), $\nabla v(y_b) = \nabla_y c(x_b, y_b)$. This implies that $x_b \in \ell_{y_b}$. Since Ω^* is uniformly c-convex, by applying Lemma 5.6 to c-convex function v and its c-support $\phi^*_{x;y_b}$ at $y_b \in \partial\Omega^*$, we see that (5.38) is strict if $y \neq y_b$. Therefore such y_b is unique. Employing (5.38) first and then (5.37), we see that

$$w_1(x) = \sup_{y \in \bar{\Omega}^*}\{c(x,y) - v(y)\} \leq c(x,y_b) - v(y_b) = \phi_{y_b;x_b}(x) + w_0(x_b), \ \forall x \in \Omega^\delta \setminus \bar{\Omega}_\delta.$$

Hence the maximum in (5.36) is attained at (x_b, y_b). The uniqueness of x_b follows by that of y_b. Consequently,

$$w_1(x) = c(x,y_b) - c(x_b,y_b) + w_0(x_b), \ \forall x \in \ell_{y_b}.$$

Therefore $\chi_{w_1}(\ell_{y_b} \setminus \{x_b\}) = \{y_b\}$ and $\chi_{w_1}(x_b) = \ell^*_{x_b}$. This proves property (i) in Lemma 5.7. From the obliqueness of ℓ_{y_b} on $\partial\Omega_\delta$, we see that the map from $x \in \Omega^r \setminus \bar{\Omega}_\delta$ to $x_b \in \partial\Omega_\delta$ is one-to-one for sufficiently small r. Recalling that x_b is the endpoint of ℓ_{y_b} (with $t = 0$ in (5.39)), we see that the mapping from $\partial\Omega_r$ to Ω_δ given by $x \mapsto x_b$ converges to the identity map as $r \to \delta$. Hence the property (ii) follows. □

5.4. Step 3: Uniformly c-convex $w_\sigma \in C^\infty(\bar{\Omega})$ with $\chi_{w_\sigma}(\Omega)$ being close to Ω^*.

Let w_1 be the c-convex function obtained in Lemma 5.7. Proposition 5.1 follows by a modification of w_1. For this purpose, we employ the argument from Section 7 in [42].

Proof of Proposition 5.1. By the diffeomorphisms in Lemma 5.7, w_1 is uniformly c-convex in tangential directions on $\partial\Omega_r$ and $\partial\Omega^r$ for all $r \in (0,\delta)$. Namely there is $\lambda_0 > 0$ so that $\mathcal{B}_{\tau\tau}[w_1]|_x \geq \lambda_0$ for all unit vector τ tangential to $\partial\Omega_r$ (or $\partial\Omega^r$) at x. Set

$$\tilde{w}_1(x) := w_1(x) + bd^2_{\partial\Omega_\delta}(x),$$

where b is a positive constant and $d_{\partial\Omega_\delta}$ is the distance from $\partial\Omega_\delta$. Then $\tilde{w}_1$ is uniformly c-convex in Ω^δ away from $\partial\Omega_\delta$ whenever $\delta > 0$ is sufficiently small. Hence we can find a further constant $\lambda_0 > 0$ (independent of δ) such that

$$\mathcal{B}_{ee}[\tilde{w}_1]|_x \geq \lambda_0, \quad \forall x \in \Omega^\delta \setminus \partial\Omega_\delta \text{ and unit } e \in T_x\mathbb{S}^n. \tag{5.40}$$

Take $0 < \varepsilon \ll \delta$ and let w_ε be the mollification of $\tilde{w}_1$, namely replacing $\tilde{w}$ in (5.7) by $\tilde{w}_1$. We verify that w_ε is uniformly c-convex in Ω^δ in what follows.

Fix $x_0 \in \Omega^\delta$ and unit vector $e \in T_{x_0}\mathbb{S}^n$. Let $x_t := \exp_{x_0}(te)$. By parallel transport, we identify each tangent space $T_{x_t}\mathbb{S}^n$ with $T_{x_0}\mathbb{S}^n$, and thus take

$$x_t^z := \exp_{x_t} z, \text{where } z \in T_{x_0}\mathbb{S}^n \simeq T_{x_t}\mathbb{S}^n.$$

Denote by $\dot{x}_t^z$ the velocity of the curve $\{x_t^z\}$ parametrized by t. Then $\dot{x}_t^z = |\dot{x}_t^z| e_t^z$ for a unit vector $e_t^z \in T_{x_t^z}\mathbb{S}^n$. Observe that $|\dot{x}_t^z| = 1 + O(|z|)$. Hence

$$\lim_{t\to 0} \frac{\tilde{w}_1(x_t^z) - \tilde{w}_1(x_0^z)}{t} = \langle \nabla \tilde{w}_1|_{x_0^z}, \dot{x}_t^z|_{t=0}\rangle = (1 + O(|z|))\nabla_{e_0^z}\tilde{w}_1(x_0^z),$$

provided $\nabla\tilde{w}_1$ exists at x_0^z. Since $\tilde{w}_1$ is Lipschitz on Ω^δ, and is smooth at all x_0^z except z belongs to a zero measure set $\mathcal{P}_{x_0} := \{z \in T_{x_0}\mathbb{S}^n : x_0^z \in \partial\Omega_\delta\}$, we obtain

$$\nabla_e w_\varepsilon(x_0) = \int_{\mathbb{R}^n} \phi_\varepsilon(z)\nabla_{e_0^z}\tilde{w}_1(x_0^z)dz + O(\varepsilon). \tag{5.41}$$

We turn to the calculation of the second derivatives. Given $x \in \partial\Omega_\delta$, let

$$\nabla^+\tilde{w}_1(x) := \lim_{y\notin\bar{\Omega}_\delta, y\to x} \nabla\tilde{w}_1(y) \quad \text{and} \quad \nabla^-\tilde{w}_1(x) := \lim_{y\in\Omega_\delta, y\to x} \nabla\tilde{w}_1(y).$$

For $z \in \mathcal{P}_{x_0}$, we denote by ν the unit outer normal of $\partial\Omega_\delta$. Then

$$\lim_{r\to 0} \frac{\tilde{w}_1(x_r^z) + \tilde{w}_1(x_{-r}^z) - 2\tilde{w}_1(x_0^z)}{r} = \langle(\nabla^+\tilde{w}_1 - \nabla^-\tilde{w}_1)|_{x_0^z}, \dot{x}_t^z|_{t=0}\rangle \mathrm{sgn}(\langle \dot{x}_t^z|_{t=0}, \nu\rangle).$$

By Lemma 5.7, $\nabla^+\tilde{w}_1$ coincides with $\nabla^-\tilde{w}_1$ in tangential directions on $\partial\Omega_\delta$. Hence

$$\begin{aligned}\lim_{r\to 0} \frac{\tilde{w}_1(x_r^z) + \tilde{w}_1(x_{-r}^z) - 2\tilde{w}_1(x_0^z)}{r} &= \langle(\nabla^+\tilde{w}_1 - \nabla^-\tilde{w}_1)|_{x_0^z}, \nu\rangle |\langle \dot{x}_t^z|_{t=0}, \nu\rangle| \\ &= (1 + O(|z|))|\langle e_0^z, \nu\rangle|\delta_\nu\tilde{w}_1(x_0^z),\end{aligned}$$

where $\delta_\nu\tilde{w}_1 := \nabla_\nu^+\tilde{w}_1 - \nabla_\nu^-\tilde{w}_1$. Let $\mathcal{Z} = [\exp_{x_0}]^{-1} : B_\pi(x_0) \to T_{x_0}\mathbb{S}^n$, $J\mathcal{Z}$ be its Jacobian, and $d\Sigma = J\mathcal{Z} d\mu_{\partial\Omega_\delta}$ with $d\mu_{\partial\Omega_\delta}$ being the area element of $\partial\Omega_\delta$. Then

$$\begin{aligned}\nabla_{ee}^2 w_\varepsilon(x_0) &= \lim_{r\to 0}\int_{\mathbb{R}^n} \phi_\varepsilon(z)\frac{\tilde{w}_1(x_r^z) + \tilde{w}_1(x_{-r}^z) - 2\tilde{w}_1(x_0^z)}{r^2}dz \\ &= \int_{\mathbb{R}^n\setminus\mathcal{P}_{x_0}} \phi_\varepsilon(z)\nabla_{e_0^z}^2\tilde{w}_1(x_0^z)dz + O(\varepsilon) + \int_{\partial\Omega_\delta} \phi_\varepsilon(\mathcal{Z}(y))\langle e_0^{\mathcal{Z}(y)}, \nu(y)\rangle^2\delta_\nu\tilde{w}_1(y)d\Sigma \\ &\ge \lambda_0 + O(\varepsilon) + \int_{\mathbb{R}^n\setminus\mathcal{P}_{x_0}} \phi_\varepsilon(z)\mathcal{A}_{e_0^z e_0^z}(x_0^z, \nabla\tilde{w}_1(x_0^z))dz \qquad (5.42) \\ &\quad + \int_{\partial\Omega_\delta} \phi_\varepsilon(\mathcal{Z}(y))\langle e_0^{\mathcal{Z}(y)}, \nu(y)\rangle^2\delta_\nu\tilde{w}_1(y)d\Sigma,\end{aligned}$$

where (5.40) is used in the last inequality. Since $\chi_{\tilde{w}_1}(\Omega) \subset\subset \Omega^*$, there is $C_0 > 0$ such that $\delta_\nu\tilde{w}_1(y) \ge C_0$ for all $y \in \partial\Omega_\delta$. Therefore, by (5.42),

$$\nabla_{ee}^2 w_\varepsilon(x_0) \ \ge \ \lambda_0 + \int_{\mathbb{R}^n\setminus\mathcal{P}_{x_0}} \phi_\varepsilon(z)\mathcal{A}_{e_0^z e_0^z}(x_0^z, \nabla\tilde{w}_1(x_0^z))dz + O(\varepsilon). \tag{5.43}$$

As in [42, Section 7], we divide Ω into three disjoint parts:

$$\begin{aligned} U_1 &= \{x \in \Omega : d_{\partial\Omega_\delta}(x) \geq \varepsilon\}, \\ U_2 &= \{x \in \Omega : d_{\partial\Omega_\delta}(x) \in (\sigma\varepsilon, \varepsilon)\}, \\ U_3 &= \{x \in \Omega : d_{\partial\Omega_\delta}(x) \leq \sigma\varepsilon\}, \end{aligned}$$

where $\sigma \in (9/10, 1)$ is a constant very close to 1. We verify the c-convexity in each U_i.

Case 1: $x_0 \in U_1$. Recall that $\mathcal{B}_{ee}[w_\varepsilon](x_0) = \nabla^2_{ee} w_\varepsilon(x_0) - \mathcal{A}_{ee}(x_0, \nabla w_\varepsilon(x_0))$. By (5.43),

$$\begin{aligned} \mathcal{B}_{ee}[w_\varepsilon](x_0) \ &\geq\ \lambda_0 - \left|\mathcal{A}_{ee}(x_0, \nabla w_\varepsilon(x_0)) - \mathcal{A}_{ee}(x_0, \nabla \tilde{w}_1(x_0))\right| \\ &\qquad - \left| \int_{\mathbb{R}^n \setminus \mathcal{P}_{x_0}} \phi_\varepsilon(z) \Big[\mathcal{A}_{e_0^z e_0^z}(x_0^z, \nabla \tilde{w}_1(x_0^z)) - \mathcal{A}_{ee}(x_0, \nabla \tilde{w}_1(x_0))\Big] dz \right| + O(\varepsilon) \end{aligned} \tag{5.44}$$

Since $x_0 \in U_1$, we have $x_0^z \notin \partial\Omega_\delta$ for all $|z| < \varepsilon$. Hence $\tilde{w}_1(x_0^z)$ is smooth for all $z \in \mathbf{B}_\varepsilon$. By (5.41), $|\nabla \tilde{w}_1(x_0) - \nabla w_\varepsilon(x_0)| \leq C\varepsilon$. It then follows by (5.44) that $\mathcal{B}_{ee}[w_\varepsilon](x_0) \geq \lambda_0/2$, provided ε is sufficiently small.

Case 2: $x_0 \in U_2$. When $x_0 \in U_2 \setminus \Omega_\delta$, we have $|Q_\varepsilon| \leq C(1-\sigma)|\mathbf{B}_\varepsilon|$ where $Q_\varepsilon := \{z \in \mathbf{B}_\varepsilon : x_0^z \in \bar{\Omega}_\delta\}$. It then follows from (5.41) that

$$\begin{aligned} |\nabla_e w_\varepsilon(x_0) - \nabla_e \tilde{w}_1(x_0)| \ &\leq\ \left\{ \int_{\mathbf{B}_\varepsilon(x_0) \setminus Q_\varepsilon} + \int_{Q_\varepsilon} \right\} \left[\phi_\varepsilon(z) \left|\nabla_{e_0^z} \tilde{w}_1(x_0^z) - \nabla_e \tilde{w}_1(x_0)\right|\right] dz + O(\varepsilon) \\ &\leq\ C(1 - \sigma + \varepsilon). \end{aligned} \tag{5.45}$$

Similarly, when $x_0 \in U_2 \cap \Omega_\delta$, we have $|\{z \in \mathbf{B}_\varepsilon : x_0^z \notin \Omega_\delta\}| \leq C(1-\sigma)|\mathbf{B}_\varepsilon|$, and thus there also holds (5.45). Hence $\nabla w_\varepsilon(x_0)$ is a small perturbation of $\nabla \tilde{w}_1(x_0)$. This together with (5.44) yields $\mathcal{B}_{ee}[w_\varepsilon](x_0) \geq \lambda_0/2$, provided $1-\sigma$ and ε is sufficiently small.

Case 3: $x_0 \in U_3$. Let $\{\boldsymbol{e}_i\}_{i=1}^n$ be a local orthonormal frame such that $\{\boldsymbol{e}_i\}_{i=1}^{n-1}$ are tangential directions and $\boldsymbol{e}_n$ is the unit outer normal vector of $\partial\Omega_r$. We show that

$$\nabla^2_{\boldsymbol{e}_1} w_\varepsilon(x_0) - \mathcal{A}_{\boldsymbol{e}_1\boldsymbol{e}_1}(x_0, \nabla w_\varepsilon(x_0)) > 0. \tag{5.46}$$

By Lemma 5.7, $\nabla_1 w_\varepsilon$ is a small perturbation of $\nabla_1 \tilde{w}_1$. Hence it suffices to show that

$$\nabla^2_{\boldsymbol{e}_1} w_\varepsilon(x_0) - \mathcal{A}_{\boldsymbol{e}_1\boldsymbol{e}_1}(x_0, \nabla_{\boldsymbol{e}_1} \tilde{w}_1(x_0), \nabla' w_\varepsilon(x_0)) > 0, \tag{5.47}$$

where $\nabla' w_\varepsilon = (\nabla_{\boldsymbol{e}_2} w_\varepsilon, \ldots, \nabla_{\boldsymbol{e}_n} w_\varepsilon)$. By the MTW condition, $\mathcal{A}_{\boldsymbol{e}_1\boldsymbol{e}_1}$ is convex in $\nabla' w_\varepsilon$. Then it follows by (5.41) and Jensen's inequality that

$$\mathcal{A}_{\boldsymbol{e}_1\boldsymbol{e}_1}(x_0, \nabla_1 \tilde{w}_1(x_0), \nabla' w_\varepsilon(x_0)) \leq \int_{\mathbb{R}^n} \phi_\varepsilon(z) \mathcal{A}_{\boldsymbol{e}_1\boldsymbol{e}_1}(x_0, \nabla_1 \tilde{w}_1(x_0), \nabla' \tilde{w}_1(x_0^z)) dz + O(\varepsilon).$$

Combining this with (5.43), we see that (5.47) holds whenever ε is sufficiently small.

On the other hand, since $x_0 \in U_3$, the second integral of (5.42) is as large as we want, provided ε sufficiently small. Therefore

$$\nabla^2_{\boldsymbol{e}_n} w_\varepsilon(x_0) - \mathcal{A}_{\boldsymbol{e}_n\boldsymbol{e}_n}(x_0, \nabla w_\varepsilon(x_0)) \geq \lambda_0. \tag{5.48}$$

By (5.46) and (5.48), we see the uniform c-convexity of w_ε in U_3. $\square$

6. Examples and applications

In this section we present some applications of Theorems 1.1 & 1.2.

6.1. **Aleksandrov problem and Gauss image problem for open hypersurfaces.**

Given two domains Ω and Ω^* in $\mathbb{S}^n$ satisfying (1.17), let $\mathcal{K}_0(\Omega,\Omega^*)$ be the collection of open convex hypersurfaces $\mathcal{M}$ in $\mathbb{R}^{n+1}$ such that

(i) the origin $O \in \mathrm{Int}\Sigma$, where Σ is the convex region of $\mathbb{R}^{n+1}$ such that $\partial\Sigma = \mathcal{M}$;
(ii) $\{tz : t > 0, z \in \mathcal{M}\} \cap \mathbb{S}^n = \Omega$;
(iii) $\mathscr{A}_{\mathcal{M}}(\Omega) = \Omega^*$, where $\mathscr{A}_{\mathcal{M}}$ is the radical Gauss map of $\mathcal{M}$.

For two Borel measures μ, μ^* on Ω and Ω^*, we are interested in if there exists $\mathcal{M} \in \mathcal{K}_0(\Omega,\Omega^*)$ so that, for every Borel set $\omega \subset \Omega$,

$$\mu^*(\mathscr{A}_{\mathcal{M}}(\omega)) = \mu(\omega). \tag{6.1}$$

When $\Omega = \Omega^* = \mathbb{S}^n$, $\mathcal{M}$ must be a closed hypersurface and it is the Gauss image problem raised in [3], which includes the Aleksandrov problem [1, 2] as a particular case. When Ω, Ω^* are spherical domains with boundary, finding $\mathcal{M}$ with property (6.1) is regarded as the Aleksandrov problem and Gauss image problem for hypersurfaces with boundary.

Denote by $r \in C(\bar\Omega)$ and $u \in C(\bar\Omega^*)$ the radical and the support function of $\mathcal{M}$. Then $w = -\log r$ and $v = \log u$ are c-transform of each other with the cost

$$c(x,y) = \log\cos d(x,y) \in \mathscr{C}_{\frac{\pi}{2}}. \tag{6.2}$$

The Gauss image problem can be reformulated as the optimal transportation problem between (Ω,μ) and (Ω^*,μ^*) with cost (6.2). More precisely, $\mathcal{M}_0 \in \mathcal{K}_0(\Omega,\Omega^*)$ satisfies (6.1) if and only if $(-\log r_0, \log u_0)$ is a pair of potential functions of (1.12) with c being (6.2), where r_0 and u_0 are respectively the radical and support functions of $\mathcal{M}_0$. One can see this fact by the discussions in [3, 40].

By Lemma 4.4 we have $\mathcal{A}_{ij}(x,Z) = -Z_iZ_j - \delta_{ij}$. Hence MTW condition follows; and $D_Z\mathcal{A}|_{(x,\nabla_x(x,y))}(\tau,\tau)\cdot\vec{\boldsymbol{n}}|_x \equiv 0$ for any tangential τ, so by Lemma 8.2 the uniform c-convexity of Ω,Ω^* is equivalent to the uniform spherical convexity of Ω,Ω^*.

We obtain the following result from Theorems 1.1 and 1.2.

Theorem 6.1. *Let Ω and Ω^* be domains in $\mathbb{S}^n$. Suppose Ω and Ω^* satisfy (1.17) with $d_0 = \pi/2$, and $f \in L^1(\Omega), g \in L^1(\Omega^*)$ are functions satisfying (1.18). The following statements hold:*

(i) *there exists a unique (up to homothetic expansion w.r.t. origin) $\mathcal{M}_0 \in \mathcal{K}_0(\Omega,\Omega^*)$ satisfying (6.1) with $\mu = f\gamma_{\mathbb{S}^n}$ and $\mu^* = g\gamma_{\mathbb{S}^n}$ if and only if (Ω, f) and (Ω^*, g) are Aleksandrov related;*
(ii) *if $\partial\Omega, \partial\Omega^*$ are moreover smooth and uniformly convex, and $f \in C^\infty(\bar\Omega), g \in C^\infty(\bar\Omega^*)$, then $\mathcal{M}_0$ is smooth and uniformly convex.*

Condition 1.17 implies that $\Omega \subset\subset \Omega^*_{\pi/2}$. It is natural to study the extremal case of Aleksandrov problem: $\Omega = \Lambda_{\pi/2}$ and $\Omega^* = \Lambda$ for some convex domain $\Lambda \subset \mathbb{S}^n$. For the extremal case, $\mathcal{M} \in \mathcal{K}_0(\Lambda_{\pi/2}, \Lambda)$ is complete and non-compact, and so the problem is of particular interest. We will deal with it in a sequential work.

6.2. **Refractor design problem.**

Let Ω, Ω^* be two spherical domains, f, g be two nonnegative integrable functions on Ω and Ω^*. The refraction problem concerns the existence of a refractor $\mathcal{M}$, which is a hypersurface in $\mathbb{R}^{n+1}$ separating two media $(\mathcal{D}_1, n_1)$ and $(\mathcal{D}_2, n_2)$, such that the light beam emanating from O with direction $x \in \Omega$ and intensity $f(x)$ is refracted by $\mathcal{M}$ into the light parallel to the light beam starting from O with direction $y \in \Omega^*$ and intensity $g(y)$. Let $\kappa = n_1/n_2$ be the quotient of the refraction indexes. It was found by Gutiérrez-Huang [20] that, if $\kappa > 1$. then the refraction problem is equivalent to the optimal transportation problem from (Ω, f) to (Ω^*, g) with cost

$$c(x, y) = \log\big(\kappa \cos d(x, y) - 1\big) \in \mathscr{C}_{\arccos \frac{1}{\kappa}}. \tag{6.3}$$

See also Gutiérrez's book [21] for a comprehensive introduction on the applications of optimal transportation in geometric optics.

Following Gutiérrez-Huang [20], we say that a surface Γ in $\mathbb{R}^{n+1}$ satisfies the uniform refraction property if any light beam starting from O is refracted by Γ into the light beam parallel to the light beam starting from O with a fixed direction y. It turns out that Γ is the sheet with opening in direction y of a hyperboloid of resolution of two sheets about the axis of direction y. More precisely, $\Gamma = \mathcal{H}(y, a)$ is radically parametrized by

$$r(x) = \frac{a}{\kappa\langle x, y\rangle - 1}, \quad \forall x \in B_{d_0}(y),$$

where $d_0 = \arccos(1/\kappa)$ and $a > 0$ is constant. A hypersurface $\mathcal{M} = \{xr(x) : x \in \Omega\}$ is called a refractor if for each $x \in \Omega$ there exists a semi-hyperboloid $\mathcal{H}(y, a)$ with $y \in \Omega^*$ supporting $\mathcal{M}$ at $r(x)x$. It follows that

$$a = \max_{x \in \bar{\Omega}}\{r(x)(\kappa\langle x, y\rangle - 1)\}.$$

That is $a = e^{v(y)}$, where $v(y)$ is the c-transform of $w(x) = -\log r(x)$. Given a refractor $\mathcal{M} = \{r(x)x : x \in \Omega\}$, let $\mathscr{N}_{\mathcal{M}}$ be a multi-valued map such that, for each $x \in \Omega$, $\mathscr{N}_{\mathcal{M}}(x)$ consists of all $y \in \Omega^*$ such that $\mathcal{H}(y, a)$ supports $\mathcal{M}$ at $r(x)x$ for some $a > 0$. The refraction problem is in fact equivalent to finding a refractor $\mathcal{M}$ such that

$$\mu_f([\mathscr{N}_{\mathcal{M}}]^{-1}(\omega)) = \mu_g^*(\omega), \text{ for every Borel set } \omega \subset \Omega^*, \tag{6.4}$$

where $\mu_f = f\gamma_{\mathbb{S}^n}$ and $\mu_g^* = g\gamma_{\mathbb{S}^n}$. The result below is a consequence of Theorem 1.1.

Theorem 6.2. *Let Ω and Ω^* be domains in $\mathbb{S}^n$ satisfying (1.17) with $d_0 = \arccos \kappa^{-1}$ ($\kappa > 1$), and $f \in L^1(\Omega), g \in L^1(\Omega^*)$ be two functions satisfying (1.18). Then there is a unique (up to homothetic expansion w.r.t. origin) refractor $\mathcal{M}$ solving (6.4) if and only if (Ω, f) and (Ω^*, g) are Aleksandrov related.*

6.3. **Reflector design problem.** The reflection problem concerns the existence of a radial graph $\mathcal{M} = \{r(x)x \in \mathbb{R}^{n+1} : x \in \Omega\}$ over $\Omega \subset \mathbb{S}^n$ such that the light beam emanatin from O with direction $x \in \Omega$ and intensity $f(x)$ is reflected by $\mathcal{M}$ into the light parallel to the light beam starting from O with direction $y \in \Omega^*$ and intensity $g(y)$. It was discovered by Wang [48] that this problem is an optimal transportation problem with cost function

$$c(x,y) = \log(1 - \cos d(x,y)). \tag{6.5}$$

For a ray from O to $r(x)x \in \mathcal{M}$, the direction of the reflected ray is

$$\mathcal{R}_{\mathcal{M}}(x) = x - 2\langle x, \nu\rangle \nu,$$

where ν is the unit outer normal of $\mathcal{M}$ at $r(x)x$. The energy conservation implies that $\mathcal{R}_{\mathcal{M}}$ is measure preserving, namely for each Borel set $\omega \subset \Omega^*$,

$$\mu_f([\mathcal{R}_{\mathcal{M}}(\omega)]^{-1}(\omega)) = \mu_g^*(\omega). \tag{6.6}$$

By replacing y with $-y$ in (6.5), Loeper [32] pointed out it is equivalent to studying

$$c(x,y) = \log(\cos d(x,y) + 1) \in \mathscr{C}_\pi^\infty.$$

Such cost satisfies the MTW condition. Since both Aleksandrov type condition (1.16) and condition (1.17) are free in the case $d_0 = \pi$, the following theorem, which was originally proved by Wang [47], can be also viewed as a result of Theorems 1.1 & 1.2.

Theorem 6.3 (Wang [47])**.** *Let Ω and Ω^* be domains in $\mathbb{S}^n$, and $f \in L^1(\Omega), g \in L^1(\Omega^*)$ be functions satisfying (1.18). The following statements hold:*

(i) *there is a unique (up to homothetic expansion w.r.t. origin) reflector $\mathcal{M}$ solving (6.6) if and only if (Ω, f) and (Ω^*, g) are Aleksandrov related;*

(ii) *if $\partial\Omega, \partial\Omega^*$ are moreover smooth and and uniformly c-convex, and $f \in C^\infty(\bar{\Omega}), g \in C^\infty(\bar{\Omega}^*)$, then the reflector $\mathcal{M}$ is smooth and uniformly c-convex.*

7. The determinant estimates

We first prove the strict obliqueness estimate, which is a result of uniform c-convexity. Let ϱ and ϱ^* be defining functions of Ω and Ω^* respectively. Set $H(x,Z) = \varrho^*(G(x,Z))$, where $G : T\mathbb{S}^n \to \mathbb{S}^n$ is given by (4.12). Given a smooth c-convex function $w(\cdot,t)$, let

$$\Theta(x,t) := \langle D_Z H(x, \nabla w), \nabla \varrho(x)\rangle,\ x \in \bar{\Omega}. \tag{7.1}$$

Let $\{\boldsymbol{e}_i\}_{i=1}^n$ and $\{\bar{\boldsymbol{e}}_i\}_{i=1}^n$ be local frames of $\mathbb{S}^n$ near x and $\chi_{w(\cdot,t)}(x)$ respectively. Then

$$Z^k \boldsymbol{e}_k = \nabla_x c|_{(x,G(x,Z))}(\boldsymbol{e}_i)\sigma^{ij}\boldsymbol{e}_j,\ \text{where } Z = Z^k\boldsymbol{e}_k.$$

Differentiating this w.r.t. Z^k, we see $\sigma_{jk} = \nabla_{x,y} c|_{x,G(x,Z)}(\boldsymbol{e}_j, D_{Z^k}G)$. Hence

$$D_{Z^k}G = \bar{\boldsymbol{e}}_j c^{\bar{j}l}\sigma_{lk}, \tag{7.2}$$

where $c^{\bar{r}l}$ is the inverse $c_{l\bar{r}}$. It follows that

$$D_{Z^k}H = \langle \nabla \varrho^*, D_{Z^k}G\rangle = \varrho^*_{\bar{j}}c^{\bar{j}l}\sigma_{lk}. \tag{7.3}$$

Remark 7.1. *By virtue of (7.1) and (7.3), we conclude that*

$$\Theta = \varrho^*_{\bar{j}}c^{\bar{j}i}\varrho_i. \tag{7.4}$$

Hence $\Theta|_{\partial\Omega} = n^*_{\bar{j}}c^{\bar{j}i}n_i$, where $\vec{\boldsymbol{n}} = n_i\sigma^{ik}\boldsymbol{e}_k$ and $\vec{\boldsymbol{n}}^* = n^*_{\bar{j}}\sigma^{\bar{j}\bar{l}}\bar{\boldsymbol{e}}_l$ are unit outer normals of $\partial\Omega$ and $\partial\Omega^*$ at x and $\chi_{w(\cdot,t)}(x)$. So $\Theta|_{\partial\Omega}$ is independent of the choice of defining functions.

Lemma 7.1. *Let $w \in C^{4,2}_{x,t}(\bar{\Omega}\times[0,T))$ be a uniformly c-convex solution of (4.18), Then*

$$\Theta|_{\partial\Omega\times[0,T)} > 0.$$

Proof. Fix $(x,t) \in \partial\Omega\times[0,T)$. Set $\bar{H}(x,t) = H(x,\nabla w(x,t))$. The boundary condition in (4.18) gives $\bar{H}_{\vec{\boldsymbol{n}}} := \nabla_{\vec{\boldsymbol{n}}}\bar{H} \geq 0$. Since $\bar{H}(\cdot,t)|_{\partial\Omega} = 0$, we see that

$$\bar{H}_i := \nabla_{\boldsymbol{e}_i}\bar{H} = \nabla_{\langle\vec{\boldsymbol{n}},\boldsymbol{e}_i\rangle\vec{\boldsymbol{n}}}\bar{H} = \varrho_i\bar{H}_{\vec{\boldsymbol{n}}},\ \forall i. \tag{7.5}$$

On the other hand, we derive from(4.4) that

$$\bar{H}_i = \langle\nabla\varrho^*, D_{\boldsymbol{e}_i}\chi_w\rangle = \varrho^*_{\bar{k}}c^{\bar{k}j}\mathcal{B}_{ji}, \tag{7.6}$$

Comparing (7.5) and (7.6), we find that

$$\varrho^*_{\bar{j}} = \bar{H}_{\vec{\boldsymbol{n}}}\mathcal{B}^{\vec{\boldsymbol{n}}i}c_{i\bar{j}}, \tag{7.7}$$

where $\mathcal{B}^{\xi\eta} = \mathcal{B}^{ij}\xi_i\eta_j$ with $\mathcal{B}^{ij}$ being the inverse of $\mathcal{B}_{ij}$. By (7.4) and (7.7), $\Theta = \bar{H}_{\vec{\boldsymbol{n}}}\mathcal{B}^{\vec{\boldsymbol{n}}\vec{\boldsymbol{n}}}$. By (7.5), we further obtain $\Theta^2 = \mathcal{B}^{\vec{\boldsymbol{n}}\vec{\boldsymbol{n}}}\bar{H}_i\mathcal{B}^{ij}\bar{H}_j$. It then follows by (7.6) that

$$\Theta^2 = \mathcal{B}^{\vec{\boldsymbol{n}}\vec{\boldsymbol{n}}}(\varrho^*_{\bar{k}}c^{\bar{k}j}\mathcal{B}_{ji}c^{\bar{l}i}\varrho^*_{\bar{l}}). \tag{7.8}$$

Strict obliqueness then follows by the positivity of $\mathcal{B}[w]$. □

We next prove the determinant estimates.

Lemma 7.2. *Let $w \in C^{4,2}_{x,t}(\bar{\Omega}\times[0,T))$ be a uniformly c-convex solution of (4.18). Suppose (1.17) holds, and (Ω,f), (Ω^*,g) are Aleksandrov related. Then there exists $C>0$ depending on $f,g,F|_{[0,d_0-\varepsilon_0]}$ (see (4.32) for ε_0) and w_0 such that*

$$C^{-1} \leq \det\mathcal{B}[w]|_{(x,t)} \leq C, \quad \forall (x,t)\in\bar{\Omega}\times[0,T). \tag{7.9}$$

Proof. Let ϱ be the defining function of Ω, and denote $\dot{w} = \partial_t w$. Consider

$$W(x,t) = 1 - \dot{w} - \varepsilon\varrho - At,$$

where $\varepsilon, A > 0$ are constants to be determined. Assume that the local smooth frame $\{\boldsymbol{e}_i\}_{i=1}^n$ is orthonormal at the point we do the calculation. Differentiating (4.18) in t,

$$\partial_t\dot{w} = \frac{\Psi}{\det\mathcal{B}}\mathcal{B}^{ij}(\dot{w}_{ij} - D_{Z_k}\mathcal{A}_{ij}\dot{w}_k) - \frac{D_{Z_k}\Psi}{\det\mathcal{B}}\dot{w}_k. \tag{7.10}$$

Fix any $t_* \in (0,T)$. Let $M = \sup_{\Omega\times[0,t_*]} \operatorname{tr}\mathcal{B}^{-1}$. In view of (7.10),

$$\begin{aligned}\frac{\Psi}{\det\mathcal{B}}\mathcal{B}^{ij}W_{ij} - \partial_t W &= -\frac{\Psi}{\det\mathcal{B}}\mathcal{B}^{ij}\dot{w}_{ij} + \partial_t\dot{w} - \varepsilon\frac{\Psi}{\det\mathcal{B}}\mathcal{B}^{ij}\varrho_{ij} + A \\ &= -\Big(\frac{\Psi}{\det\mathcal{B}}\mathcal{B}^{ij}D_{Z_k}\mathcal{A}_{ij} + \frac{D_{Z_k}\Psi}{\det\mathcal{B}}\Big)\dot{w}_k - \varepsilon\frac{\Psi}{\det\mathcal{B}}\mathcal{B}^{ij}\varrho_{ij} + A \\ &\geq \Big(\frac{\Psi}{\det\mathcal{B}}\mathcal{B}^{ij}D_{Z_k}\mathcal{A}_{ij} + \frac{D_{Z_k}\Psi}{\det\mathcal{B}}\Big)W_k - \varepsilon C(1+M^{n+1}) + A,\end{aligned}$$

where C^1-estimates (Lemma 4.9, Corollary 4.1) are used. Taking $A = \varepsilon C(2+M^{n+1})$,

$$\frac{\Psi}{\det\mathcal{B}}\mathcal{B}^{ij}W_{ij} - \Big(\frac{\Psi}{\det B}\mathcal{B}^{ij}D_{Z_k}\mathcal{A}_{ij} + \frac{D_{Z_k}\Psi}{\det\mathcal{B}}\Big)W_k - \partial_t W \geq 0,$$

which implies that $\max_{\bar{\Omega}\times[0,t_*]} W$ is achieved on the parabolic boundary of $\Omega\times(0,t_*)$.

Suppose $\max_{\bar{\Omega}\times[0,t_*]} W = W(x_0,\tau_0)$ with $(x_0,\tau_0) \in \partial\Omega\times(0,t_*]$. By Lemma 7.1,

$$\langle\nabla W, D_Z H\rangle|_{(x_0,\tau_0)} = \langle a\nabla\varrho, D_Z H\rangle|_{(x_0,\tau_0)} = a\Theta|_{(x_0,\tau_0)} \geq 0, \quad \text{where } a \geq 0. \tag{7.11}$$

On the other hand, the boundary condition in (4.18) implies that $H(x,\nabla w(x,t)) = 0$ whenever $x \in \partial\Omega$. By differentiating it in t, we obtain $\dot{w}_k D_{Z^k}H = 0$. Therefore

$$\langle\nabla W, D_Z H\rangle|_{\partial\Omega\times t} = -\varepsilon\langle D_Z H, \nabla\varrho\rangle|_{\partial\Omega\times t} = -\varepsilon\Theta|_{\partial\Omega\times t} < 0, \tag{7.12}$$

where the inequality follows by Lemma 7.1 again. Putting (7.11) and (7.12) together, we arrive a contradiction. Hence $\max_{\bar{\Omega}\times[0,t_*]} W$ must be attained on $\bar{\Omega}\times\{0\}$. So

$$\partial_t w(x,t) \geq \min_{\bar{\Omega}} \partial_t w(\cdot,0) - \varepsilon\max_{\bar{\Omega}}|\varrho| - \varepsilon C(2+M^{n+1})t_* \quad \forall (x,t) \in \bar{\Omega}\times[0,t_*].$$

Sending $\varepsilon \to 0$, we see $\min_{\bar{\Omega}\times[0,t_*]} \partial_t w \geq \min_{\bar{\Omega}} \partial_t w(\cdot,0)$. By the arbitrariness of $t_* \in (0,T]$ and using (4.18), we obtain the lower bound in (7.9). The upper bound follows by considering $\widetilde{W} = 1 - \dot{w} + \varepsilon\varrho - At$ similarly. We omit the remaining details. $\square$

8. Uniform c-convexity of domains

In this section, we discuss the notion of uniform c-convexity of domains on $\mathbb{S}^n$ and give some applications for boundary estimates.

Definition 8.1. *Suppose Ω and Ω^* are domains of $\mathbb{S}^n$ satisfying (1.17). Let*

$$\begin{aligned}(\Omega)^\wedge_y &= \nabla_y c(\Omega_y, y) \subset T_y\mathbb{S}^n, \quad \text{where } y \in \bar{\Omega}^* \text{ and } \Omega_y = \Omega\cap B_{d_0}(y), \\ (\Omega^*)^\wedge_x &= \nabla_x c(x,\Omega^*_x) \subset T_x\mathbb{S}^n, \ \text{where } x \in \bar{\Omega} \text{ and } \Omega^*_x = \Omega^*\cap B_{d_0}(x).\end{aligned}$$

We say that Ω is (strictly) c-convex w.r.t. Ω^* if $(\Omega)^\wedge_y$ is (strictly) convex in $\mathbb{R}^n \simeq T_y\mathbb{S}^n$ for each $y \in \bar{\Omega}^*$. Analogously Ω^* is (strictly) c-convex w.r.t. Ω if $(\Omega^*)^\wedge_x$ is (strictly) convex in $\mathbb{R}^n \simeq T_x\mathbb{S}^n$ for each $x \in \bar{\Omega}$.

Given $y \in \bar{\Omega}^*$ and $x \in \Omega_y$, let $\xi(x) \in T_y\mathbb{S}^n$ be the unique unit vector such that $x = \exp_y \xi(x)$. It is easily checked that

$$\nabla_y c(x,y) = -F'(d(x,y))\xi(x). \tag{8.1}$$

By the first condition in (1.11), $(\Omega)^\wedge_y$ is unbounded in $T_y\mathbb{S}^n$. Given $y \in \bar{\Omega}^*$, let

$$(\partial\Omega)_y := \partial\Omega \cap B_{d_0}(y) \quad \text{and} \quad (\partial\Omega)^\wedge_y := \nabla_y c((\partial\Omega)_y, y).$$

Lemma 8.1. *Suppose Ω and Ω^* are C^2-smooth domains of $\mathbb{S}^n$ satisfying (1.17). Define*

$$\hat{X}_y(\cdot) = \nabla_y c(\cdot, y) : \Omega_y \to (\Omega)^\wedge_y \subset \mathbb{R}^n \simeq T_y\mathbb{S}^n, \tag{8.2}$$

where the given point y satisfies $(\partial\Omega)_y \neq \emptyset$. Let $\hat{I\!I}_y$ and $I\!I$ be the second fundamental forms of $(\partial\Omega)^\wedge_y$ and $(\partial\Omega)_y$, respectively. Then for each $x \in (\partial\Omega)_y$, the differential $D_x\hat{X}_y$ is an isomorphism from $T_x(\partial\Omega)$ to $T_{\hat{X}_y(x)}((\partial\Omega)^\wedge_y)$, and for each unit $\tau \in T_x(\partial\Omega)$,

$$\hat{I\!I}_y|_{\hat{X}_y(x)}(\hat{\tau},\hat{\tau}) = \nabla^2_{x,y}c(\vec{\boldsymbol{n}}|_x, \hat{\nu}|_{\hat{X}_y(x)}) \cdot I\!I|_x(\tau,\tau) - \nabla^2_x\nabla_y c(\tau,\tau;\hat{\nu}|_{\hat{X}_y(x)}), \tag{8.3}$$

where $\hat{\tau} = D_x\hat{X}_y(\tau)$ and $\vec{\boldsymbol{n}}, \hat{\nu}$ are respectively the unit outer normals of $\partial\Omega, (\partial\Omega)^\wedge_y$.

Proof. Let $\{\boldsymbol{e}_i\}_{i=1}^n$ be a local frame of $\mathbb{S}^n$ around $x \in (\partial\Omega)_y$ such that $\{\boldsymbol{e}_i(x)\}_{i=1}^{n-1}$ is an orthonormal basis of $\mathbb{R}^{n-1} \simeq T_x(\partial\Omega)$ and $\boldsymbol{e}_n(x) = \vec{\boldsymbol{n}}|_x$. Let $\{\bar{\boldsymbol{e}}_i\}_{i=1}^n$ be a local frame of $\mathbb{S}^n$ around y such that $\{\bar{\boldsymbol{e}}_i(y)\}_{i=1}^n$ is an orthonormal basis of $\mathbb{R}^n \simeq T_y\mathbb{S}^n$. Since $\{\boldsymbol{e}_i(x)\}_{i=1}^{n-1} \subset T_x((\partial\Omega)_y)$, we have $\{\hat{X}_i(x)\}_{i=1}^{n-1} \subset T_{\hat{X}_y(x)}((\partial\Omega)^\wedge_y)$, where

$$\hat{X}_i := D_x\hat{X}_y(\boldsymbol{e}_i) = c_{i\bar{j}}(x,y)\bar{\boldsymbol{e}}_j. \tag{8.4}$$

Then the non-degeneracy of $\nabla^2_{x,y}c$ implies $D_x\hat{X}_y$ is an isomorphism. For $i,j \neq n-1$,

$$(\hat{I\!I}_y)_{ij} := \hat{I\!I}_y(\hat{X}_i,\hat{X}_j) = \langle D_{\boldsymbol{e}_i}\hat{X}_y, D_{\boldsymbol{e}_j}\hat{\nu}\rangle = -\langle D_{\boldsymbol{e}_j}D_{\boldsymbol{e}_i}\hat{X}_y, \hat{\nu}\rangle.$$

Denote $c_{ij,\bar{k}} = \nabla^2_x\nabla_y c(\boldsymbol{e}_i,\boldsymbol{e}_j;\bar{\boldsymbol{e}}_k)$. It follows by (8.4) that

$$\begin{aligned}
D_{\boldsymbol{e}_j}D_{\boldsymbol{e}_i}\hat{X}_y &= D_{\boldsymbol{e}_j}(\nabla_x\nabla_y c(\boldsymbol{e}_i,\bar{\boldsymbol{e}}_k))\bar{\boldsymbol{e}}_k \\
&= \nabla^2_x\nabla_y c(\boldsymbol{e}_i,\boldsymbol{e}_j;\bar{\boldsymbol{e}}_k)\bar{\boldsymbol{e}}_k + \nabla^2_{x,y}c(\nabla_{\boldsymbol{e}_j}\boldsymbol{e}_i;\bar{\boldsymbol{e}}_k)\bar{\boldsymbol{e}}_k \\
&= c_{ij,\bar{k}}\bar{\boldsymbol{e}}_k - \nabla^2_{x,y}c(I\!I_{ij}\vec{\boldsymbol{n}}|_x;\bar{\boldsymbol{e}}_k)\bar{\boldsymbol{e}}_k + \nabla^2_{x,y}c((\nabla_{\boldsymbol{e}_j}\boldsymbol{e}_i)^{\tan};\bar{\boldsymbol{e}}_k)\bar{\boldsymbol{e}}_k \\
&= c_{ij,\bar{k}}\bar{\boldsymbol{e}}_k - I\!I_{ij}c_{n,\bar{k}}\bar{\boldsymbol{e}}_k + \langle(\nabla_{\boldsymbol{e}_j}\boldsymbol{e}_i)^{\tan},\boldsymbol{e}_k\rangle\hat{X}_k. \qquad (8.5)
\end{aligned}$$

Multiplying $\hat{\nu}|_{\hat{X}_y(x)}$ at both sides of (8.5), we have

$$(\hat{I\!I}_y)_{ij} = I\!I_{ij}c_{n,\bar{k}}\hat{\gamma}_{\bar{k}} - c_{ij,\bar{k}}\hat{\gamma}_{\bar{k}}, \quad \text{where } \hat{\gamma}_{\bar{k}} = \langle\hat{\nu},\bar{\boldsymbol{e}}_k\rangle. \tag{8.6}$$

Write $\tau = \tau^i\boldsymbol{e}_i$. Multiplying $\tau^i\tau^j$ at both sides of (8.6), we obtain (8.3). $\square$

Remark 8.1. *By exchanging the role of x and y, we obtain the similar relation between the second fundamental forms of $(\partial\Omega^*)^\wedge_x$ and $(\partial\Omega^*)_x$.*

We also give an alternative formulation of the uniform c-convexity, which is useful in finding defining functions with some needed properties.

Lemma 8.2. *Let Ω and Ω^* be C^2-smooth domains of $\mathbb{S}^n$ satisfying (1.17). Suppose (1.28) and (1.29) hold with $\delta_0 > 0$. Given $y \in \bar{\Omega}^*$ with $(\partial\Omega)_y \neq \emptyset$, let II be the second fundamental form of $(\partial\Omega)_y$. Then for each $x \in (\partial\Omega)_y$ and each unit vector $\tau \in T_x(\partial\Omega)$*

$$II|_x(\tau,\tau) - D_Z\mathcal{A}|_{(x,\nabla_x c(x,y))}(\tau,\tau)\cdot \vec{\boldsymbol{n}}|_x \geq \delta', \text{ if } d(x,y) < d_0 - \delta. \tag{8.7}$$

Analogously given $x \in \bar{\Omega}$ with $(\partial\Omega^*)_x \neq \emptyset$, let II^* be the second fundamental form of $(\partial\Omega^*)_x$. Then for each $y \in (\partial\Omega^*)_x$ and each unit vector $\tau^* \in T_y(\partial\Omega^*)$

$$II^*|_y(\tau^*,\tau^*) - D_Z\mathcal{A}|_{(y,\nabla_y c(x,y))}(\tau^*,\tau^*)\cdot \vec{\boldsymbol{n}}^*|_y \geq \delta', \text{ if } d(x,y) < d_0 - \delta. \tag{8.8}$$

Proof. By the duality of Ω and Ω^*, it suffices to prove (8.7). Let $\{\boldsymbol{e}_i\}_{i=1}^n$ be a local frame of $\mathbb{S}^n$ around $x \in (\partial\Omega)_y$ so that $\{\boldsymbol{e}_i(x)\}_{i=1}^{n-1}$ is an orthonormal basis of $T_x(\partial\Omega)$ and $\boldsymbol{e}_n(x) = \vec{\boldsymbol{n}}|_x$. Let $\hat{X}_y(\cdot) = \nabla_y c(\cdot, y)$. Consider the vector field

$$\hat{\eta} = \hat{X}_n - \sum_{1\leq i,j\leq n-1} \langle \hat{X}_n, \hat{X}_i\rangle (\hat{g}_y)^{ij}\hat{X}_j, \tag{8.9}$$

where $(\hat{g}_y)^{ij}$ is the inverse of $(\hat{g}_y)_{ij} = \langle \hat{X}_i, \hat{X}_j\rangle$. Since $\hat{X}_i$ $(i \neq n)$ is tangential to $(\partial\Omega)^\wedge_y$, we see $\hat{\eta} = |\hat{\eta}|\hat{\nu}$, where $\hat{\nu}$ is the unit outer normal of $(\partial\Omega)^\wedge_y$. By (8.4), we have

$$\nabla^2_{x,y} c(\vec{\boldsymbol{n}}|_x, \hat{\nu}|_{\hat{X}_y(x)}) = \langle \hat{X}_n, \hat{\nu}\rangle|_x = \langle \hat{\eta}, \hat{\nu}\rangle|_x = |\hat{\eta}|(x) > 0. \tag{8.10}$$

Employing (7.2) and (8.4), we see $\langle D_{Z^i}G, \hat{X}_j\rangle = \delta_{ij}$. Hence $\langle D_{Z^n}G, \hat{X}_i\rangle = 0$ for all $i \neq n$ and $\langle D_{Z^n}G, \hat{X}_n\rangle = 1$. So $D_{Z^n}G = \mu\hat{\nu}$ for some $\mu > 0$ at x. By (8.9), we deduce $\langle D_{Z^n}G, \hat{\eta}\rangle = 1$. Hence $D_{Z^n}G = \hat{\nu}/|\hat{\eta}|$. Differentiating $\mathcal{A}_{ij}|_{(x,Z)} = c_{ij}|_{(x,G(x,Z))}$ gives

$$D_{Z^n}\mathcal{A}|_{(x,\nabla_x c(x,y))}(\boldsymbol{e}_i, \boldsymbol{e}_j) = \nabla^2_x\nabla_y c(\boldsymbol{e}_i, \boldsymbol{e}_j; D_{Z^n}G) = \frac{\nabla^2_x\nabla_y c(\boldsymbol{e}_i, \boldsymbol{e}_j; \hat{\nu})}{\nabla^2_{x,y} c(\vec{\boldsymbol{n}}|_x, \hat{\nu}|_{\hat{X}(x)})}, \tag{8.11}$$

where (8.10) is used in the last equality. By (1.28) and (8.10), for all $i, j \neq n$,

$$\delta_0(\hat{g}_y|_{\hat{X}_y(x)})_{ij} \leq \frac{\hat{II}_y|_{\nabla_y c(x,y)}(\hat{X}_i, \hat{X}_j)}{\nabla^2_{x,y} c(\vec{\boldsymbol{n}}|_x, \hat{\nu}|_{\hat{X}(x)})}.$$

In view of Lemma 8.1 and (8.11), we further obtain

$$\delta_0(\hat{g}_y|_{\hat{X}_y(x)})_{ij} \leq II|_x(\boldsymbol{e}_i, \boldsymbol{e}_j) - D_{Z^n}\mathcal{A}|_{(x,\nabla_x c(x,y))}(\boldsymbol{e}_i, \boldsymbol{e}_j). \tag{8.12}$$

Write $\tau = \tau^i\boldsymbol{e}_i$. Multiplying $\tau^i\tau^j$ at both sides of (8.12), we obtain (8.7). □

By Lemma 8.2, one can follow the argument in [29] to find nice auxiliary functions.

Lemma 8.3 ([29], Theorem A.1)**.** *Let Ω and Ω^* be C^2 domains of $\mathbb{S}^n$ satisfying (1.17). Suppose (1.28) and (1.29) hold with $\delta_0 > 0$. Then there exist defining functions ϱ of Ω and ϱ^* of Ω^* such that (i) for any $y \in \bar{\Omega}^*$, $x \in (\partial\Omega)_y \neq \emptyset$ and unit $\xi \in T_x\mathbb{S}^n$, there holds*

$$\nabla^2\varrho|_x(\xi,\xi) - D_Z\mathcal{A}|_{(x,\nabla_x c(x,y))}(\xi,\xi)\cdot \vec{\boldsymbol{n}}|_x \geq \frac{\delta'}{2}, \text{ if } d(x,y) < d_0 - \delta. \tag{8.13}$$

(ii) for any $x \in \bar{\Omega}$, $y \in (\partial\Omega^*)_x \neq \emptyset$ and unit $\xi^* \in T_y\mathbb{S}^n$ there holds

$$\nabla^2 \varrho^*|_y(\xi^*, \xi^*) - D_Z\mathcal{A}|_{(y,\nabla_y c(x,y))}(\xi^*, \xi^*) \cdot \vec{\boldsymbol{n}}^*|_y \geq \frac{\delta'}{2}, \text{ if } d(x,y) < d_0 - \delta. \tag{8.14}$$

Remark 8.2. *For the later boundary estimates, it suffices to take δ in (8.13) and (8.14) to be ε_0 in Lemma 4.9. Therefore we can fix a uniformly positive δ' in Lemma 8.3.*

Lemma 8.4 ([29], Corollary A.2)**.** *Under the same assumption of Lemma 8.3, given a constant $R_0 > 0$, there exist a C^3-smooth function $\widetilde{H}$ on $\Gamma_{R_0} := \{(x, Z) : x \in \bar{\Omega},\ |Z| \leq R_0\}$ and $\delta_0^* > 0$ such that, for all $(x, Z) \in \Gamma_{R_0}$,*

- (i) *$D^2_{ZZ}\widetilde{H}|_{(x,Z)}(\xi, \xi) \geq \delta_0^*|\xi|^2$, where $\xi \in T_{G(x,Z)}\mathbb{S}^n$, (see (4.12) for G);*
- (ii) *$\widetilde{H}(x, Z) \leq 0$ if $G(x, Z)$ lies in Ω^*, and vanishes if $G(x, Z) \in \partial\Omega^*$;*
- (iii) *$D_Z\widetilde{H}(x, Z) = D_Z H(x, Z)$ whenever $G(x, Z) \in \partial\Omega^*$, where $H(x, Z) = \varrho^*(G(x, Z))$ with ϱ^* being a defining function of Ω^* in Lemma 8.3.*

We next present a maximum principle.

Lemma 8.5. *Let $w \in C^{4,2}_{x,t}(\bar{\Omega} \times [0, T))$ be a uniformly c-convex solution of (4.18). Define*

$$\mathscr{L}\phi := \mathcal{B}^{ij}\Big(\nabla^2_{ij}\phi - D_{Z^k}\mathcal{A}_{ij}|_{(x,\nabla w)}\nabla_k\phi\Big) - \frac{\partial_t\phi}{1 - \partial_t w},\ \forall \phi \in C^{4,2}_{x,t}(\bar{\Omega} \times [0, T)). \tag{8.15}$$

Suppose (1.17) holds, and (Ω, f), (Ω^*, g) are Aleksandrov related. Given $V \in C^2(T\bar{\Omega})$, take $\bar{V}(x, t) = V(x, \nabla w(x, t))$. Then the following statements hold:

- (i) *There exists $C > 0$ depending on $f, g, w_0, F|_{[0,d_0-\varepsilon_0]}$ and $\|V\|_{C^2(T\bar{\Omega})}$ such that*

$$\mathscr{L}\bar{V} - D_{Z^k}\Phi\nabla_k\bar{V} \leq \mathcal{B}_{ij}\nabla^2_{Z^iZ^j}V + C\,\mathrm{tr}\,\mathcal{B}^{-1}, \tag{8.16}$$

  where $\Phi(x, Z) = \log\Psi(x, Z)$ with Ψ being in (4.17), $tr\mathcal{B}^{-1} = \mathcal{B}^{ij}\sigma_{ij}$, and σ_{ij} is the spherical metric.
- (ii) *Suppose Ω is uniformly c-convex w.r.t. Ω^*, and there is $C' > 0$ such that*

$$\mathscr{L}\bar{V} - D_{Z^k}\Phi\nabla_k\bar{V} \leq C' tr\mathcal{B}^{-1},\ \forall (x, t) \in \bar{\Omega} \times [0, T), \tag{8.17}$$

  where $tr\mathcal{B}^{-1} = \mathcal{B}^{ij}\sigma_{ij}$. For $t_* \in (0, T)$, if $\min_{\partial\Omega\times[0,t_*]}\bar{V} \geq 0$ is attained at $(x_0, t_0) \in \partial\Omega \times [0, t_*]$, then there is $C > 0$ (independent of t_*) so that

$$\nabla_{\vec{\boldsymbol{n}}}\bar{V}(x_0, t_0) \leq C, \tag{8.18}$$

  where $\vec{\boldsymbol{n}}$ is the unit outer normal of $\partial\Omega$ at x_0.

Proof. Since (8.16) is independent of coordinates, we take normal coordinate centered at where we do the calculation, and assume $\mathcal{B}_{ij}$ is also diagonalized. Differentiate (4.18),

$$\mathcal{B}^{ii}\mathcal{B}_{ii;k} - \frac{\partial_t w_k}{1 - \partial_t w} = \Phi_k + w_{lk}D_{Z^l}\Phi.$$

By the Ricci identity, we commutate the indices of $w_{ij;k}$ and thus obtain

$$\mathcal{B}^{ii}(w_{ki;i} - D_{Z_l}\mathcal{A}_{ii}w_{lk}) - \frac{\partial_t w_k}{1-\partial_t w} \leq \mathcal{B}_{lk}D_{Z_l}\Phi + C(1+\operatorname{tr}\mathcal{B}^{-1}). \tag{8.19}$$

On the other hand, straightforward computation gives

$$\mathscr{L}\bar{V} \leq D_{Z_k}V\Big(\mathcal{B}^{ii}(w_{ki;i} - D_{Z_l}\mathcal{A}_{ii}w_{ik}) - \frac{\partial_t w_k}{1-\partial_t w}\Big) + \mathcal{B}_{kl}D^2_{Z_kZ_l}V + C(1+\operatorname{tr}\mathcal{B}^{-1}).$$

Plugging (8.19) into the above, we obtain

$$\mathscr{L}\bar{V} - D_{Z^k}\Phi\nabla_k\bar{V} \leq \mathcal{B}_{ij}\nabla^2_{Z^iZ^j}V + C(1+\operatorname{tr}\mathcal{B}^{-1}).$$

By Lemma 7.2, $\operatorname{tr}\mathcal{B}^{-1} \geq n(\det\mathcal{B})^{-1/n} \geq 1/C$. Hence (8.16) follows.

We then turn to (8.18). The proof follows by the argument in [29, Theorem 9.2]. Fix $y_0 \in \bar{\Omega}^*$ so that $d(x_0,y_0) < d_0$. Let X_{y_0} be as in (8.2). For $(x,t) \in \bar{\Omega}_{y_0} \times [0,t_*]$, let

$$U(x,t) = \bar{V}(x,t) - \bar{V}(x_0,t_0) + \langle \xi, \hat{X}_{y_0}(x) - \hat{X}_{y_0}(x_0)\rangle + Ad^2(x,x_0) - B\varrho(x),$$

where ϱ is a defining function of Ω satisfying (8.13) in Lemma 8.3, while $\xi \in T_{y_0}\mathbb{S}^n$ and constants $A, B > 0$ are to be determined. Set $\Omega_{y_0}(x_0;\varepsilon) = B_\varepsilon(x_0) \cap \Omega_{y_0}$. We claim that if ε, A, B and ξ are selected appropriately, then

(a) $U \geq 0$ on $\Omega_{y_0}(x_0;\varepsilon) \times \{0\}$;
(b) $U \geq 0$ on $\partial\big(\Omega_{y_0}(x_0;\varepsilon)\big) \times [0,t_*]$;
(c) $\mathscr{L}U - \nabla_{Z_k}\Phi \cdot \nabla_k U \leq 0$ in $\Omega_{y_0}(x_0;\varepsilon) \times [0,t_*]$.

Once (a)-(c) are proved, we conclude from $U(x_0,t_0) = 0$ and the maximum principle

$$0 \geq \nabla_{\vec{n}}U(x_0,t_0) \geq \nabla_{\vec{n}}\bar{V}(x_0,t_0) - C.$$

This shows (8.18). It remains to verify (a)-(c).

By the c-convexity, $(\Omega)^\wedge_{y_0}$ is a strictly convex in $T_{y_0}\mathbb{S}^n$. Hence there exists a unique supporting hyperplane Π to $(\Omega)^\wedge_{y_0}$ at $\hat{x}_0 := \hat{X}_{y_0}(x_0)$. Let $\{\bar{\boldsymbol{e}}_i\}_{i=1}^n$ be an orthonormal basis of $T_{y_0}\mathbb{S}^n$ such that $\bar{\boldsymbol{e}}_n$ is the unit outer normal of $(\Omega)^\wedge_{y_0}$ at $\hat{x}_0$. Take $x_{y_0} = \hat{X}^{-1}_{y_0}(\hat{x})$ and

$$\hat{V}(\hat{x},t) = \bar{V}(x_{y_0},t), \text{ for } \hat{x} \in (\Omega)^\wedge_{y_0}.$$

For $\hat{x} \in (\Omega)^\wedge_{y_0}$, let $\pi'(\hat{x})$ and $\pi(\hat{x})$ be the intersections of $\{\hat{x} + t\bar{\boldsymbol{e}}_n\}_{t\geq 0}$ with $(\partial\Omega)^\wedge_{y_0}$ and Π respectively. Then, since $\hat{V}(\hat{x}_0,t_0) = \min\hat{V}(\cdot,t_0)$,

$$\begin{aligned}\hat{V}(\hat{x},0) - \hat{V}(\hat{x}_0,t_0) &\geq \hat{V}(\pi'(\hat{x}),0) - \hat{V}(\hat{x}_0,t_0) - \Big[\sup_{(\Omega)^\wedge_{y_0}}|D\hat{V}(\cdot,0)|\Big]|\hat{x} - \pi(\hat{x})| \\ &\geq -\langle \xi, \hat{x} - \hat{x}_0\rangle, \quad \forall \hat{x} \in (\Omega)^\wedge_{y_0},\end{aligned}$$

provided that $\xi = -M\bar{\boldsymbol{e}}_n$ with large multiplier M. Since $-\varrho \geq 0$, we see that (a) holds.

We next show (b). Note that $\langle \xi, \hat{x} - \hat{x}_0\rangle \geq 0$, $\forall \hat{x} \in (\Omega)^\wedge_{y_0}$. When $x \in \partial\Omega_{y_0}(x_0;\varepsilon) \cap \partial\Omega$, (b) is obvious. When $x \in \partial\Omega_{y_0}(x_0;\varepsilon) \cap \Omega$, we take A large and conclude $U(x,t) \geq 0$.

We infer from the assumption (8.17) and the fact $\mathrm{tr}\mathcal{B}^{-1} \geq 1/C$ that

$$\mathscr{L}U - \nabla_{Z_k}\Phi \cdot \nabla_k U \leq -B\mathscr{L}\varrho + C\mathrm{tr}\mathcal{B}^{-1}.$$

In view of (8.13) in Lemma 8.3, by taking small $\varepsilon > 0$, there exists a time-independent positive constant δ_0 so that $\mathscr{L}\varrho \geq 4^{-1}\delta_0\mathrm{tr}\mathcal{B}^{-1}$, $\forall x \in \Omega_{y_0}(x_0;\varepsilon)$. Choosing B large, we obtain (c) by combining the above two estimates. □

9. The uniform obliqueness estimates

This section is devoted to the uniform obliqueness of the boundary condition.

Lemma 9.1. *Let $w \in C^{4,2}_{x,t}(\bar{\Omega} \times [0,T))$ be a uniformly c-convex solution of (4.18). Suppose (1.17) holds, and (Ω, f), (Ω^*, g) are Aleksandrov related. Then there exists $\delta_1 > 0$ depending on $f, g, F|_{[0,d_0-\varepsilon_0]}, w_0$ and the modulus of c-convexity of Ω, Ω^*, such that*

$$\inf_{\partial\Omega\times[0,T)} \Theta(x,t) \geq \delta_1.$$

By Remark 7.1, $\Theta|_{\partial\Omega}(\cdot,t)$, as given by (7.1), is independent of the choice of defining functions. In the proof below, we use the defining functions in Lemma 8.3.

Proof. By Lemma 4.9, $R_0 := \sup_{\bar{\Omega}\times[0,T)} |\nabla w| < \infty$. Let $\bar{V}(x,t) = V(x,\nabla w)$, where

$$V(x,Z) = \widetilde{\Theta}(x,Z) - A\widetilde{H}(x,Z), \tag{9.1}$$

where $\widetilde{\Theta}(x,Z) = \langle D_Z H, \nabla\varrho\rangle$ such that $\Theta(x,t) = \widetilde{\Theta}(x,\nabla w)$, $\widetilde{H}$ is given in Lemma 8.4 and $A > 0$ is to be determined. Then, by (i) in Lemma 8.4,

$$D^2_{Z_iZ_j}V = D^2_{Z_iZ_j}\widetilde{\Theta} - AD^2_{Z_iZ_j}\widetilde{H} \leq (C - A\delta_0^*)\sigma_{ij} \leq 0$$

for large A. This together with (i) in Lemma 8.5 implies

$$\mathscr{L}\bar{V} - D_{Z_k}\Phi \cdot \nabla_k\bar{V} \leq \mathcal{B}_{ij}D^2_{Z_iZ_j}V + C\mathrm{tr}\mathcal{B}^{-1} \leq C\mathrm{tr}\mathcal{B}^{-1}.$$

Fix $t_* < T$. Suppose $\min_{\partial\Omega\times[0,t_*]}\Theta = \Theta(x_0,t_0)$ for some $(x_0,t_0) \in \partial\Omega\times[0,t_*]$. By (ii) in Lemma 8.4 and Lemma 7.1, $\min_{\partial\Omega\times[0,t_*]}\bar{V} = \bar{V}(x_0,t_0) > 0$. Using (ii) in Lemma 8.5,

$$\nabla\bar{V}(x_0,t_0) = \kappa\vec{\boldsymbol{n}}|_{x_0} \quad \text{with } \kappa \leq C, \tag{9.2}$$

where $C > 0$ is independent of t_*.

Let $\{\boldsymbol{e}_i\}_{i=1}^n$ and $\{\bar{\boldsymbol{e}}_i\}_{i=1}^n$ be local frames around x_0 and $y_0 = \chi_{w(\cdot,t_0)}(x_0)$ respectively, such that $\{\boldsymbol{e}_i(x_0)\}_{i=1}^{n-1}$ is orthonormal and $\boldsymbol{e}_n(x_0)$ is the unit outer normal of $\partial\Omega$ at x_0. By (7.4), we have

$$\nabla_i\Theta = \varrho^*_{\bar{k}}c^{\bar{k}j}\varrho_{ij} - \varrho^*_{\bar{k}}c^{\bar{k}l}\nabla_i c_{l\bar{m}}c^{\bar{m}j}\varrho_j + \varrho^*_{\bar{k}\bar{l}}\langle\bar{e}_l, D_{\boldsymbol{e}_i}\chi_w\rangle c^{\bar{k}j}\varrho_j.$$

Employing (4.4), we further obtain

$$\nabla_i\Theta = \varrho^*_{\bar{k}}c^{\bar{k}l}(\varrho_{li} - c_{il;\bar{m}}c^{\bar{m}j}\varrho_j) + \mathcal{B}_{ij}c^{\bar{k}j}c^{\bar{m}l}\varrho_l(\varrho^*_{\bar{k}\bar{m}} - \varrho^*_{\bar{r}}c^{\bar{r}s}c_{s;\bar{k}\bar{m}}). \tag{9.3}$$

Since $\bar{H}(x,t) = \varrho^*(G(x,\nabla w))$ vanishes on $\partial\Omega$, there is $\kappa' \geq 0$ such that

$$\kappa' \varrho_j|_{x_0} = \bar{H}_j|_{(x_0,t_0)} = \langle \nabla \varrho^*, D_{\boldsymbol{e}_j}\chi_w\rangle|_{(x_0,t_0)} = \varrho^*_{\bar{p}} c^{\bar{p}i}\mathcal{B}_{ij}|_{(x_0,t_0)}. \tag{9.4}$$

Combining this with (9.3), we conclude, at (x_0,t_0),

$$\varrho^*_{\bar{p}} c^{\bar{p}i}\nabla_i\Theta = \varrho^*_{\bar{p}} c^{\bar{p}i}\varrho^*_{\bar{k}} c^{\bar{k}l}(\varrho_{li} - c_{il;\bar{m}}c^{\bar{m}j}\varrho_j) + \kappa' c^{\bar{k}j}\varrho_j c^{\bar{m}l}\varrho_l(\varrho^*_{\bar{k}\bar{m}} - \varrho^*_{\bar{r}}c^{\bar{r}s}c_{s;\bar{k}\bar{m}}).$$

Since Ω,Ω^* are uniformly c-convex, combining Lemma 4.9 and Lemma 8.3,

$$\varrho^*_{\bar{p}} c^{\bar{p}i}\nabla_i\Theta|_{(x_0,t_0)} \geq \frac{\delta'}{2}\left(|\varrho^*_{\bar{p}} c^{\bar{p}i}\boldsymbol{e}_i|^2 + \kappa'|c^{\bar{k}j}\varrho_j\bar{\boldsymbol{e}}_k|^2\right) \geq \delta'_0, \tag{9.5}$$

for some $\delta'_0 > 0$ (independent of t_*). In view of (7.4) and (9.2), we see that

$$\kappa\Theta|_{(x_0,t_0)} = \kappa\varrho^*_{\bar{p}} c^{\bar{p}m}\varrho_m = \varrho^*_{\bar{p}} c^{\bar{p}n}\bar{V}_n|_{(x_0,t_0)}.$$

Since $\bar{V}_k|_{(x_0,t_0)} = 0$ for all $k \neq n$, we further obtain by (9.1), (iii) in Lemma 8.4 and the last equality in (9.4),

$$\kappa\Theta|_{(x_0,t_0)} = \varrho^*_{\bar{p}} c^{\bar{p}i}\bar{V}_i|_{(x_0,t_0)} = \varrho^*_{\bar{p}} c^{\bar{p}i}\nabla_i\Theta|_{(x_0,t_0)} - A\varrho^*_{\bar{p}} c^{\bar{p}i}\varrho^*_{\bar{q}} c^{\bar{q}j}\mathcal{B}_{ij}|_{(x_0,t_0)}.$$

It then follows by and (9.5) that

$$\kappa\Theta|_{(x_0,t_0)} \geq \delta'_0 - A\varrho^*_{\bar{p}} c^{\bar{p}i}\varrho^*_{\bar{q}} c^{\bar{q}j}\mathcal{B}_{ij}|_{(x_0,t_0)}. \tag{9.6}$$

Case 1: $\kappa > 0$. Assume $\kappa\Theta|_{(x_0,t_0)} < \frac{1}{2}\delta'_0$. Otherwise we are done by $\kappa \leq C$. Using (9.6),

$$\varrho^*_{\bar{p}} c^{\bar{p}i}\varrho^*_{\bar{q}} c^{\bar{q}j}\mathcal{B}_{ij}[w]|_{(x_0,t_0)} \geq \frac{\delta'_0}{2A} =: \delta'_1 \text{ (independent of } t_*). \tag{9.7}$$

Case 2: $\kappa \leq 0$. We again obtain (9.7) by throwing $\kappa\Theta$ in (9.6).

We next complete the proof by the dual argument used in [43]. Let $v(y,t)$ be the c-transform of $w(x,t)$. By Lemma 4.6, $v \in C^{4,2}_{y,t}(\bar{\Omega}^* \times [0,T))$ is uniformly c-convex and satisfies (4.20). Consider $\Theta^*(y,t) = \Theta(\chi_v(y,t),t)$. It attains minimum on $\partial\Omega^* \times [0,t_*]$ at (y_0,t_0) with $y_0 = \chi_w(x_0,t_0)$. Applying the above argument to Θ^*, we conclude that either $\Theta^*(y_0,t_0)$ is bounded from below by a positive constant independent of t_*; or

$$c^{\bar{k}i}\varrho_i c^{\bar{l}j}\varrho_j\mathcal{B}_{\bar{k}\bar{l}}[v]|_{(y_0,t_0)} \geq \delta''_1 \text{ (independent of } t_*). \tag{9.8}$$

In the first case, we are done. For the second, since $\chi_{v(\cdot,t)} \circ \chi_{w(\cdot,t)} = id$,

$$\boldsymbol{e}_k = D\chi_v(D\chi_w(\boldsymbol{e}_k)) = D\chi_v(\bar{\boldsymbol{e}}_l c^{\bar{l}j}\mathcal{B}_{jk}[w]) = \mathcal{B}_{\bar{l}\bar{p}}[v]c^{\bar{p}i}\boldsymbol{e}_i c^{\bar{l}j}\mathcal{B}_{jk}[w], \tag{9.9}$$

where we also use (4.4). It follows that

$$\mathcal{B}^{\vec{n}\vec{n}}[w] = \mathcal{B}^{ij}[w]\varrho_i\varrho_j = c^{\bar{k}i}\varrho_i c^{\bar{l}j}\varrho_j\mathcal{B}_{\bar{k}\bar{l}}[v]. \tag{9.10}$$

By virtue of (7.8), the uniform obliqueness is a result of (9.7), (9.8) and (9.10). $\square$

10. A priori second derivative estimates

This part of our argument is similar to the treatment of the derivative estimates in [28, 29, 42, 43]. Most barrier functions come from these references. But for readers' convenience, we still give complete proofs.

10.1. Interior C^2-estimates.

Lemma 10.1. *Let $\mathcal{A}_{ij;k} = \nabla_k[\mathcal{A}_{ij}(x, \nabla w)]$ and $\mathcal{A}_{ij;kl} = \nabla_l\nabla_k[\mathcal{A}_{ij}(x, \nabla w)]$. Then*

$$\mathcal{B}_{ij;k} = \mathcal{B}_{ik;j} - \mathcal{A}_{ij;k} + \mathcal{A}_{ik;j} + \sigma_{ik}w_j - \sigma_{ij}w_k, \tag{10.1}$$

$$\mathcal{B}_{ii;11} = \mathcal{B}_{11;ii} + 2\mathcal{B}_{ii}\sigma_{11} - 2\mathcal{B}_{11}\sigma_{ii} - \mathcal{A}_{ii;11} + \mathcal{A}_{11;ii} - 2\mathcal{A}_{11}\sigma_{ii} + 2\mathcal{A}_{ii}\sigma_{11}. \tag{10.2}$$

Proof. The Ricci's identity yields that

$$w_{ij;k} = w_{ik;j} + \sigma_{ik}w_j - \sigma_{ij}w_k, \tag{10.3}$$

which yields (10.1). Using Ricci's identity repeatedly, (10.2) is a result of identity below

$$w_{ij;11} = w_{11;ij} + 2w_{ij}\sigma_{11} - 2\sigma_{ij}w_{11} - w_{i1}\sigma_{1j} + \sigma_{i1}w_{1j}.$$

□

Lemma 10.2. *Let $w \in C^{4,2}_{x,t}(\bar{\Omega} \times [0, T))$ be a uniformly c-convex solution of (4.18). Given $(x, t) \in \Omega \times [0, T)$, let $\{\boldsymbol{e}_i\}_{i=1}^n$ be a local smooth frame of $\mathbb{S}^n$ such that $\{\boldsymbol{e}_i(x)\}_{i=1}^n$ is an orthonormal basis of $T_x\mathbb{S}^n$ and the matrix $\mathcal{B}_{ij}(x, t)$ is diagonalized with $\mathcal{B}_{11}$ being its maximal eigenvalue. Then there exists $C > 0$ depending on $f, g, F|_{[0,d_0-\varepsilon_0]}$ (see (4.32) for ε_0) and w_0, but independent of T, such that at (x, t) there holds*

$$\mathscr{L}\mathcal{B}_{11} \geq \mathcal{B}^{ii}\mathcal{B}^{jj}\mathcal{B}^2_{ij;1} + \nabla^2_{11}\Phi - C\mathcal{R}, \tag{10.4}$$

where $\mathcal{R} := (1 + \mathcal{B}_{11})(1 + \operatorname{tr}\mathcal{B}^{-1})$, $\mathscr{L}$ and Φ are defined in (8.15) and (i) of Lemma 8.5.

Proof. The MTW condition (see (1.27)) yields that $D^2_{Z_1Z_1}\mathcal{A}_{ii} \geq 0$ for any $i \neq 1$. Then

$$\begin{aligned}\mathcal{B}^{ii}(D^2_{Z_kZ_l}\mathcal{A}_{ii}w_{k1}w_{l1} - D^2_{Z_kZ_l}\mathcal{A}_{11}w_{ki}w_{li}) &\geq \sum_{i\neq 1}\mathcal{B}^{ii}(D^2_{Z_1Z_1}\mathcal{A}_{ii}w^2_{11} - D^2_{Z_iZ_i}\mathcal{A}_{11}w^2_{ii}) - C\mathcal{R}\\ &\geq -C\mathcal{R}.\end{aligned} \tag{10.5}$$

Using (10.1), we have

$$\mathcal{B}^{ii}(D_{Z_k}\mathcal{A}_{ii}w_{k1;1} - D_{Z_k}\mathcal{A}_{11}w_{ki;i}) \geq \mathcal{B}^{ii}D_{Z_k}\mathcal{A}_{ii}\mathcal{B}_{11;k} - \mathcal{B}^{ii}D_{Z_k}\mathcal{A}_{11}\mathcal{B}_{ii;k} - C\mathcal{R}. \tag{10.6}$$

Differentiating the differential equation in (4.18), we obtain

$$\mathcal{B}^{ii}\mathcal{B}_{ii;k} - \frac{\partial_t w_k}{1 - \partial_t w} = \nabla_k\Phi, \tag{10.7}$$

$$\mathcal{B}^{ii}\mathcal{B}_{ii;11} - \frac{\partial_t w_{11}}{1 - \partial_t w} = \frac{(\partial_t w_1)^2}{(1 - \partial_t w)^2} + \mathcal{B}^{ii}\mathcal{B}^{jj}\mathcal{B}^2_{ij;1} + \nabla^2_{11}\Phi, \tag{10.8}$$

where $\Phi = \log \Psi(x, \nabla w)$. On the other hand, (10.2) gives

$$\begin{aligned}\mathscr{L}\mathcal{B}_{11} &= \mathcal{B}^{ii}(\mathcal{B}_{11;ii} - D_{Z_k}\mathcal{A}_{ii}\mathcal{B}_{11;k}) - \frac{\partial_t \mathcal{B}_{11}}{1-\partial_t w}\\ &\geq \mathcal{B}^{ii}\mathcal{B}_{ii;11} - \frac{\partial_t w_{11}}{1-\partial_t w} + \mathcal{B}^{ii}(\mathcal{A}_{ii;11} - \mathcal{A}_{11;ii})\\ &\quad -\mathcal{B}^{ii}D_{Z_k}\mathcal{A}_{ii}\mathcal{B}_{11;k} + D_{Z_k}\mathcal{A}_{11}\frac{\partial_t w_k}{1-\partial_t w} - C\mathcal{R}.\end{aligned}$$

Recall that (4.15). Employing (10.5)–(10.8), we further obtain

$$\begin{aligned}\mathscr{L}\mathcal{B}_{11} &\geq \mathcal{B}^{ii}\mathcal{B}^{jj}\mathcal{B}^2_{ij;1} + \nabla^2_{11}\Phi - \mathcal{B}^{ii}D_{Z_k}\mathcal{A}_{11}\mathcal{B}_{ii;k} + D_{Z_k}\mathcal{A}_{11}\frac{\partial_t w_k}{1-\partial_t w} - C\mathcal{R}\\ &\geq \mathcal{B}^{ii}\mathcal{B}^{jj}\mathcal{B}^2_{ij;1} + \nabla^2_{11}\Phi - C\mathcal{R}.\end{aligned}$$

$\square$

Lemma 10.3. *Let $w \in C^{4,2}_{x,t}(\bar\Omega \times [0,T))$ be a uniformly c-convex solution of (4.18). Suppose (1.17) holds, and (Ω, f), (Ω^*, g) are Aleksandrov related. Then there exists $C > 0$ depending on $f, g, F|_{[0,d_0-\varepsilon_0]}$ and w_0, but independent of T, such that*

$$\sup_{\Omega}\big(\max_{\xi\in\mathbb{S}^n}\mathcal{B}_{\xi\xi}(\cdot,t)\big) \leq C\Big(1 + \sup_{\partial\Omega}\big(\max_{\xi\in\mathbb{S}^n}\mathcal{B}_{\xi\xi}(\cdot,t)\big)\Big), \quad \forall t\in[0,T). \tag{10.9}$$

Proof. Consider the auxiliary function

$$\widetilde{W}(x,\xi,t) = \log\mathcal{B}_{\xi\xi} + \frac{A}{2}|\nabla w|^2,\ (x,\xi,t)\in T\mathbb{S}^n\times[0,T),\ |\xi|=1,$$

where $A \geq 1$ is a constant to be specified later. Given a $t_* < T$, suppose $\max_{\bar\Omega\times[0,t_*]} W$ attains at $(x_0,t_0)\in \operatorname{Int}\Omega\times(0,t_*]$. By a rotation of coordinates, we assume $\mathcal{B}_{ij}$ is diagonal and $\mathcal{B}_{11}\geq\mathcal{B}_{ii}$ for any $i\geq 2$ at (x_0,t_0). Hence $W(x,t) = \log\mathcal{B}_{11} + \frac{A}{2}|\nabla w|^2$ achieve its maximum at (x_0,t_0). By (10.1),

$$\mathcal{B}_{1i;1} = \mathcal{B}_{11;i} + \mathcal{X}_i,$$

where $\mathcal{X}_i = D_{Z_k}\mathcal{A}_{11}w_{ki} - D_{Z_k}\mathcal{A}_{1i}w_{k1} + \sigma_{11}w_i - \sigma_{1i}w_1$. Note that (4.15) is used. Then

$$\begin{aligned}\mathcal{B}^{11}\mathcal{B}^{ii}\mathcal{B}^{jj}\mathcal{B}^2_{ij;1} - (\mathcal{B}^{11})^2\mathcal{B}^{ii}\mathcal{B}^2_{11;i} &\geq (\mathcal{B}^{11})^2\sum_{i\neq1}\mathcal{B}^{ii}(2\mathcal{B}^2_{1i;1} - \mathcal{B}^2_{11;i})\\ &= \mathcal{B}^{-2}_{11}\sum_{i\neq1}\mathcal{B}^{ii}\big((\mathcal{B}_{11;i}+2\mathcal{X}_i)^2 - 2\mathcal{X}_i^2\big)\\ &\geq -C - C\operatorname{tr}\mathcal{B}^{-1}.\end{aligned}\tag{10.10}$$

Using (10.3), we have

$$\begin{aligned}\mathcal{B}^{ii}(w_{ik;i}w_k + w^2_{ik}) &= \mathcal{B}^{ii}w_{ii;k}w_k + \mathcal{B}^{ii}(|\nabla w|^2 - w_i^2) + \mathcal{B}^{ii}(\mathcal{B}_{ik}+\mathcal{A}_{ik})^2\\ &\geq \mathcal{B}^{ii}\mathcal{B}_{ii;k}w_k + \mathcal{B}^{ii}D_{Z_l}\mathcal{A}_{ii}w_{lk}w_k + \operatorname{tr}\mathcal{B} - C.\end{aligned}\tag{10.11}$$

Note that $\nabla W(x_0,t_0)=0$. Therefore $\mathcal{B}^{11}\mathcal{B}_{11;i}+Aw_kw_{ki}=0$. Hence, at (x_0,t_0),

$$\begin{aligned}\mathcal{B}^{11}\nabla^2_{11}\Phi+A\langle\nabla\Phi,\nabla w\rangle &\geq D_{Z_i}\Phi(\mathcal{B}^{11}\mathcal{B}_{11;i}+Aw_kw_{ki})-C\mathcal{B}_{11}-CA\\ &\geq -C\mathcal{B}_{11}-CA.\end{aligned}\tag{10.12}$$

By Lemma 10.2, we have

$$\begin{aligned}0\geq \mathscr{L}W|_{(x_0,t_0)} &= \mathcal{B}^{11}\mathscr{L}\mathcal{B}_{11}-(\mathcal{B}^{11})^2\mathcal{B}^{ii}\mathcal{B}^2_{11;i}+\frac{A}{2}\mathscr{L}|\nabla w|^2\\ &\geq \mathcal{B}^{11}\mathcal{B}^{ii}\mathcal{B}^{jj}\mathcal{B}^2_{ij;1}-(\mathcal{B}^{11})^2\mathcal{B}^{ii}\mathcal{B}^2_{11;i}+\mathcal{B}^{11}\nabla^2_{11}\Phi-C-C\operatorname{tr}\mathcal{B}^{-1}\\ &\quad +A\mathcal{B}^{ii}(w_{ik;i}w_k+w^2_{ik})-A\mathcal{B}^{ii}D_{Z_k}\mathcal{A}_{ii}w_{kl}w_l-\frac{A\langle\nabla w,\nabla w_t\rangle}{1-\partial_t w}.\end{aligned}$$

This together with (10.10) and (10.11) yields

$$0\geq \mathcal{B}^{11}\nabla^2_{11}\Phi+A\operatorname{tr}\mathcal{B}+A\mathcal{B}^{ii}\mathcal{B}_{ii;k}w_k-A\frac{\langle\nabla w,\nabla w_t\rangle}{1-\partial_t w}-CA-C\operatorname{tr}\mathcal{B}^{-1}.$$

By (10.7) and (10.12), we further obtain, at (x_0,t_0),

$$\begin{aligned}0 &\geq \mathcal{B}^{11}\nabla^2_{11}\Phi+A\langle\nabla\Phi,\nabla w\rangle+A\operatorname{tr}\mathcal{B}-CA-C\operatorname{tr}\mathcal{B}^{-1}\\ &\geq (A-C)\operatorname{tr}\mathcal{B}-CA-C\operatorname{tr}\mathcal{B}^{-1}.\end{aligned}$$

Taking A sufficiently large, we have for any $\varepsilon\in(0,1)$,

$$\operatorname{tr}\mathcal{B}[w]|_{(x_0,t_0)}\leq \varepsilon\sup_{\Omega\times[0,t_*]}\operatorname{tr}\mathcal{B}^{-1}[w]+C_\varepsilon,\tag{10.13}$$

where C_ε is a positive constant depending on ε. Applying (10.13) to $v(y,t)$, the c-transform of $w(x,t)$, and using (9.9), we deduce that

$$\operatorname{tr}\mathcal{B}^{-1}[w]\leq C\operatorname{tr}\mathcal{B}[v]\leq C\varepsilon\sup_{\Omega^*\times[0,t_*]}\operatorname{tr}\mathcal{B}^{-1}[v]+C_\varepsilon\leq C\varepsilon\sup_{\Omega\times[0,t_*]}\operatorname{tr}\mathcal{B}[w]+C_\varepsilon.$$

Combining this with (10.13) and taking ε small, we obtain $\sup_{\Omega\times[0,t_*]}\operatorname{tr}\mathcal{B}[w]\leq C$. Since t_* is arbitrary, we see the interior bound of $\mathcal{B}_{\xi\xi}$. As a result, (10.9) holds. □

10.2. **Boundary C^2-estimates.** For simplicity of notations, we denote

$$\begin{aligned}M &= \sup_{\Omega\times[0,T)}\max_{\xi\in\mathbb{S}^n}\mathcal{B}_{\xi\xi}(x,t),\\ M_1 &= \sup_{\partial\Omega\times[0,T)}\max_{\xi\in\mathbb{S}^n}\mathcal{B}_{\xi\xi}(x,t),\\ M_2 &= \sup_{\partial\Omega\times[0,T)}\max_{\xi\in\mathbb{S}^n,\xi\cdot\vec{\boldsymbol{n}}=0}\mathcal{B}_{\xi\xi}(x,t),\end{aligned}$$

where $\vec{\boldsymbol{n}}$ is the unit outer normal of $\partial\Omega$. Recall $H(x,Z)=\varrho^*(G(x,Z))$ in Section 7. Let

$$\beta(x,t)=D_ZH(x,\nabla w).$$

By (7.1), $\Theta=\langle\beta,\vec{\boldsymbol{n}}\rangle$. Recall $\Theta|_{\partial\Omega}$ is independent of the choice of defining functions.

Lemma 10.4. *Let $w \in C^{4,2}_{x,t}(\bar{\Omega}\times[0,T))$ be uniformly c-convex solution of (4.18). Suppose (1.17) holds, and (Ω, f), (Ω^*, g) are Aleksandrov related. Then there exists $C > 0$ depending on f, g, w_0, $F|_{[0,d_0-\varepsilon_0]}$ (see (4.32) for ε_0) and the modulus of c-convexity of Ω, Ω^*, but independent of T, such that*

$$\mathcal{B}_{\beta\beta}(x,t) \leq C(1+M)^{\frac{n-2}{n-1}}, \quad \forall (x,t) \in \partial\Omega \times [0,T). \tag{10.14}$$

Proof. Given $(x,t) \in \partial\Omega \times [0,T)$, let $\{\boldsymbol{e}_i\}_{i=1}^n$ be a local smooth frame of $\mathbb{S}^n$ so that $\{\boldsymbol{e}_i(x_0)\}_{i=1}^n$ is orthonormal and $\boldsymbol{e}_n(x)$ is the unit outer normal of $\partial\Omega$ at x. Let $\bar{H}(x,t) = H(x, \nabla w)$. For any $\xi = \xi^i \boldsymbol{e}_i$, using (7.6) and (7.3), we have

$$\nabla_\xi \bar{H} = \xi^i \bar{H}_i = \mathcal{B}_{\beta\xi}. \tag{10.15}$$

The boundary condition in (4.18) gives $\bar{H}|_{\partial\Omega} = 0$. It then follows that

$$\mathcal{B}_{\beta\tau} = 0 \ \forall \tau \in T_x\partial\Omega, \text{ and } \mathcal{B}_{\beta\vec{\boldsymbol{n}}} \geq 0. \tag{10.16}$$

By Lemma 7.2 and the arithmetic geometric mean inequality, we have

$$[\operatorname{tr}\mathcal{B}]^{\frac{1}{n-1}} \leq C[\frac{\operatorname{tr}\mathcal{B}}{\det\mathcal{B}}]^{\frac{1}{n-1}} \leq C \operatorname{tr}\mathcal{B}^{-1}.$$

Combining this with (i) in Lemma 8.5, we have

$$\mathscr{L}\bar{H} - D_{Z_k}\Phi \cdot \nabla_k \bar{H} \leq C(\operatorname{tr}\mathcal{B} + \operatorname{tr}\mathcal{B}^{-1}) \leq C(1+M)^{\frac{n-2}{n-1}} \operatorname{tr}\mathcal{B}^{-1}.$$

Applying (ii) of Lemma 8.5 to $(1+M)^{-\frac{n-2}{n-1}}\bar{H}$, we see $\bar{H}_{\vec{\boldsymbol{n}}} \leq C(1+M)^{\frac{n-2}{n-1}}$. By (10.16), (10.15) and $\Theta = \langle \beta, \vec{\boldsymbol{n}}\rangle$, we obtain

$$\mathcal{B}_{\beta\beta}(x,t) = \Theta\mathcal{B}_{\beta\vec{\boldsymbol{n}}} = \Theta\bar{H}_{\vec{\boldsymbol{n}}} \leq C(1+M)^{\frac{n-2}{n-1}}.$$

□

Remark 10.1. *Given $x \in \partial\Omega$ and a unit vector $\xi \in T_x\mathbb{S}^n$, write*

$$\xi = \tau + \frac{\xi \cdot \vec{\boldsymbol{n}}}{\Theta}\beta, \quad \textit{with } \tau \in T_x\partial\Omega. \tag{10.17}$$

It follows by (10.16), Lemma 9.1, Lemma 10.3 and Lemma 10.4 that

$$\mathcal{B}_{\xi\xi}|_{\partial\Omega} = \mathcal{B}_{\tau\tau} + \frac{(\xi\cdot\vec{\boldsymbol{n}})^2}{\Theta^2}\mathcal{B}_{\beta\beta} \leq CM_2 + C(1+M_1)^{\frac{n-2}{n-1}}. \tag{10.18}$$

By the arbitrariness of (x,t), it is easily verified that

$$\frac{M_1}{M_2} \leq C. \tag{10.19}$$

Lemma 10.5. *Let $w \in C^{4,2}_{x,t}(\bar{\Omega}\times[0,T))$ be uniformly c-convex solution of (4.18). Then there exists $C > 0$ independent of T such that*

$$\sup_{\partial\Omega\times[0,T)} \max_{\xi\in\mathbb{S}^n} \mathcal{B}_{\xi\xi} \leq C.$$

Proof. Fix $t_* < T$. Assume $\sup_{\partial\Omega\times[0,t_*]}\sup_{|\xi|=1,\xi\cdot\vec{\boldsymbol{n}}=0}\mathcal{B}_{\xi\xi}$ occurs at (x_0,t_0) and $\xi_0\in T_{x_0}\partial\Omega$. Suppose $\{\boldsymbol{e}_i(x_0)\}_{i=1}^n$ is orthonormal, $\xi_0=\boldsymbol{e}_1(x_0)$ and $\mathcal{B}_{ij}|_{(x_0,t_0)}$ is diagonal. Let $\tau(x,t)$ be as in (10.17) with ξ replaced by $\boldsymbol{e}_1(x)$. Then $\tau|_{\partial\Omega}$ is tangential to $\partial\Omega$. By Lemma 9.1,

$$|\tau|^2\le 1-\frac{2\boldsymbol{e}_1\cdot\vec{\boldsymbol{n}}}{\Theta}\beta\cdot\boldsymbol{e}_1+C(\boldsymbol{e}_1\cdot\vec{\boldsymbol{n}})^2.$$

By the identity in (10.18), the estimate of $|\tau|^2$, and Lemmas 9.1 & 10.4 that, for $t\le t_*$,

$$\mathcal{B}_{11}|_{\partial\Omega}\le\big(1-\frac{2\boldsymbol{e}_1\cdot\vec{\boldsymbol{n}}}{\Theta}\beta\cdot\boldsymbol{e}_1+C(\boldsymbol{e}_1\cdot\vec{\boldsymbol{n}})^2\big)\mathcal{B}_{11}|_{(x_0,t_0)}+C(1+M)^{\frac{n-2}{n-1}}(\boldsymbol{e}_1\cdot\vec{\boldsymbol{n}})^2. \tag{10.20}$$

With $\widetilde{H}$ in Lemma 8.4, we consider the auxiliary function

$$Q(x,t)=\frac{\mathcal{B}_{11}(x,t)}{\mathcal{B}_{11}(x_0,t_0)}+2\frac{\boldsymbol{e}_1\cdot\vec{\boldsymbol{n}}}{\Theta}\langle\beta,\boldsymbol{e}_1\rangle+A\widetilde{H}(x,\nabla w),$$

where $A>0$ is to be determined. By (10.20), we see that, for $t\in[0,t_*]$,

$$Q|_{\partial\Omega}\le 1+C(1+M)^{\frac{n-2}{n-1}}d^2(x,x_0). \tag{10.21}$$

In view of Lemma 10.2, we obtain

$$\begin{aligned}\mathscr{L}\mathcal{B}_{11} &\ge \nabla^2_{11}\Phi-C(1+\operatorname{tr}\mathcal{B})(1+\operatorname{tr}\mathcal{B}^{-1})\\ &\ge D_{Z_k}\Phi\cdot\mathcal{B}_{11;k}-C[\operatorname{tr}\mathcal{B}]^2-C(1+\operatorname{tr}\mathcal{B})(1+\operatorname{tr}\mathcal{B}^{-1}).\end{aligned}$$

By Lemma 10.3 and Remark 10.1, we have $\sup_{\Omega\times[0,t_*]}\operatorname{tr}\mathcal{B}\le C\mathcal{B}_{11}(x_0,t_0)$. Therefore

$$\frac{\mathscr{L}\mathcal{B}_{11}(x,t)}{\mathcal{B}_{11}(x_0,t_0)}\ge\frac{D_{Z_k}\Phi\cdot\mathcal{B}_{11;k}}{\mathcal{B}_{11}(x_0,t_0)}-C(1+\operatorname{tr}\mathcal{B}+\operatorname{tr}\mathcal{B}^{-1}). \tag{10.22}$$

Appying (i) in Lemma 8.5 to $-V$, where $V(x,Z)=2(\boldsymbol{e}_1\cdot\vec{\boldsymbol{n}})\frac{\langle\nabla_Z H,\boldsymbol{e}_1\rangle}{\langle\nabla_Z H,\nabla\varrho\rangle}+A\widetilde{H}(x,Z)$, and using (i) in Lemma 8.4, we obtain

$$\mathscr{L}\bar{V}-D_{Z_k}\Phi\cdot\nabla_k\bar{V}\ge D^2_{Z_iZ_j}V\cdot\mathcal{B}_{ij}-C\operatorname{tr}\mathcal{B}^{-1}\ge(A\delta_0^*-C)\operatorname{tr}\mathcal{B}-C\operatorname{tr}\mathcal{B}^{-1}. \tag{10.23}$$

Since $Q=\frac{\mathcal{B}_{11}(x,t)}{\mathcal{B}_{11}(x_0,t_0)}+\bar{V}(x,t)$ we conclude by (10.22) and (10.23) that

$$\mathscr{L}Q-D_{Z_k}\Phi\cdot\nabla_kQ\ge -C\operatorname{tr}\mathcal{B}^{-1}, \tag{10.24}$$

where we take A large and use $1\le C\operatorname{tr}\mathcal{B}^{-1}$ (by Lemma 7.2). By virtue of (10.21) and (10.24), a slight modification of the proof of (ii) in Lemma 8.5 gives

$$-\nabla_{\vec{\boldsymbol{n}}}Q|_{(x_0,t_0)}\le C.$$

By Lemma 9.1, Lemma 10.3 and Lemma 10.4, we further obtain

$$-\nabla_\beta\mathcal{B}_{11}|_{(x_0,t_0)}\le C(1+\mathcal{B}_{\beta\beta})\mathcal{B}_{11}(x_0,t_0)\le C(1+M_1)^{\frac{2n-3}{n-1}}. \tag{10.25}$$

Since $H(x,\nabla w)|_{\partial\Omega}\equiv 0$, we have, at (x_0,t_0),

$$0=\nabla^2_{11}H\ge\mathcal{B}_{11;k}D_{Z_k}H+\mathcal{B}_{k1}\mathcal{B}_{l1}D^2_{Z_kZ_l}H-C(1+M_1).$$

This together with (10.25) yields, at (x_0, t_0),

$$\mathcal{B}_{k1}\mathcal{B}_{l1}D^2_{Z_kZ_l}H \leq C(1+M_1)^{\frac{2n-3}{n-1}}. \tag{10.26}$$

Using (i) in Lemma 8.4, (10.26) and (10.19), we deduce that

$$\delta_0^* M_2^2 \leq C(1+M_1)^{\frac{2n-3}{n-1}} \leq C(1+M_2)^{\frac{2n-3}{n-1}}.$$

Therefore, $M_2 \leq C$. We complete the proof by using (10.19) again. $\square$

11. Appendix: Strict convexity and $C^{1,\alpha}$-regularity

The main purpose of this appendix is to show the strict convexity and differentiability of potential functions under MTW condition.

Theorem 11.1. *Let c be a cost function in $\mathscr{C}^\infty_{d_0}$ satisfying the MTW condition, Ω and Ω^* be two domains in $\mathbb{S}^n$ that are uniformly c-convex to each other. Assume f and g are two functions on Ω and Ω^* respectively such that*

$$\frac{1}{\Lambda} \leq f \leq \Lambda \quad \textit{and} \quad \frac{1}{\Lambda} \leq g \leq \Lambda, \tag{11.1}$$

for some $\Lambda > 0$. Suppose also Ω, Ω^* satisfy condition (1.17), and (Ω, f), (Ω^*, g) are Aleksandrov related. Then the potential function w of (1.14) is strictly c-convex and $w \in C^{1,\alpha}(\Omega)$ for some $\alpha > 0$.

Such result was proved by many authors [10, 16, 19, 44]. Our situation is a little bit different from theirs, as the cost function $c(x, y)$ in $\mathscr{C}^\infty_{d_0}$ is locally bounded and smooth only when $d(x, y) < d_0$ and diverges to infinity as $d(x, y) \to d_0$. Hence $c(\cdot, y)$ cannot be smoothly defined on the whole Ω for fixed y in general. But the ideas in the above-mentioned literatures are still available after some modifications. For readers' convenience, we sketch the proof of Theorem 11.1 in this appendix.

11.1. Loeper's maximum principle and some outcomes.

Suppose Ω and Ω^* are c-convex w.r.t. each other and satisfy (1.17). Given $y_0 \in \Omega^*$, let $\{x_t\}_{t\in[0,1]} \in \Omega_{y_0}$ be a c-segment w.r.t. y_0, namely

$$\nabla_y c(x_t, y_0) = \eta_t := (1-t)\eta_0 + t\eta_1 \in T_{y_0}\mathbb{S}^n.$$

Consider the c-functions

$$\phi_t^*(y) := c(x_t, y) - c(x_t, y_0), \ y \in \mathbb{S}^n. \tag{11.2}$$

Since $c \in \mathscr{C}^\infty_{d_0}$ satisfies the MTW condition, we have the following Loeper's maximum principle [26, 31, 41]

$$\phi_t^*(y) \leq \max\{\phi_0^*(y), \phi_1^*(y)\}, \quad \forall y \in \Omega^*. \tag{11.3}$$

Remark 11.1. *Fix any $y \in \Omega^*$ and consider the function $\mathcal{E}(t) := \phi_t^*(y)$. The c-convexity of Ω^* w.r.t. Ω and the MTW condition yield a crucial property: $\ddot{\mathcal{E}}(t) \geq 0$ whenever $\dot{\mathcal{E}}(t) = 0$ on any subinterval of $[0,1]$ with $d(x_t, y) < d_0$. Hence there is no peak on the graph of $\mathcal{E}$ along the entire $[0,1]$. This gives (11.3). See the proof of Lemma 5.6 for further details.*

Since $c \in \mathscr{C}^\infty_{d_0}$ is symmetric in x and y, the above result still hold when the roles of x and y are exchanged.

Let $u \in C(\bar{\Omega})$ be a c-convex function. Its subdifferential is defined for each $x \in \Omega$ by

$$\partial u(x) = \{\xi \in T_x\mathbb{S}^n : u(x') \geq u(x) + \langle \xi, \exp_x^{-1} x' \rangle + o(d(x, x'))\}.$$

We say the c-convex function u is indexed by Ω^* if $\chi_u(x) \subset \bar{\Omega}^*$ for all $x \in \Omega$. Thus

$$\chi_u(x) = \{y \in B_{d_0}(x) \cap \bar{\Omega}^* : u(x') \geq c(x', y) - c(x, y) + u(x), \ \forall x' \in \Omega\}.$$

Recall that the potential functions (w, v) of (1.14) are both c-convex indexed by Ω^* and Ω respectively. We also define the c-subdifferential $\partial^c u(x)$ for each $x \in \Omega$ by

$$\partial^c u(x) = \{\xi \in T_x\mathbb{S}^n : \xi = \nabla_x c(x, y), \ y \in \chi_u(x)\}.$$

Note that the subdifferential is always a convex set and $\partial^c u(x) \subseteq \partial u(x)$. In this section, we always assume that c-convex function $u \in C(\bar{\Omega})$ is indexed with Ω^*.

By exploiting the MTW condition and using Loeper's maximum principle, the following properties were known in [25, 26, 31, 41].

Proposition 11.1. *Let $c \in \mathscr{C}^\infty_{d_0}$ satisfy MTW condition and Ω, Ω^* be two domains that are c-convex to each other. If $u \in C(\bar{\Omega})$ is c-convex (indexed by Ω^*), then for all $x \in \Omega$,*

(i) *$\partial^c u(x) = \partial u(x)$;*
(ii) *$\chi_u(x)$ is connected and is c-convex w.r.t. x.*

Note that (i) in Proposition 11.1 implies that if $\phi_y(\cdot) := c(\cdot, y) + a$ is a local c-support of u at $x_0 \in \Omega$, namely $u(x) \geq \phi_y(x)$ for x in a neighbourhood of x_0 and $u(x_0) = \phi_y(x_0)$, then ϕ_y must be a (global) c-support, that is $u(x) \geq \phi_y(x)$ for all $x \in \Omega$.

Given $x_0 \in \Omega$ and $y \in B_{d_0}(x_0)$, let $\phi_{y;x_0}$ be the c-function normalized at x_0 as in (5.22). If $y_0 \in \chi_u(x_0)$, then $\phi_{y_0;x_0}(\cdot) + u(x_0)$ supports u at x_0. The contact set $\Sigma = \Sigma_u(x_0)$ is

$$\Sigma_u(x_0) = \{x \in \Omega : u(x) = u(x_0) + \phi_{y_0;x_0}(x)\}.$$

For $h > 0$, let $S_h = S_{h,u}(x_0)$ be the sub-level set of u:

$$S_{h,u}(x_0) = \{x \in \Omega : u(x) - u(x_0) - \phi_{y_0;x_0}(x) < h\}. \tag{11.4}$$

Let $u^c \in C(\bar{\Omega}^*)$ be the c-transform of u. We have $\Sigma_u(x_0) = \chi_{u^c}(y_0)$, and hence, by (ii) in Proposition 11.1, $\Sigma_u(x_0)$ is c-convex w.r.t. y_0. Since $S_{h,u}(x_0)$ is continuous in h w.r.t. the metric given by Hausdorff distance, and $S_{h,u}(x_0)$ converges to $\Sigma_u(x_0)$ in Hausdorff distance as $h \to 0$, we find that $S_{h,u}(x_0)$ is a connected and open set.

By (11.3), we further see that $S_{h,u}(x_0)$ is c-convex w.r.t. y_0. For this end, consider the following coordinate system around any given $\bar{x}_0 \in \bar{\Omega} \cap B_{d_0}(y_0)$: for every $x \in B_{d_0}(y_0)$,

$$x \mapsto \eta(x) := \nabla_y c(x, y_0) - \nabla_y c(\bar{x}_0, y_0) \in T_{y_0}\mathbb{S}^n \simeq \mathbb{R}^n, \tag{11.5}$$

Denote by $x(\eta)$ the inverse map of (11.5). Then $\bar{x}_0 = x(0)$. Set

$$\hat{\phi}_{y_0;\bar{x}_0}(\eta) := \phi_{y_0;\bar{x}_0}(x(\eta)),\ \ \eta \in T_{y_0}\mathbb{S}^n, \tag{11.6}$$

and consider the normalization of u at $\bar{x}_0$,

$$\hat{u}(\eta) = u(x(\eta)) - \hat{\phi}_{y_0;\bar{x}_0}(\eta) - u(\bar{x}_0), \quad \eta \in (\Omega)^\wedge_{y_0;\bar{x}_0}, \tag{11.7}$$

where $(\Omega)^\wedge_{y_0;\bar{x}_0} := (\Omega)^\wedge_{y_0} - \nabla_y c(\bar{x}_0, y_0)$ with $(\Omega)^\wedge_{y_0}$ being in Definition 8.1, which would be two unbounded convex domains in $T_{y_0}\mathbb{S}^n \simeq \mathbb{R}^n$.

We also introduce the normalized cost function at $\bar{x}_0$ via

$$\hat{c}(\eta, y) = \hat{c}(\eta, y; \bar{x}_0) := \hat{\phi}_{y;\bar{x}_0}(\eta),\ \ (\eta, y) \in T_{y_0}\mathbb{S}^n \times B_{d_0}(\bar{x}_0). \tag{11.8}$$

It is not hard to check that $\hat{u}$ is $\hat{c}$-convex and its contact set and sub-level set

$$\begin{aligned} \hat{\Sigma} = \hat{\Sigma}_{\hat{u}}(\eta(\bar{x}_0)) &:= \{\eta \in (\Omega)^\wedge_{y_0;\bar{x}_0} : \hat{u}(\eta) = 0\}, \\ \hat{S}_h = \hat{S}_{h,\hat{u}}(\eta(\bar{x}_0)) &:= \{\eta \in (\Omega)^\wedge_{y_0;\bar{x}_0} : \hat{u}(\eta) < h\}, \end{aligned} \tag{11.9}$$

are respectively related to Σ and S_h by

$$\hat{\Sigma} = (\Sigma)^\wedge_{y_0;\bar{x}_0} \quad \text{and} \quad \hat{S}_h = (S_h)^\wedge_{y_0;\bar{x}_0}.$$

Lemma 11.1. *Let $c \in \mathscr{C}^\infty_{d_0}$ satisfy MTW condition and Ω, Ω^* be two domains that are c-convex to each other. If $u \in C(\bar{\Omega})$ is c-convex (indexed by Ω^*), then for every $x_0 \in \Omega$ and $y_0 \in \chi_u(x_0)$, $\hat{S}_h$ is convex in $T_{y_0}\mathbb{S}^n \simeq \mathbb{R}^n$. Equivalently, S_h is c-convex w.r.t. y_0.*

Proof. Assume by contradiction $\hat{S}_h$ is not convex. Then there exist $\eta_0, \eta_1 \in \hat{S}_h$ such that the line segment $[\eta_0, \eta_1]$ does not entirely lies in $\hat{S}_h \cup \partial\hat{S}_h$. Let $\eta_t = (1-t)\eta_0 + t\eta_1$ and $x_t = x(\eta_t)$ for $t \in [0, 1]$. Take $t_0 \in (0, 1)$ so that $\eta_{t_0} \notin \hat{S}_h \cup \partial\hat{S}_h$. Let $\hat{\phi}_{y_{t_0}}(\eta) = \hat{c}(\eta, y_{t_0}) + a_{t_0}$ be a $\hat{c}$-support of $\hat{u}$ at η_{t_0}. By applying Loeper's maximum principle (11.3) to $y = y_{t_0}$,

$$h < \hat{\phi}_{y_{t_0}}(\eta_{t_0}) \le \max\{\hat{\phi}_{y_{t_0}}(\eta_0), \hat{\phi}_{y_{t_0}}(\eta_1)\} \le \max\{\hat{u}(\eta_0), \hat{u}(\eta_1)\} < h,$$

which leads to a contradiction. □

11.2. **Aleskandrov type estimates and c-cones.**

In this subsection we follow the ideas in [10], though adapted to our case. Given $\bar{x}_0, y_0$ with $d(\bar{x}_0, y_0) < d_0$, let $D \subset\subset B_{d_0}(y_0)$ be a domain which contains $\bar{x}_0$ and is c-convex w.r.t. y_0. We define for $h > 0$ a c-cone $\vee = \vee^{D,h}_{\bar{x}_0}$ with vertex $\bar{x}_0$ and base $\partial D \times h$ by

$$\vee^{D,h}_{\bar{x}_0}(x) := \sup\{\phi_{y;\bar{x}_0}(x) : \phi_{y;\bar{x}_0}(\cdot)|_{\partial D} \le \phi_{y_0;\bar{x}_0}(\cdot)|_{\partial D} + h,\ y \in B_{d_0}(\bar{x}_0)\},\ x \in D, \tag{11.10}$$

where $\phi_{y;\bar{x}_0}$ is given by (5.22). Let $\hat{D} = (D)^{\wedge}_{y_0;\bar{x}_0}$. In the coordinate (11.5), $\vee$ is written as a $\hat{c}$-cone $\hat{\vee}$ with base $\partial\hat{D}$ and vertex at the origin

$$\hat{\vee}(\eta) = \sup\{\hat{\phi}_{y;\bar{x}_0}(\eta) : \hat{\phi}_{y;\bar{x}_0}(\cdot) \le \hat{\phi}_{y_0;\bar{x}_0}(\cdot) + h \text{ on } \partial\hat{D}\},\ \eta \in \hat{D}. \tag{11.11}$$

where $\hat{\phi}_{y;\bar{x}_0}$ is given by (11.6).

To study the c-cone, we consider the coordinate around y_0 indicated below

$$y \mapsto \xi(y), \text{ where } L\xi(y) = \nabla_x c(\bar{x}_0, y) - \nabla_x c(\bar{x}_0, y_0), \tag{11.12}$$

and L is a linear transformation of $T_{\bar{x}_0}\mathbb{S}^n$ such that $L\xi := \xi^j L^i_j E_i$ with $\{E_j\}$ being the coordinate frame $T_{\bar{x}_0}\mathbb{S}^n$. Denote by $\{\bar{E}_i\}$ the coordinate frame of (11.5) in $T_{y_0}\mathbb{S}^n$. Take

$$L^i_j := \nabla^2_{x,y} c|_{(\bar{x}_0,y_0)}(E_i, \bar{E}_j). \tag{11.13}$$

Consider the modified cost function with reference $\bar{x}_0$,

$$\tilde{c}(\eta,\xi) = \tilde{c}(\eta,\xi;\bar{x}_0) := \hat{\phi}_{y(\xi);\bar{x}_0}(\eta) - \hat{\phi}_{y_0;\bar{x}_0}(\eta), \tag{11.14}$$

where $(\eta,\xi) \in T_{y_0}\mathbb{S}^n \times T_{\bar{x}_0}\mathbb{S}^n$ and $y(\xi)$ is the inverse of (11.12).

Lemma 11.2. *Let $c \in \mathscr{C}^\infty_{d_0}$. Then*

$$\tilde{c}(\eta,\xi) = \eta\cdot\xi + a_{ij,kl}\eta^i\eta^j\xi^k\xi^l, \tag{11.15}$$

where $a_{ij,kl} = a_{ij,kl}(\eta,\xi)$ are functions on $\mathbb{R}^{2n}$. Moreover, there exists $C = C_\varepsilon > 0$ such that $|a_{ij,kl}(\eta,\xi)| \le C_\varepsilon$ whenever $\sup\{d(x(s\eta), y(t\xi)) : (s,t) \in [0,1]^2\} \le d_0 - \varepsilon$.

Proof. Firstly note that

$$\tilde{c}(\eta,0) = \tilde{c}(0,\xi) = 0. \tag{11.16}$$

Differentiating (11.5) and (11.12) w.r.t. η and ξ respectively gives

$$\delta^i_j = \nabla^2_{x,y} c|_{(x,y_0)}(\boldsymbol{e}_j, \bar{E}_i) \quad \text{and} \quad L^i_j = \nabla^2_{x,y} c|_{(\bar{x}_0,y)}(E_i, \bar{\boldsymbol{e}}_j), \tag{11.17}$$

where $\boldsymbol{e}_j = \partial_{\eta^j} x(\eta)$ and $\bar{\boldsymbol{e}}_j = \partial_{\xi^j} y(\xi)$. From (11.6) and (11.14), we see that

$$\begin{aligned} D_{\eta^i}\tilde{c}(\eta,\xi) &= \langle \nabla_x c|_{(x,y)}, \boldsymbol{e}_i(x)\rangle - \langle \nabla_x c|_{(x,y_0)}, \boldsymbol{e}_i(x)\rangle, \\ D_{\xi^j}\tilde{c}(\eta,\xi) &= \langle \nabla_y c|_{(x,y)}, \bar{\boldsymbol{e}}_j(y)\rangle - \langle \nabla_y c|_{(\bar{x}_0,y)}, \bar{\boldsymbol{e}}_j(y)\rangle, \end{aligned} \tag{11.18}$$

and

$$D^2_{\eta^i,\xi^j}\tilde{c}(\eta,\xi) = \nabla^2_{x,y} c|_{(x,y)}(\boldsymbol{e}_i(x), \bar{\boldsymbol{e}}_j(y)). \tag{11.19}$$

Assume $\boldsymbol{e}_j(\bar{x}_0) = A^k_j E_k$ and $\bar{\boldsymbol{e}}_j(y_0) = \bar{A}^l_j \bar{E}_l$. By (11.17),

$$\delta^i_j = A^k_j \nabla^2_{x,y} c|_{(\bar{x}_0,y_0)}(E_k, \bar{E}_i) \quad \text{and} \quad L^i_j = \bar{A}^l_j \nabla^2_{x,y} c|_{(\bar{x}_0,y_0)}(E_i, \bar{E}_l) \tag{11.20}$$

The first formula in (11.20) together with (11.19) shows

$$D^2_{\eta^i,\xi^j}\tilde{c}|_{(0,0)} = A^k_i \bar{A}^l_j \nabla^2_{x,y} c|_{(\bar{x}_0,y_0)}(E_k, \bar{E}_l) = \bar{A}^i_j;$$

while the second one in (11.20) together with (11.13) gives $\bar{A}^i_j = \delta^i_j$. Hence

$$D^2_{\eta^i,\xi^j}\tilde{c}|_{(0,0)} = \delta^i_j. \tag{11.21}$$

Since (11.18) implies $D_{\eta^i}\tilde{c}(0,\xi) = \langle \xi, \boldsymbol{e}_i(\bar{x}_0)\rangle$ and $D_{\xi^j}\tilde{c}(\eta,0) = \langle \eta, \boldsymbol{e}_j(y_0)\rangle$, we also have

$$D^3_{\eta,\xi\xi}\tilde{c}(0,\xi) = 0 \text{ and } D^3_{\eta\eta,\xi}\tilde{c}(\eta,0) = 0. \tag{11.22}$$

Direct computation gives

$$\begin{aligned}
\int_0^1\int_0^1 D^2_{\eta^i,\xi^j}\tilde{c}(s\eta,t\xi)\eta^i\xi^j dtds &= \int_0^1 \frac{d}{ds}\Big(\int_0^1 \frac{d}{dt}\tilde{c}(s\eta,t\xi)dt\Big)ds \\
&= \int_0^1 \frac{d}{ds}\Big(\tilde{c}(s\eta,\xi) - \tilde{c}(s\eta,0)\Big)ds \\
&= \tilde{c}(\eta,\xi) - \tilde{c}(0,\xi) - \tilde{c}(\eta,0) + \tilde{c}(0,0) \\
&= \tilde{c}(\eta,\xi). \qquad (11.23)
\end{aligned}$$

In the last line, (11.16) is employed. On the other hand, we have

$$\begin{aligned}
D^2_{\eta^i,\xi^j}\tilde{c}(s\eta,t\xi) &= D^2_{\eta^i,\xi^j}\tilde{c}(0,t\xi) + \int_0^s D^3_{\eta^i\eta^k,\xi^j}\tilde{c}(s'\eta,t\xi)\eta^k ds' \\
&= D^2_{\eta^i,\xi^j}\tilde{c}(0,0) + \int_0^t D^3_{\eta^i,\xi^j\xi^k}\tilde{c}(0,t'\xi)\xi^k dt' \\
&\quad + \int_0^s\int_0^t D^4_{\eta^i\eta^k,\xi^j\xi^l}\tilde{c}(s'\eta,t'\xi)\eta^k\xi^l dt'ds' + \int_0^s D^3_{\eta^i\eta^k,\xi^j}\tilde{c}(s'\eta,0)\eta^k ds' \\
&= \delta^i_j + \int_0^s\int_0^t D^4_{\eta^i\eta^k,\xi^j\xi^l}\tilde{c}(s'\eta,t'\xi)\eta^k\xi^l dt'ds', \qquad (11.24)
\end{aligned}$$

where we use (11.21) and (11.22). Inserting (11.24) into (11.23), we obtain (11.15) with

$$a_{ij,kl}(\eta,\xi) = \int_0^1\int_0^1\int_0^s\int_0^t D^4_{\eta^i\eta^j,\xi^k\xi^l}\tilde{c}(s'\eta,t'\xi)dt'ds'dtds.$$

$\square$

Lemma 11.3. *If $c \in \mathscr{C}^\infty_{d_0}$ satisfies the MTW condition, then so does $\tilde{c}(\eta,\xi)$.*

Proof. It follows by (11.18) that

$$D^2_{\eta^i\eta^j}\tilde{c}(\eta,\xi) = \nabla^2_x c|_{(x,y)}(\boldsymbol{e}_i,\boldsymbol{e}_j) - \nabla^2_x c|_{(x,y_0)}(\boldsymbol{e}_i,\boldsymbol{e}_j).$$

As a result, the $\mathcal{A}$-tensor associated to the modified cost $\tilde{c}$ is given by

$$\widetilde{\mathcal{A}}_{ij}(\eta,Z) = \nabla^2_x c|_{(x,y)}(\boldsymbol{e}_i,\boldsymbol{e}_j) - \nabla^2_x c|_{(x,y_0)}(\boldsymbol{e}_i,\boldsymbol{e}_j)$$

where $y = y(\xi)$ is determined by Z via

$$Z_i = D_{\eta^i}\tilde{c}(\eta,\xi) = \langle \nabla_x c|_{(x,y)}, \boldsymbol{e}_i(x)\rangle - \langle \nabla_x c|_{(x,y_0)}, \boldsymbol{e}_i(x)\rangle.$$

Therefore

$$\widetilde{\mathcal{A}}_{ij}(\eta,Z) = \mathcal{A}_{ij}\big(x(\eta), Z + \nabla_x c(x,y_0)\big) + \nabla^2_x c|_{(x,y_0)}(\boldsymbol{e}_i,\boldsymbol{e}_j),$$

which leads to $D^2_Z\widetilde{\mathcal{A}}|_{(\eta,Z)} = D^2_Z\mathcal{A}|_{(x(\eta),Z+\nabla_x c(x,y_0))}$. Hence $\tilde{c}$ satisfies MTW condition. $\square$

The $\hat{c}$-cone (11.11) can be also expressed as the modified $\tilde{c}$-cone

$$\tilde{\vee}(\eta) = \sup\{\tilde{c}(\eta, \xi) : \tilde{c}(\cdot, \xi) \le h \text{ on } \partial\hat{D}\},\ \eta \in \hat{D}. \tag{11.25}$$

For each $\theta \in \mathbb{S}^{n-1}$, we take $m_\theta(t) = \max_{\partial\hat{D}} \tilde{c}(\cdot, t\theta)$. Since $c(\bar{x}_0, y) \to -\infty$ as $d(\bar{x}_0, y) \nearrow d_0$, $\lim_{t\to+\infty} m_\theta(t) = +\infty$. Hence $t_\theta = \sup\{t > 0 : m_\theta(t) = h\}$ exists. Since $\tilde{c}$ satisfies the MTW condition (Lemma 11.3), it follows by Loeper's maximum principle, for $t \in (0, t_\theta)$,

$$\tilde{c}(\eta, t\theta) \le \max\{\tilde{c}(\eta, 0), \tilde{c}(\eta, \xi_\theta)\},\ \forall\, \eta \in \hat{D},$$

where $\xi_\theta := t_\theta\theta$. As a result, $\tilde{c}(\eta, t\theta) \le h$ for all $\eta \in \partial\hat{D}$. By (11.25),

$$\tilde{\vee}(\eta) = \sup\left\{\tilde{c}(\eta, \xi_\theta) : \theta \in \mathbb{S}^{n-1}\right\},\ \eta \in \hat{D}. \tag{11.26}$$

For $\eta \in \hat{D} \setminus \{0\}$, let $\eta_b = s_b\eta \in \partial\hat{D}$ (for some unique $s_b > 0$). Since segment $[0, \eta_b]$ is $\tilde{c}$-convex w.r.t. $\xi = 0$, another use of Loeper's maximum principle and $\tilde{c}(\cdot, 0) \equiv 0$ shows that, for any given ξ,

$$\tilde{c}(\eta, \xi) \le \max\{\tilde{c}(0, \xi), \tilde{c}(\eta_b, \xi)\},\ \forall\, \eta \in \hat{D}.$$

As $\tilde{c}(\eta_b, \xi_\theta) \le h$, we find $\hat{D} \subset \{\eta : \tilde{c}(\eta, \xi_\theta) < h\}$ for all ξ_θ. Consequently

$$\hat{D} = \cap_{\theta\in\mathbb{S}^n}\{\eta : \tilde{c}(\eta, \xi_\theta) < h\} \quad \text{and} \quad \tilde{\vee}(\cdot) \le h \text{ on } \hat{D}.$$

It is not hard to see $\chi_{\tilde{\vee}}(0) = \{s\xi_\theta : \theta \in \mathbb{S}^{n-1},\ 0 \le s \le 1\}$ and is convex. We assert

$$\chi_{\tilde{\vee}}(0) = \chi_{\tilde{\vee}}(\hat{D}). \tag{11.27}$$

Fix any $\eta_0 \in \hat{D} \setminus \{0\}$. Suppose $\tilde{\vee}$ is differentiable at η_0. Then $\chi_{\tilde{\vee}}(\eta_0)$ is singleton. By (11.26), there exists $\theta_0 \in \mathbb{S}^{n-1}$ such that $\tilde{c}(\cdot, \xi_{\theta_0})$ supports $\tilde{\vee}$ at η_0. Hence $\chi_{\tilde{\vee}}(\eta_0) = \{\xi_{\theta_0}\}$. Next assume $\chi_{\tilde{\vee}}(\eta_0)$ is not singleton. By Proposition 11.1, $\partial\tilde{\vee}(\eta_0) = D_\eta\tilde{c}(\eta_0, \chi_{\tilde{\vee}}(\eta_0))$ is convex. Denote by $\mathrm{extr}\partial\tilde{\vee}(\eta_0)$ the extreme point of $\partial\tilde{\vee}(\eta_0)$. Since for every $p \in \mathrm{extr}\partial\tilde{\vee}(\eta_0)$, there exists a sequence $\eta_k \to \eta_0$ such that $\tilde{\vee}$ is differentiable at η_k and $D_\eta\tilde{c}(\eta_k, \xi_{\theta_k}) \to p$, where $\theta_k \in \mathbb{S}^{n-1}$ is so that $\tilde{c}(\cdot, \xi_{\theta_k})$ supports $\tilde{\vee}$ at η_k. As a result, $\mathrm{extr}\partial\tilde{\vee}(\eta_0) \subseteq \partial\tilde{\vee}(0)$. By the convexity, $\partial\tilde{\vee}(\eta_0) = \partial\tilde{\vee}(0)$. This proves (11.27). Hence, each $\tilde{c}$-support of $\tilde{\vee}$ at some point in the closure of $\hat{D}$ must also be a $\tilde{c}$-support at 0.

Suppose there is a positive constant $\Lambda > 0$ such that

$$\frac{1}{\Lambda}\gamma_{\mathbb{S}^n} \le \mu_u \le \Lambda\gamma_{\mathbb{S}^n} \quad \text{in } \Omega, \tag{11.28}$$

where μ_u is the nonnegative Radon measure associated to the c-convex function u (indexed by Ω^*) defined as (see [38]),

$$\mu_u(\omega) = \gamma_{\mathbb{S}^n}(\chi_u(\omega)),\ \text{for any Borel set } \omega \subset \Omega.$$

Given $x_0 \in \Omega$, $y_0 \in \chi_u(x_0)$, $h > 0$, and a subset $D \subset\subset \Omega_{y_0}$ which contains x_0 and is c-convex w.r.t. y_0, we aim to estimate $\phi_{y_0} + h - u$ in D, where

$$\phi_{y_0}(\cdot) = \phi_{y_0;x_0}(\cdot) + u(x_0) \tag{11.29}$$

is the c-support of u at x_0. For this end, a major tool is the c-cone. Given $\bar{x}_0 \in D$ (not necessarily $\bar{x}_0 = x_0$), consider the c-cone function $\vee_{\bar{x}_0} = \vee_{\bar{x}_0}^{D,h'_{\bar{x}_0}}$ defined by (11.10), where $h'_{\bar{x}_0} = \phi_{y_0}(\bar{x}_0) + h - u(\bar{x}_0)$. Thanks to (11.27), each c-support of $\vee_{\bar{x}_0}$ at some point in $\bar{D}$ must also be a c-support at $\bar{x}_0$. By c-convexity, we find

$$\chi_{\vee_{\bar{x}_0}}(D) \subset \chi_u(D), \tag{11.30}$$

provided (i) $\bar{x}_0 = x_0$ and $\phi_{y_0}(\cdot) + h \leq u(\cdot)$ on ∂D, or (ii) $\phi_{y_0}(\cdot) + h = u(\cdot)$ on ∂D. By (11.28), the size of right-hand side is estimated by $\gamma_{\mathbb{S}^n}(D)$. Thanks to Lemma 11.2, after some suitable changes (11.5) and (11.12), $\vee_{\bar{x}_0}$ becomes $\tilde{\vee}$ given by (11.25) in the new coordinate, which can be approximated by the standard Euclidean cone

$$\vee^* = \vee^{*\hat{D},h'_{\bar{x}_0}}(\eta) = \sup\{\eta \cdot \xi : \ \bar{\eta} \cdot \xi \leq h'_{\bar{x}_0} \text{ for all } \bar{\eta} \in \partial \hat{D}\}.$$

Therefore the ideas of Caffarelli [6] for studying cost $x \cdot y$ in $\mathbb{R}^n$ can be used. Since $\max\{d(\bar{x}_0, \chi_u(D)), d(y_0, D)\} \leq d_0 - \varepsilon_0$ provided diam(D) and h is small,

$$\max\Big\{\max_{\chi_u(D)} |\nabla_x c(\bar{x}_0, \cdot)|, \ \max_D |\nabla_y c(\cdot, y_0)|, \ |\nabla^2_{x,y} c(\bar{x}_0, y_0)|\Big\} \leq C = C_{\varepsilon_0}.$$

Hence the size of left-hand side of (11.30) can be estimated by that of $\vee^*$ and is given in terms of $h'_{\bar{x}_0}, D$ and the distance from $\bar{x}_0$ to ∂D. The upper bound of $h'_{\bar{x}_0}$ thus follows by (11.30). If $\phi_{y_0}(\cdot) + h \geq u(\cdot)$ on ∂D, by constructing a suitable barrier, the positive lower bound of h can be obtained.

More precisely, the following was proved in e.g. [10, Section 3] and [16, Section 6].

Proposition 11.2. *Let c, Ω, Ω^* be as in Theorem 11.1. Supose $u \in C(\bar{\Omega})$ is c-convex (indexed by Ω^*), and (11.28) holds. Let $x_0 \in \Omega$, $y_0 \in \chi_u(x_0)$, and $D \subset\subset \Omega_{y_0}$ be c-convex (w.r.t. y_0) containing x_0. Then there exist small constants $h_0, \delta_0 > 0$ and constant $C = C_{n,h_0,\delta_0,\Lambda} > 0$, such that the estimates below hold when diam$(D) < \delta_0$ and $h < h_0$:*

(i) *if $\phi_{y_0}(\cdot) + h \geq u(\cdot)$ on ∂D with ϕ_{y_0} given by (11.29), then $\gamma_{\mathbb{S}^n}(D) \leq C h^{n/2}$;*
(ii) *if $\phi_{y_0}(\cdot) + h \leq u(\cdot)$ on ∂D, then $h^{n/2} \leq C \gamma_{\mathbb{S}^n}(D)$;*
(iii) *if $\bar{x}_0 \in D$ is close to ∂D in the sense that there exist $\boldsymbol{e} \in \mathbb{S}^{n-1}$ and $\varepsilon, l > 0$ with*

$$\sup_{\eta \in \hat{D}} \langle \eta - \eta(\bar{x}_0), \boldsymbol{e} \rangle = \varepsilon l$$

and $\hat{D} = (D)^{\wedge}_{y_0;x_0}$ contains a line segment of length l parallel to $\boldsymbol{e}$, then

$$|\phi_{y_0}(\bar{x}_0) + h - u(\bar{x}_0)|^{n/2} \leq C \varepsilon \gamma_{\mathbb{S}^n}(D),$$

provided $\phi_{y_0}(\cdot) + h = u(\cdot)$ on ∂D. Note that C is independent of l.

11.3. **Proof of Theorem 11.1.**

In this subsection we follow the arguments in [10, 16] and sketch the proof of Theorem 11.1. Let w be the potential function. By Theorem 3.1 and (11.1), w satisfies (11.28) for another Λ. By the c-convexity, $\chi_w(\bar{\Omega}) = \bar{\Omega}^*$.

Strict convexity

For a contradiction we suppose there is a c-support ϕ_{y_0} such that

$$\Sigma = \{x \in \bar{\Omega} : w(x) = \phi_{y_0}(x)\}$$

contains more than one point in Ω. By Proposition 11.1, Σ is c-convex w.r.t. y_0. Making the coordinate change (11.5) (for some $\bar{x}_0 \in \bar{\Omega} \cap B_{d_0}(y_0)$ to be specified later), let again

$$\hat{\Omega} = (\Omega)^{\wedge}_{y_0;\bar{x}_0} \quad \text{and} \quad \hat{\Sigma} = (\Sigma)^{\wedge}_{y_0;\bar{x}_0}.$$

These two sets are convex in $T_{y_0}\mathbb{S}^n$. Since w is bounded while $\phi_{y_0}(x) \to -\infty$ as $d(x, y_0) \to d_0$, we see that $\hat{\Sigma}$ is closed and bounded.

Step1. Extreme points cannot be in the interior of $\hat{\Omega}$

Let $\bar{\eta}_0 \in \hat{\Omega}$ be an extreme point of $\hat{\Sigma}$. Take $\bar{x}_0$ in (11.5) such that $\bar{\eta}_0 = 0$. Note that the change of $\bar{x}_0$ only gives a translation in η-coordinate. Fix such $\bar{x}_0$ and make another coordinate change (11.12). Let $\hat{w}(\eta) = w(x(\eta)) - \hat{\phi}_{y_0;\bar{x}_0}(\eta) - w(\bar{x}_0)$. Then

$$\hat{w} \geq 0 \text{ in } \hat{\Omega},\ \hat{w}(0) = 0, \text{ and } \hat{\Sigma} = \{\hat{w} = 0\}.$$

Let $\Sigma^h = \{x \in \Omega : w(x) < \phi_{y_0}(x)\}$ be the sub-level set of w, which is c-convex w.r.t. y_0 by Lemma 11.1. Let $D = D_r \subset \Omega_{y_0}$ be an open set containing $\bar{x}_0$ such that $\mathbb{B}_r(0) = (D)^{\wedge}_{y_0;\bar{x}_0}$ is an open ball of $\Omega^{\wedge}_{y_0;\bar{x}_0}$ centered at $\bar{x}_0 = x(0)$ with radius r. Clearly $D_r \cap \Sigma^h$ is c-convex w.r.t. y_0 and

$$\phi_{y_0}(\cdot) + h \geq w(\cdot) \text{ on } \partial(D_r \cap \Sigma^h).$$

By (i) in Proposition 11.2, if $h < h_0$ and $r < r_0$ then

$$\begin{aligned} h^{n/2} &\geq \gamma_{\mathbb{S}^n}(D_r \cap \Sigma^h)/C_{h_0,r_0} \\ &\geq \gamma_{\mathbb{S}^n}(\Sigma^h)/C_{h_0,r}, \end{aligned} \tag{11.31}$$

where we use $\gamma_{\mathbb{S}^n}(\Sigma^h) \leq C_r \gamma_{\mathbb{S}^n}(D_r \cap \Sigma^h)$ for some constant C_r depending on r.

By a rotation of coordinate, $\hat{\Sigma}$ is contained in $\{(\eta_1, \cdots, \eta_n) : \eta_1 < 0\}$. Take a small $a^- > 0$ such that $\hat{\Sigma} \cap \{\eta_1 > -a^-\}$ is compactly contained in $\hat{\mathcal{U}}$. As shown in [10, Lemma 4.1], Lemma 11.2 yields the existence of a sequence ξ_t, $\xi_t \to 0$ as $t \to 0$, such that

$$\hat{D}_t = \{\eta \in \hat{\mathcal{U}} : \hat{w}(\eta) < \tilde{c}(\eta, \xi_t) + \lambda_t\}, \text{ where } \lambda_t = -\tilde{c}(-\frac{1}{2}a^- e_1, \xi_t),$$

enjoys the following properties

(i) $\hat{D}_t$ is compactly contained in $\hat{\Omega}$;
(ii) $\inf\{\eta_1 : \eta = (\eta_1, \cdots, \eta_n) \in D_t\} \leq -\frac{1}{2}a^-$;
(iii) $a_t^+ = \sup\{\eta_1 : \eta = (\eta_1, \cdots, \eta_n) \in D_t\} \to 0$ as $t \to 0$.

It is not hard to see from (11.14) that $\hat{D}_t = (D_t)^{\wedge}_{y_0;x_0}$, where

$$D_t = \{x \in \Omega : w(x) < \phi_{y_t;\bar{x}_0}(x) + \lambda_t\},\ y_t = y(\xi_t).$$

Hence we can apply (iii) in Proposition 11.2 to $D = D_t$, $\boldsymbol{e} = e_1$, $\varepsilon = a_t^+/a^-$, $h = a_t$ and $u = w$, thus conclude there exists a small $t_0 > 0$ so that if $t \in (0, t_0)$ then

$$\lambda_t^{n/2} \le C_{t_0} \frac{a_t^+}{a^-} \gamma_{\mathbb{S}^n}(D_t).$$

By the construction of ξ_t in [10] and using Lemma 11.2, $\lambda_t \approx \frac{1}{2} a^- t$ for small t. Hence

$$t^{n/2} \le C_{t_0} a_t^+ \gamma_{\mathbb{S}^n}(D_t),\ \text{for all } t \in (0, t_0). \tag{11.32}$$

If $h = a^- t$ then $D_t \subset \Sigma^h$ when t is small. It then follows by (11.31) and (11.32),

$$a_t^+ \ge C_{t_0},\ \forall t \to 0,$$

for another positive constant C_{t_0} which is independent of t. We arrive a contradiction.

Step2. Extreme points cannot be on the boundary of $\hat{\Omega}$

In [16], the authors showed that χ_w does not mix the interiors and boundaries of two uniformly c-convex domains: namely

(a) if $\mu_w \ge \Lambda^{-1}\gamma_{\mathbb{S}^n}$ on $\bar{\Omega}$ and Ω^* is c-convex w.r.t. Ω, then $\chi_w(x) \subset \Omega^*$, $\forall x \in \Omega$;

(b) if $\mu_w \le \Lambda\gamma_{\mathbb{S}^n}$ on $\bar{\Omega}$ and Ω is c-convex w.r.t. Ω^*, then $\chi_w(x) \cap \Omega^* = \emptyset$, $\forall x \in \partial\Omega$.

For (a), assume by contradiction $x_0 \in \Omega$ and $y_0 \in \chi_w(x_0) \cap \partial\Omega^*$. In the coordinate systems (11.5) and (11.12) (with $\bar{x}_0 = x_0$), let us consider the cost $\tilde{c}(\eta, \xi)$ defined by (11.14) and $\tilde{c}$-convex function $\hat{w}(\eta) = w(x(\eta)) - \hat{\phi}_{y_0;x_0}(\eta) - w(x_0)$. Suppose e is the unit outer normal of $\hat{\Omega}^* = (\Omega^*)^{\wedge}_{x_0;y_0}$ at the origin in $T_{x_0}\mathbb{S}^n$. Consider the cones

$$\mathcal{C} = \{t\eta \in T_{y_0}\mathbb{S}^n :\ t \in [0, \varepsilon) \text{ and } \eta \in B'_\theta(e)\},$$

where $B'_\theta(e)$ is the geodesic ball of $\mathbb{S}^{n-1}$ with radius $\theta > 0$ and center $e \in \mathbb{S}^{n-1}$, and

$$\mathcal{C}^* = \{\xi \in T_{x_0}\mathbb{S}^n :\ -e \cdot \xi \le \theta|\xi| + C_0\varepsilon|\xi|^2\}.$$

Employing Lemma 11.2, $\tilde{c}$ is close to $\eta \cdot \xi$, it was shown in [16, Section 5] that $\chi_{\tilde{w}}(\mathcal{C}) \subset \mathcal{C}^* \cap \hat{\Omega}^*$ when θ and ε are small. This together with the uniform c-convexity of Ω^* yields

$$\varepsilon O(\theta^{n-1}) = |\mathcal{C}| \le C|\mathcal{C}^* \cap \hat{\Omega}^*| \le C\theta^{n+1}.$$

We arrive a contradiction by fixing ε small and taking $\theta \to 0$. Conclusion (b) then follows by applying (a) to the dual function $\hat{w}^{\tilde{c}}$.

Let us turn to the strict convexity of w. Assume to the contrary Σ contains more than one point. By Step1, we have $0 = x^{-1}(\bar{x}_0)$, an extreme point $\hat{\Sigma}$, is on $\partial\hat{\Omega}$. Since Σ is c-convex w.r.t. y_0, Σ contains an interior point of Ω, say x_1. By conclusion (a) above, $y_0 \in \chi_w(x_1)$ is in the interior of Ω^*. However, as $\bar{x}_0 \in \partial\Omega$, we find from conclusion (b) that $y_0 \in \partial\Omega^*$, a contradiction.

$C^{1,\alpha}$-regularity

The proof of C^1-regularity is essentially the same as that of strict convexity. The readers are referred to [10, Corollary 4.1] and [16, Theorem 8.3]. Since the regularity is a local property, by the coordinates (11.5) and (11.12), it suffices to study the $C^{1,\alpha}$-smoothness of the potential function $\hat{w}(\eta) = w(x(\eta)) - \hat{\phi}_{y_0;\bar{x}_0}(\eta) - w(\bar{x}_0)$ related to cost (11.14) in the Euclidean ambient space. Therefore the regularity result follows exactly by the same argument in [10, 16], see also [19, 44]. The readers may consult the references for details, and we do not sketch the proof here.

References

[1] Aleksandrov, A.-D.: An application of the domain invariance theorem to the proofs of existence. Izv. Akad. Nauk SSSR Ser. Mat. 3 (1939), 243-255.

[2] Aleksandrov, A.-D: Existence and uniqueness of a convex surface with a given integral curvature. C. R. (Doklady) Acad. Sci. URSS (N.S.) 35 (1942), 131-134.

[3] Böröczky, K.J.; Lutwak, E.; Yang, D.; Zhang, G.Y.; Zhao. Y.M.: The Gauss image problem. Comm. Pure Appl. Math. 73 (2020), 1406-1452.

[4] Brenier, Y.: Polar factorization and monotone rearrangement of vector-valued functions. Comm. Pure Appl. Math. 44 (1991), no. 4, 375–417.

[5] Caffarelli, L. A.: Boundary regularity of maps with convex potentials. Comm. Pure Appl. Math. 45 (1992), 1141–1151.

[6] Caffarelli, L. A.: The regularity of mappings with a convex potential. J. Amer. Math. Soc. 5 (1992), 99–104.

[7] Caffarelli, L. A.: Boundary regularity of maps with convex potentials. II. Ann. of Math. (2) 144 (1996), 453–496.

[8] Caffarelli, L. A.: Allocation maps with general cost functions. Partial differential equations and applications, 29–35, Lecture Notes in Pure and Appl. Math., 177, Dekker, New York, 1996.

[9] Chen, S.; Liu, J.; Wang, X.-J.: Global regularity for the Monge-Ampère equation with natural boundary condition. Ann. of Math. (2) 194 (2021), 745–793.

[10] Chen, S.; Wang, X.-J.: Strict convexity and $C^{1,\alpha}$ regularity of potential functions in optimal transportation under condition A3w. J. Differential Equations 260 (2016), 1954–1974.

[11] De Philippis, G.; Figalli, A.: The Monge-Ampère equation and its link to optimal transportation. Bull. Amer. Math. Soc. 51 (2014), 527–580.

[12] Delanoë, P.: Classical solvability in dimension two of the second boundary-value problem associated with the Monge-Ampère operator. Ann. Inst. H. Poincaré Anal. Non Linéaire 8 (1991), 443–457.

[13] Delanoë, P.: On the smoothness of the potential function in Riemannian optimal transport. Comm. Anal. Geom. 23 (2015), 11–89.

[14] Delanoë, P.; Ge, Y.: Regularity of optimal transport on compact, locally nearly spherical, manifolds. J. Reine Angew. Math. 646 (2010), 65–115.

[15] Evans, L. C.: Partial differential equations and Monge-Kantorovich mass transfer, in Current Developments in Mathematics, 1997 (Cambridge, MA), Int. Press, Boston, MA, 1999, pp. 65–126.

[16] Figalli, A.; Kim, Y.-H.; McCann, R. J.: Hölder continuity and injectivity of optimal maps. Arch. Ration. Mech. Anal. 209 (2013), no. 3, 747–795.

[17] Figalli, A.; Kim, Y.-H.; McCann, R. J.: Regularity of optimal transport maps on multiple products of spheres. J. Eur. Math. Soc. 15 (2013), 1131–1166.

[18] Guan, P.; Wang, X.-J.: On a Monge-Ampère equation arising in geometric optics. J. Differential Geom.48 (1998), 205–223.

[19] Guillen, N.; Kitagawa, J.: On the local geometry of maps with c-convex potentials. Calc. Var. Partial Differential Equations52(2015), 345–387.

[20] Gutiérrez, C.E.; Huang, Q.B.: The refractor problem in reshaping light beams. Arch. Ration. Mech. Anal. 193 (2009), 423–443.

[21] Gutiérrez, C.E.: Optimal transport and applications to geometric optics. SpringerBriefs PDEs Data Sci. Springer, Singapore, 2023, x+135 pp.

[22] Gangbo, W., McCann, R.J.: Optimal maps in Monge's mass transport problem. C.R. Acad. Sci. Paris, Series I, Math. 321 (1995), 1653–1658.

[23] Kantorovich, L.: On the transfer of masses. Dokl. Acad. Nauk. USSR **37** (1942), 227–229.

[24] Kantorovich, L.: On a problem of Monge. Uspekhi Mat. Nauk. **3** (1948), 225–226.

[25] Kim, Y.-H.; McCann, R.J.: On the cost-subdifferentials of cost-convex functions. Available at ArXiv:0706.1226v1.

[26] Kim,Y.-H.; McCann, R.J.: Continuity, curvature, and the general covariance of optimal transportation. J. Eur. Math. Soc. (JEMS) 12 (2010), 1009–1040.

[27] Kim, Y.-H.; McCann, R. J.: Towards the smoothness of optimal maps on Riemannian submersions and Riemannian products (of round spheres in particular). J. Reine Angew. Math. 664 (2012), 1–27.

[28] Kim, Y.-H.; Streets, J.; Warren, M.: Parabolic optimal transport equations on manifolds. Int. Math. Res. Not. IMRN (2012), 4325–4350.

[29] Kitagawa, J.: A parabolic flow toward solutions of the optimal transportation problem on domains with boundary. J. Reine Angew. Math. 672 (2012), 127–160.

[30] Krylov, N. V.: Nonlinear elliptic and parabolic equations of the second order. D. Reidel Publishing Co., Dordrecht (1987).

[31] Loeper, G.: On the regularity of solutions of optimal transportation problems. Acta Math. 202 (2009), no. 2, 241–283.

[32] Loeper, G.: Regularity of optimal maps on the sphere: the quadratic cost and the reflector antenna. Arch. Ration. Mech. Anal. 199 (2011), 269–289.

[33] Loeper, G.; Villani, C.: Regularity of optimal transport in curved geometry: the nonfocal case. Duke Math. J. 151 (2010), no. 3, 431–485.

[34] Li, Q.-R.; Wang, X.-J.: A class of optimal transportation problems on the sphere (in Chinese). Sci. Sin. Math. 48:1 (2018), 181–200.

[35] Lieberman, G. M.: Second order parabolic differential equations. World Scientific, Singapore, 1996.

[36] Liu, J.; Trudinger, N. S.; Wang, X.-J.: Interior $C^{2,\alpha}$ regularity for potential functions in optimal transportation. Comm. Partial Differential Equations 35 (2010), 165–184.

[37] Liu, J.; Trudinger, N. S.; Wang, X.-J.: On asymptotic behaviour and $W^{2,p}$ regularity of potentials in optimal transportation. Arch. Ration. Mech. Anal. 215 (2015), 867–905.

[38] Ma, X.N.; Trudinger, N.S.; Wang, X.J.: Regularity of potential functions of the optimal transportation problem. Arch. Ration. Mech. Anal. 177 (2005), 151-183.

[39] Monge, G.: Mémoire sur la Theorie des Déblais et des Remblais, In: Historie de l'Académie Royale des Sciences de Paris, avec les Mémories de Mathématique et de Physique pour la Même année, 1781, 666–740.

[40] Oliker, V.: Embedding $\mathbb{S}^n$ into $\mathbb{R}^{n+1}$ with given integral Gauss curvature and optimal mass transport on $\mathbb{S}^n$. Adv. Math. 213 (2007), 600-620.

[41] Trudinger, N.S.; Wang, X.-J.: On strict convexity and continuous differentiability of potential functions in optimal transportation. Arch. Ration. Mech. Anal. 192 (2009), 403-418.

[42] Trudinger, N. S.; Wang, X.-J.: On the second boundary value problem for Monge-Ampère type equations and optimal transportation. Ann. Scuola Norm. Sup. Pisa Cl. Sci. (5) Vol. VIII (2009), 143-174.

[43] Urbas, J.: On the second boundary value problem for equations of Monge-Ampère type. J. Reine Angew. Math. 487 (1997), 115–124.

[44] Vétois, J.: Continuity and injectivity of optimal maps. Calc. Var. Partial Differential Equations 52 (2015), 587–607.
[45] Villani, C.: Topics in Optimal Transportation. Grad. Stud. Math. 58, Amer. Math. Soc., Providence, RI, 2003.
[46] Villani, C.: Optimal Transport, Old and New. Grundlehren Math. Wissen., Springer-Verlag, Berlin, 2009.
[47] Wang, X.-J.: On the design of reflector antenna. Inverse Problems 12 (1996), 351–375.
[48] Wang, X.-J.: On the design of a reflector antenna II. Calc. Var. Partial Differential Equations 20 (2004), 329–341.

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